



Polytechnic University of the Philippines
College of Engineering
Computer Engineering Department

Solved Problems in Feedback and Control Systems

(Electrical Network Transfer Functions, Translational Mechanical System Transfer Functions, Rotational Mechanical System Transfer Functions, Transfer Function involving Gears, Electromechanical System Transfer Function, Block Diagram Algebra, and Signal Flow Graphs)

Completion requirement for the subject
CMPE 309 - Feedback and Control Systems

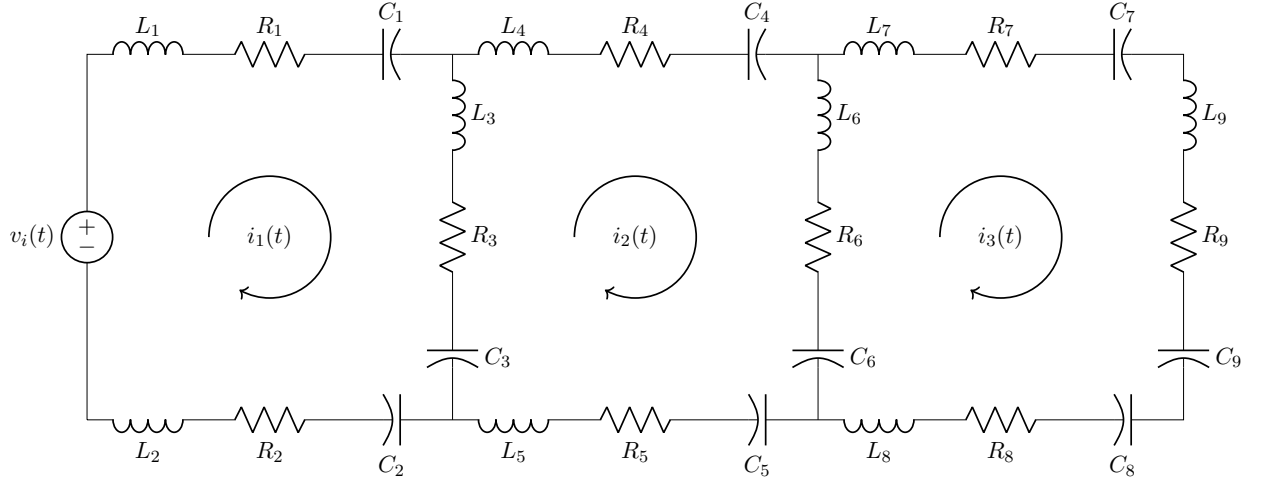
YAMBOT, Cedric G.
2023-18849-MN-1
Bachelor of Science in Computer Engineering

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Problem 1

Given the electrical network diagram below, determine the transfer function $G(s) = I_3(s)/V_i(s)$.



Given:

$$\begin{aligned} R_1 &= 1 \Omega & L_1 &= 1 \text{ H} & C_1 &= 1 \text{ F} \\ R_2 &= 1 \Omega & L_2 &= 1 \text{ H} & C_2 &= 1 \text{ F} \\ R_3 &= 1 \Omega & L_3 &= 1 \text{ H} & C_3 &= 1 \text{ F} \\ R_4 &= 1 \Omega & L_4 &= 1 \text{ H} & C_4 &= 1 \text{ F} \\ R_5 &= 1 \Omega & L_5 &= 1 \text{ H} & C_5 &= 1 \text{ F} \\ R_6 &= 1 \Omega & L_6 &= 1 \text{ H} & C_6 &= 1 \text{ F} \\ R_7 &= 1 \Omega & L_7 &= 1 \text{ H} & C_7 &= 1 \text{ F} \\ R_8 &= 1 \Omega & L_8 &= 1 \text{ H} & C_8 &= 1 \text{ F} \\ R_9 &= 1 \Omega & L_9 &= 1 \text{ H} & C_9 &= 1 \text{ F} \end{aligned}$$

Required:

$$G(s) = \frac{I_3(s)}{V_i(s)}$$

Solution:

System of Equations:

$$\begin{aligned} (3K)I_1(s) - (K)I_2(s) &= V_i(s) \\ -(K)I_1(s) + (4K)I_2(s) - (K)I_3(s) &= 0 \\ -(K)I_2(s) + (4K)I_3(s) &= 0 \end{aligned}$$

where $K = \frac{s^2+s+1}{s}$.

Matrix Form:

$$\begin{bmatrix} 3K & -K & 0 \\ -K & 4K & -K \\ 0 & -K & 4K \end{bmatrix} \begin{bmatrix} I_1(s) \\ I_2(s) \\ I_3(s) \end{bmatrix} = \begin{bmatrix} V_i(s) \\ 0 \\ 0 \end{bmatrix}$$

Determinant Δ :

$$\Delta = \begin{vmatrix} 3K & -K & 0 \\ -K & 4K & -K \\ 0 & -K & 4K \end{vmatrix} = 41K^3$$

Using Cramer's Rule for $I_3(s)$:

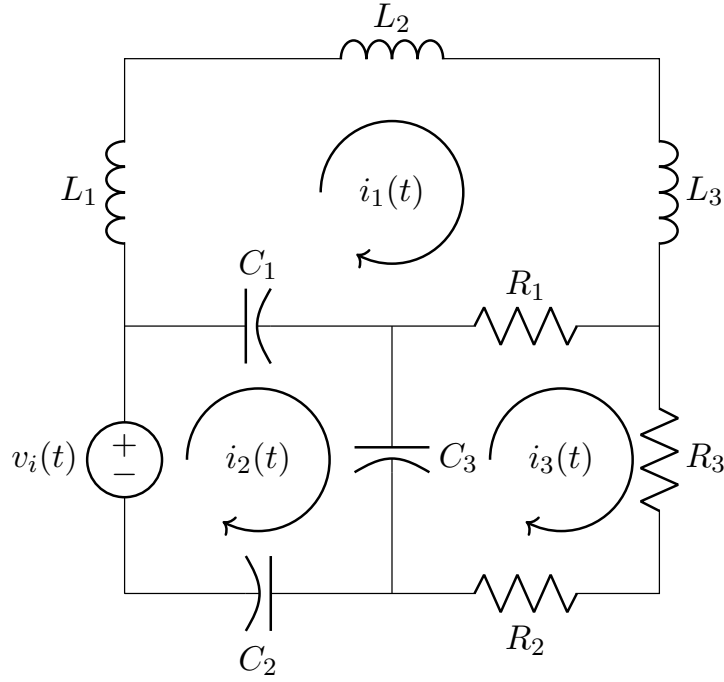
$$I_3(s) = \frac{\begin{vmatrix} 3K & -K & V_i(s) \\ -K & 4K & 0 \\ 0 & -K & 0 \end{vmatrix}}{\Delta} = \frac{K^2 V_i(s)}{41K^3} = \frac{V_i(s)}{41K}$$

Transfer Function:

$$G(s) = \frac{I_3(s)}{V_i(s)} = \frac{s}{41s^2 + 41s + 41}$$

Problem 2

Given the electrical network diagram below, determine the transfer function $G(s) = I_3(s)/V_i(s)$.



Given:

$$\begin{aligned} R_1 &= 1 \Omega & L_1 &= 1 \text{ H} & C_1 &= 1 \text{ F} \\ R_2 &= 1 \Omega & L_2 &= 1 \text{ H} & C_2 &= 1 \text{ F} \\ R_3 &= 1 \Omega & L_3 &= 1 \text{ H} & C_3 &= 1 \text{ F} \end{aligned}$$

Required:

$$G(s) = \frac{I_3(s)}{V_i(s)}$$

Solution:

System of Equations:

$$\begin{aligned} (3s^2 + s + 1)I_1(s) - I_2(s) - sI_3(s) &= 0 \\ -I_1(s) + 3I_2(s) - I_3(s) &= sV_i(s) \\ -sI_1(s) - I_2(s) + (2s + 1)I_3(s) &= 0 \end{aligned}$$

Matrix Form:

$$\begin{bmatrix} 3s^2 + s + 1 & -1 & -s \\ -1 & 3 & -1 \\ -s & -1 & 2s + 1 \end{bmatrix} \begin{bmatrix} I_1(s) \\ I_2(s) \\ I_3(s) \end{bmatrix} = \begin{bmatrix} 0 \\ sV_i(s) \\ 0 \end{bmatrix}$$

Determinant Δ :

$$\Delta = \begin{vmatrix} 3s^2 + s + 1 & -1 & -s \\ -1 & 3 & -1 \\ -s & -1 & 2s + 1 \end{vmatrix} = 18s^3 + 9s^2 + 6s + 1$$

Using Cramer's Rule for $I_3(s)$:

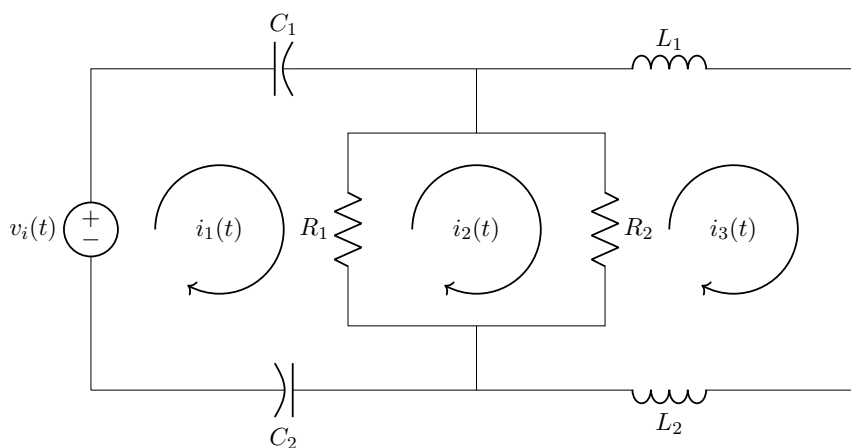
$$I_3(s) = \frac{\begin{vmatrix} 3s^2 + s + 1 & -1 & 0 \\ -1 & 3 & sV_i(s) \\ -s & -1 & 0 \end{vmatrix}}{\Delta} = \frac{sV_i(s)(3s^2 + 2s + 1)}{18s^3 + 9s^2 + 6s + 1}$$

Transfer Function:

$$G(s) = \frac{I_3(s)}{V_i(s)} = \frac{3s^3 + 2s^2 + s}{18s^3 + 9s^2 + 6s + 1}$$

Problem 3

Given the electrical network diagram below, determine the transfer function $G(s) = I_3(s)/V_i(s)$.



Given:

$$\begin{aligned} R_1 &= 1 \, \Omega & L_1 &= 1 \, \text{H} & C_1 &= 1 \, \text{F} \\ R_2 &= 1 \, \Omega & L_2 &= 1 \, \text{H} & C_2 &= 1 \, \text{F} \end{aligned}$$

Required:

$$G(s) = \frac{I_3(s)}{V_i(s)}$$

Solution:

System of Equations:

$$\begin{aligned} (s+2)I_1(s) - sI_2(s) &= sV_i(s) \\ -I_1(s) + 2I_2(s) - I_3(s) &= 0 \\ -I_2(s) + (2s+1)I_3(s) &= 0 \end{aligned}$$

Matrix Form:

$$\begin{bmatrix} s+2 & -s & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2s+1 \end{bmatrix} \begin{bmatrix} I_1(s) \\ I_2(s) \\ I_3(s) \end{bmatrix} = \begin{bmatrix} sV_i(s) \\ 0 \\ 0 \end{bmatrix}$$

Determinant Δ :

$$\Delta = \begin{vmatrix} s+2 & -s & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2s+1 \end{vmatrix} = 2s^2 + 8s + 2$$

Using Cramer's Rule for $I_3(s)$:

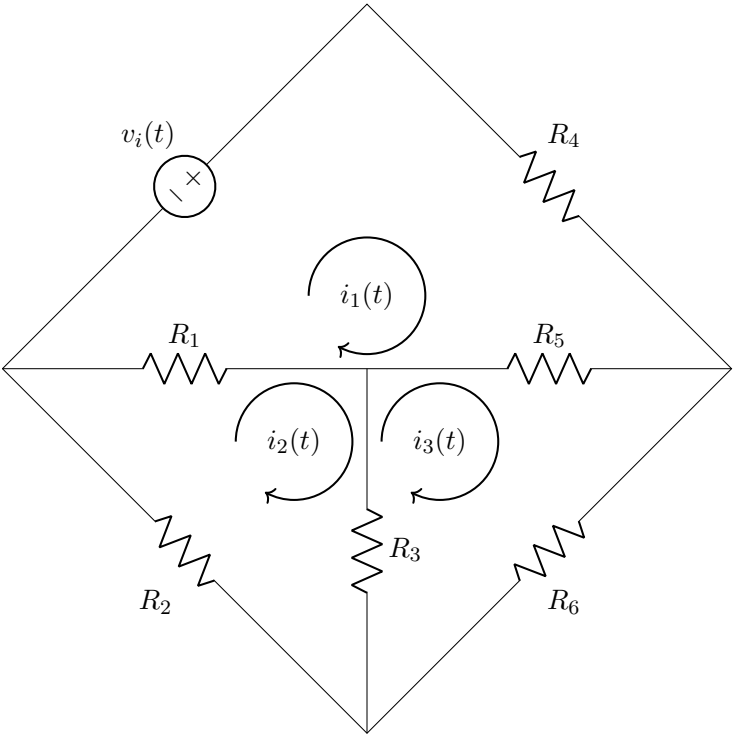
$$I_3(s) = \frac{\begin{vmatrix} s+2 & -s & sV_i(s) \\ -1 & 2 & 0 \\ 0 & -1 & 0 \end{vmatrix}}{\Delta} = \frac{sV_i(s)}{2s^2 + 8s + 2}$$

Transfer Function:

$$G(s) = \frac{I_3(s)}{V_i(s)} = \frac{s}{2s^2 + 8s + 2}$$

Problem 4

Given the electrical network diagram below, determine the transfer function $G(s) = I_3(s)/V_i(s)$.



Given:

$$R_1 = 1 \, \Omega$$

$$R_2 = 1 \, \Omega$$

$$R_3 = 1 \, \Omega$$

$$R_4 = 1 \, \Omega$$

$$R_5 = 1 \, \Omega$$

$$R_6 = 1 \, \Omega$$

Required:

$$G(s) = \frac{I_3(s)}{V_i(s)}$$

Solution:

System of Equations:

$$3I_1(s) - I_2(s) - I_3(s) = V_i(s)$$

$$-I_1(s) + 3I_2(s) - I_3(s) = 0$$

$$-I_1(s) - I_2(s) + 3I_3(s) = 0$$

Matrix Form:

$$\begin{bmatrix} 3 & -1 & -1 \\ -1 & 3 & -1 \\ -1 & -1 & 3 \end{bmatrix} \begin{bmatrix} I_1(s) \\ I_2(s) \\ I_3(s) \end{bmatrix} = \begin{bmatrix} V_i(s) \\ 0 \\ 0 \end{bmatrix}$$

Determinant Δ :

$$\Delta = \begin{vmatrix} 3 & -1 & -1 \\ -1 & 3 & -1 \\ -1 & -1 & 3 \end{vmatrix} = 16$$

Using Cramer's Rule for $I_3(s)$:

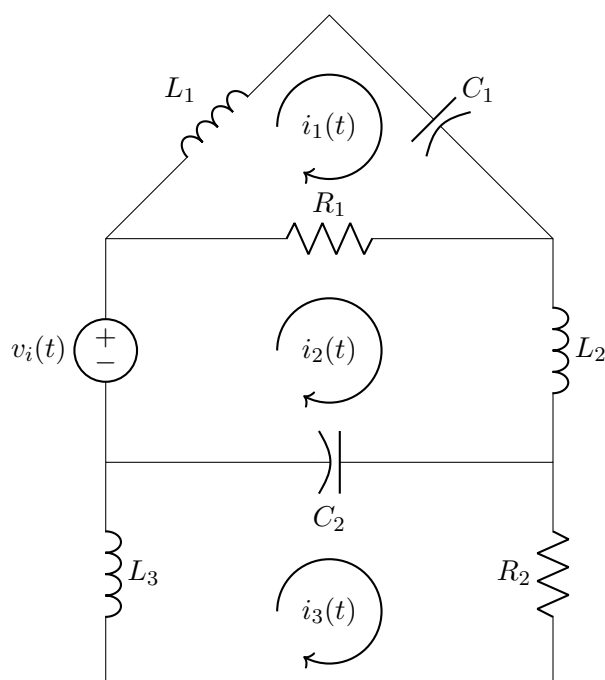
$$I_3(s) = \frac{\begin{vmatrix} 3 & -1 & V_i(s) \\ -1 & 3 & 0 \\ -1 & -1 & 0 \end{vmatrix}}{\Delta} = \frac{4V_i(s)}{16} = \frac{V_i(s)}{4}$$

Transfer Function:

$$G(s) = \frac{I_3(s)}{V_i(s)} = \frac{1}{4}$$

Problem 5

Given the electrical network diagram below, determine the transfer function $G(s) = I_3(s)/V_i(s)$.



Given:

$$R_1 = 1 \Omega \quad L_1 = 1 \text{ H} \quad C_1 = 1 \text{ F}$$

$$R_2 = 1 \Omega \quad L_2 = 1 \text{ H} \quad C_2 = 1 \text{ F}$$

$$L_3 = 1 \text{ H}$$

Required:

$$G(s) = \frac{I_3(s)}{V_i(s)}$$

Solution:

System of Equations:

$$\begin{aligned} (s^2 + s + 1)I_1(s) - sI_2(s) &= 0 \\ -sI_1(s) + (s^2 + s + 1)I_2(s) - I_3(s) &= sV_i(s) \\ -I_2(s) + (s^2 + s + 1)I_3(s) &= 0 \end{aligned}$$

Matrix Form:

$$\begin{bmatrix} s^2 + s + 1 & -s & 0 \\ -s & s^2 + s + 1 & -1 \\ 0 & -1 & s^2 + s + 1 \end{bmatrix} \begin{bmatrix} I_1(s) \\ I_2(s) \\ I_3(s) \end{bmatrix} = \begin{bmatrix} 0 \\ sV_i(s) \\ 0 \end{bmatrix}$$

Determinant Δ :

$$\Delta = \begin{vmatrix} s^2 + s + 1 & -s & 0 \\ -s & s^2 + s + 1 & -1 \\ 0 & -1 & s^2 + s + 1 \end{vmatrix} = s^6 + 3s^5 + 5s^4 + 6s^3 + 4s^2 + 2s$$

Using Cramer's Rule for $I_3(s)$:

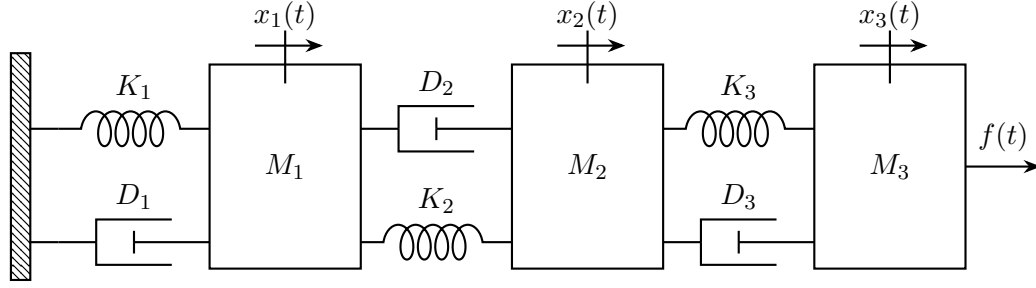
$$I_3(s) = \frac{\begin{vmatrix} s^2 + s + 1 & -s & 0 \\ -s & s^2 + s + 1 & sV_i(s) \\ 0 & -1 & 0 \end{vmatrix}}{\Delta} = \frac{sV_i(s)(s^2 + s + 1)}{s^6 + 3s^5 + 5s^4 + 6s^3 + 4s^2 + 2s}$$

Transfer Function:

$$G(s) = \frac{I_3(s)}{V_i(s)} = \frac{s^2 + s + 1}{s^5 + 3s^4 + 5s^3 + 6s^2 + 4s + 2}$$

Problem 6

For the translational mechanical system diagram shown below, determine the transfer function $G(s) = X_1(s)/F(s)$.



Given:

As per drawing

$$\begin{aligned} M_1 &= 1 \text{ kg} & K_1 &= 6 \text{ N/m} & D_1 &= 3 \text{ N} \cdot \text{s/m} \\ M_2 &= 2 \text{ kg} & K_2 &= 4 \text{ N/m} & D_2 &= 2 \text{ N} \cdot \text{s/m} \\ M_3 &= 2 \text{ kg} & K_3 &= 4 \text{ N/m} & D_3 &= 2 \text{ N} \cdot \text{s/m} \end{aligned}$$

Required:

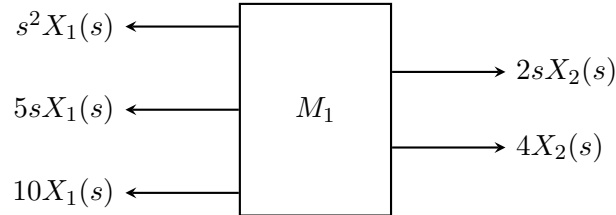
$$G(s) = \frac{X_1(s)}{F(s)}$$

Solution:

Deriving the Equations of Motion

We apply Newton's second law ($\sum F = ma$) to each mass, with the positive direction of motion to the right.

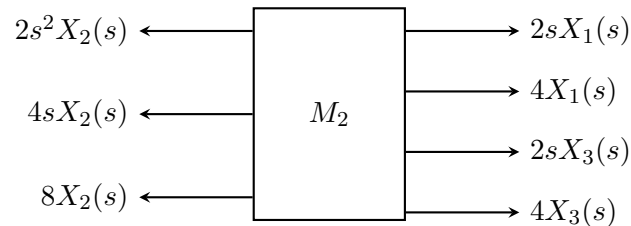
For Mass M_1 :



The resulting equation in the Laplace domain is:

$$(s^2 + 5s + 10)X_1(s) - (2s + 4)X_2(s) = 0 \quad (1)$$

For Mass M_2 :



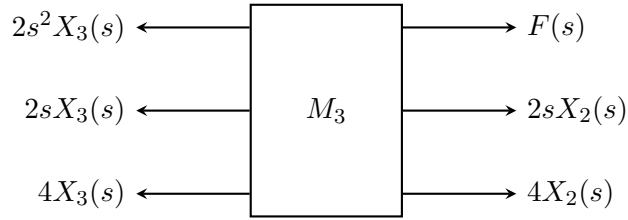
The resulting equation in the Laplace domain is:

$$-(2s + 4)X_1(s) + (2s^2 + 4s + 8)X_2(s) - (2s + 4)X_3(s) = 0 \quad (2)$$

For Mass M_3 :

The resulting equation in the Laplace domain is:

$$-(2s + 4)X_2(s) + (2s^2 + 2s + 4)X_3(s) = F(s) \quad (3)$$



Solving the System using Cramer’s Rule

The system can be expressed in matrix form $A\mathbf{x} = \mathbf{b}$:

$$\begin{bmatrix} s^2 + 5s + 10 & -(2s + 4) & 0 \\ -(2s + 4) & 2s^2 + 4s + 8 & -(2s + 4) \\ 0 & -(2s + 4) & 2s^2 + 2s + 4 \end{bmatrix} \begin{bmatrix} X_1(s) \\ X_2(s) \\ X_3(s) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ F(s) \end{bmatrix}$$

The solution for $X_1(s)$ is $G(s) = \frac{X_1(s)}{F(s)} = \frac{\det(A_1)}{\det(A)} = \frac{\Delta_1}{\Delta}$.

Calculating the Numerator Determinant, Δ_1 :

$$\Delta_1 = \det \begin{bmatrix} 0 & -(2s + 4) & 0 \\ 0 & 2s^2 + 4s + 8 & -(2s + 4) \\ F(s) & -(2s + 4) & 2s^2 + 2s + 4 \end{bmatrix} = 4(s^2 + 4s + 4)F(s)$$

Calculating the Denominator Determinant, Δ :

$$\Delta = \det(A) = 4(s^6 + 8s^5 + 30s^4 + 59s^3 + 74s^2 + 36s + 24)$$

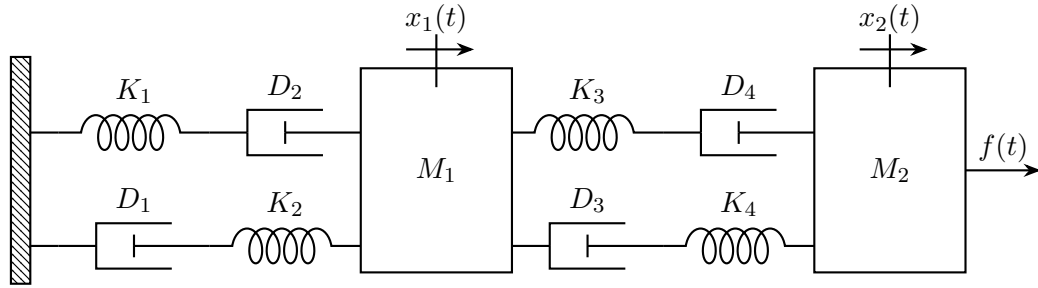
Constructing the Transfer Function

$$G(s) = \frac{X_1(s)}{F(s)} = \frac{\Delta_1/F(s)}{\Delta}$$

Expanding the denominator yields the simplified standard form:

$$G(s) = \frac{s^2 + 4s + 4}{s^6 + 8s^5 + 30s^4 + 59s^3 + 74s^2 + 36s + 24}$$

Problem 7



Given:

$$\begin{aligned}
 M_1 &= 1 \text{ kg} & K_1 &= 2 \text{ N/m} & D_1 &= 2 \text{ N} \cdot \text{s/m} \\
 M_2 &= 2 \text{ kg} & K_2 &= 3 \text{ N/m} & D_2 &= 1 \text{ N} \cdot \text{s/m} \\
 & & K_3 &= 1 \text{ N/m} & D_3 &= 1 \text{ N} \cdot \text{s/m} \\
 & & K_4 &= 1 \text{ N/m} & D_4 &= 1 \text{ N} \cdot \text{s/m}
 \end{aligned}$$

Required:

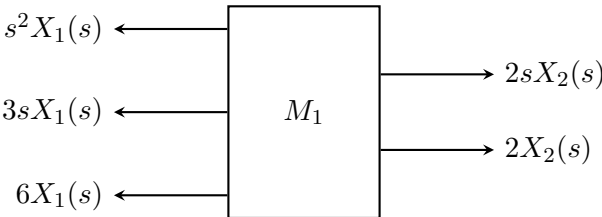
$$G(s) = \frac{X_1(s)}{F(s)}$$

Solution:

Deriving the Equations of Motion

We apply Newton's second law ($\sum F = ma$) to each mass, with the positive direction of motion to the right.

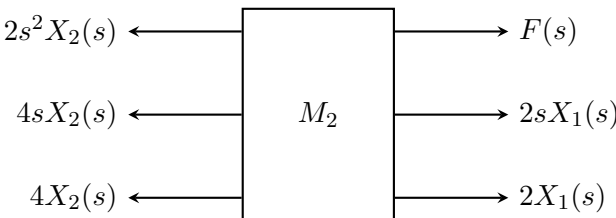
For Mass M_1 :



The resulting equation in the Laplace domain is:

$$(s^2 + 3s + 6)X_1(s) - (2s + 2)X_2(s) = 0 \tag{1}$$

For Mass M_2 :



The resulting equation in the Laplace domain is:

$$-(2s + 2)X_1(s) + (2s^2 + 4s + 4)X_2(s) = F(s) \tag{2}$$

Solving the System using Cramer's Rule

The system can be expressed in matrix form $A\mathbf{x} = \mathbf{b}$:

$$\begin{bmatrix} s^2 + 3s + 6 & -(2s + 2) \\ -(2s + 2) & 2s^2 + 4s + 4 \end{bmatrix} \begin{bmatrix} X_1(s) \\ X_2(s) \end{bmatrix} = \begin{bmatrix} 0 \\ F(s) \end{bmatrix}$$

The solution for $X_1(s)$ is $G(s) = \frac{X_1(s)}{F(s)} = \frac{\det(A_1)}{\det(A)} = \frac{\Delta_1}{\Delta}$.

Calculating the Numerator Determinant, Δ_1 :

$$\Delta_1 = \det \begin{bmatrix} 0 & -(2s + 2) \\ F(s) & 2s^2 + 4s + 4 \end{bmatrix} = (2s + 2)F(s)$$

Calculating the Denominator Determinant, Δ :

$$\Delta = \det(A) = 2s^4 + 10s^3 + 26s^2 + 36s + 20$$

Constructing the Transfer Function

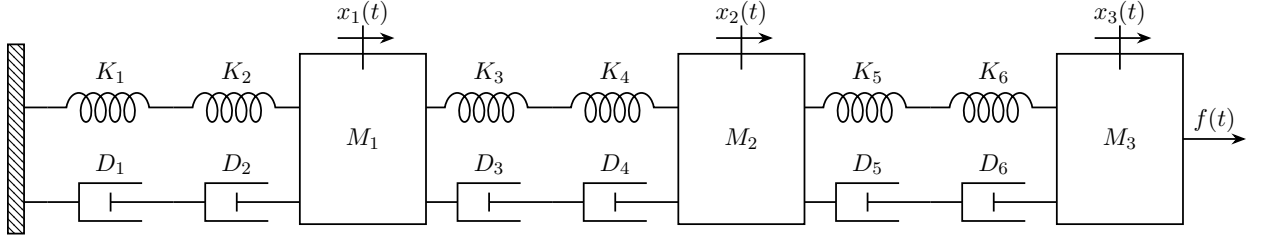
$$G(s) = \frac{X_1(s)}{F(s)} = \frac{\Delta_1/F(s)}{\Delta}$$

Expanding the denominator yields the simplified standard form:

$$G(s) = \frac{2s + 2}{2s^4 + 10s^3 + 26s^2 + 36s + 20}$$

Problem 8

For the translational mechanical system diagram shown below, determine the transfer function $G(s) = X_1(s)/F(s)$.



Given:

$$\begin{aligned} M_1 &= 1 \text{ kg} & K_1 &= 2 \text{ N/m} & D_1 &= 1 \text{ N} \cdot \text{s/m} \\ M_2 &= 2 \text{ kg} & K_2 &= 2 \text{ N/m} & D_2 &= 1 \text{ N} \cdot \text{s/m} \\ M_3 &= 1 \text{ kg} & K_3 &= 1 \text{ N/m} & D_3 &= 1 \text{ N} \cdot \text{s/m} \\ & & K_4 &= 1 \text{ N/m} & D_4 &= 1 \text{ N} \cdot \text{s/m} \end{aligned}$$

Required:

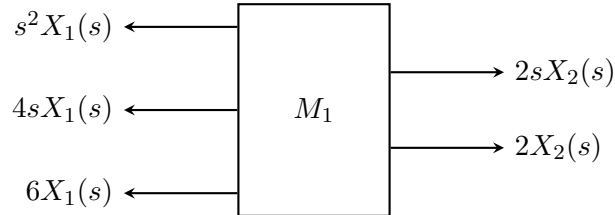
$$G(s) = \frac{X_1(s)}{F(s)}$$

Solution:

Deriving the Equations of Motion

We apply Newton's second law ($\sum F = ma$) to each mass, with the positive direction of motion to the right.

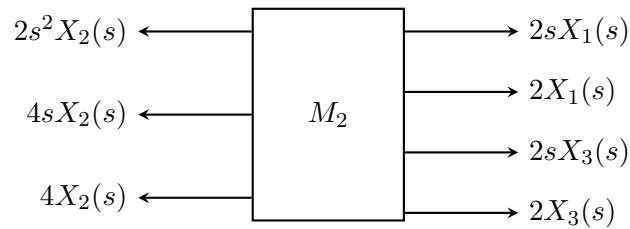
For Mass M_1 :



The resulting equation in the Laplace domain is:

$$(s^2 + 4s + 6)X_1(s) - (2s + 2)X_2(s) = 0 \quad (1)$$

For Mass M_2 :



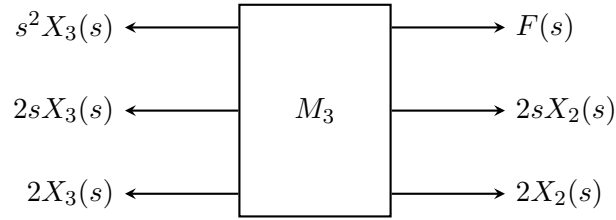
The resulting equation in the Laplace domain is:

$$-(2s + 2)X_1(s) + (2s^2 + 4s + 4)X_2(s) - (2s + 2)X_3(s) = 0 \quad (2)$$

For Mass M_3 :

The resulting equation in the Laplace domain is:

$$-(2s + 2)X_2(s) + (s^2 + 2s + 2)X_3(s) = F(s) \quad (3)$$



Solving the System using Cramer's Rule

The system can be expressed in matrix form $A\mathbf{x} = \mathbf{b}$:

$$\begin{bmatrix} s^2 + 4s + 6 & -(2s + 2) & 0 \\ -(2s + 2) & 2s^2 + 4s + 4 & -(2s + 2) \\ 0 & -(2s + 2) & s^2 + 2s + 2 \end{bmatrix} \begin{bmatrix} X_1(s) \\ X_2(s) \\ X_3(s) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ F(s) \end{bmatrix}$$

The solution for $X_1(s)$ is $G(s) = \frac{X_1(s)}{F(s)} = \frac{\det(A_1)}{\det(A)} = \frac{\Delta_1}{\Delta}$.

Calculating the Numerator Determinant, Δ_1 :

$$\Delta_1 = \det \begin{bmatrix} 0 & -(2s + 2) & 0 \\ 0 & 2s^2 + 4s + 4 & -(2s + 2) \\ F(s) & -(2s + 2) & s^2 + 2s + 2 \end{bmatrix} = (s^2 + 4s + 4)F(s)$$

Calculating the Denominator Determinant, Δ :

$$\Delta = \det(A) = s^6 + 8s^5 + 28s^4 + 52s^3 + 52s^2 + 24s + 8$$

Constructing the Transfer Function

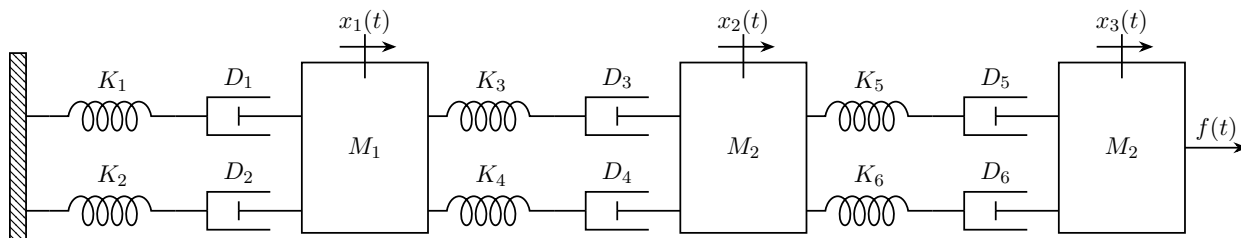
$$G(s) = \frac{X_1(s)}{F(s)} = \frac{\Delta_1/F(s)}{\Delta}$$

Expanding the denominator yields the simplified standard form:

$$G(s) = \frac{s^2 + 4s + 4}{s^6 + 8s^5 + 28s^4 + 52s^3 + 52s^2 + 24s + 8}$$

Problem 9

For the translational mechanical system diagram shown below, determine the transfer function $G(s) = X_1(s)/F(s)$.



Given:

$$\begin{aligned}
 M_1 &= 1 \text{ kg} & K_1 &= 1 \text{ N/m} & D_1 &= 1 \text{ N} \cdot \text{s/m} \\
 M_2 &= 2 \text{ kg} & K_2 &= 1 \text{ N/m} & D_2 &= 1 \text{ N} \cdot \text{s/m} \\
 M_3 &= 1 \text{ kg} & K_3 &= 1 \text{ N/m} & D_3 &= 1 \text{ N} \cdot \text{s/m} \\
 & & K_4 &= 2 \text{ N/m} & D_4 &= 1 \text{ N} \cdot \text{s/m} \\
 & & K_5 &= 1 \text{ N/m} & D_5 &= 1 \text{ N} \cdot \text{s/m} \\
 & & K_6 &= 2 \text{ N/m} & D_6 &= 1 \text{ N} \cdot \text{s/m}
 \end{aligned}$$

Required:

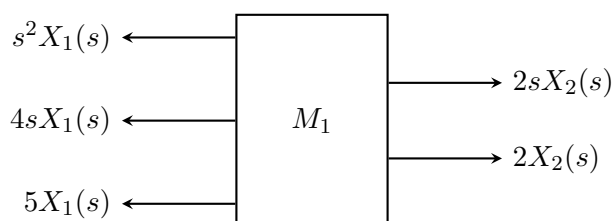
$$G(s) = \frac{X_1(s)}{F(s)}$$

Solution:

Deriving the Equations of Motion

We apply Newton's second law ($\sum F = ma$) to each mass, with the positive direction of motion to the right.

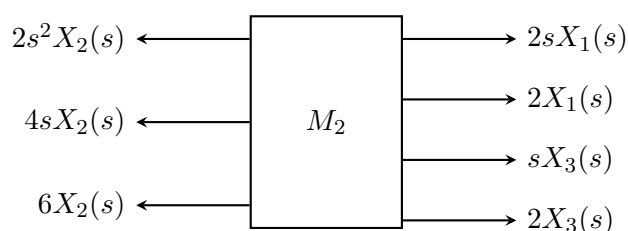
For Mass M_1 :



The resulting equation in the Laplace domain is:

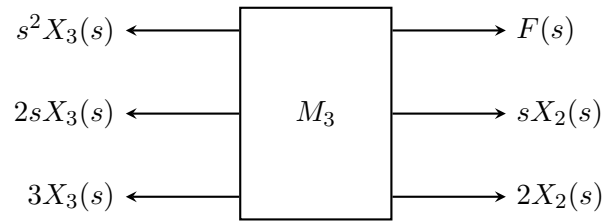
$$(s^2 + 4s + 5)X_1(s) - (2s + 2)X_2(s) = 0 \quad (1)$$

For Mass M_2 :



The resulting equation in the Laplace domain is:

$$-(2s + 2)X_1(s) + (2s^2 + 4s + 6)X_2(s) - (s + 2)X_3(s) = 0 \quad (2)$$



For Mass M_3 :

The resulting equation in the Laplace domain is:

$$-(s + 2)X_2(s) + (s^2 + 2s + 3)X_3(s) = F(s) \quad (3)$$

Solving the System using Cramer's Rule

The system can be expressed in matrix form $A\mathbf{x} = \mathbf{b}$:

$$\begin{bmatrix} s^2 + 4s + 5 & -(2s + 2) & 0 \\ -(2s + 2) & 2s^2 + 4s + 6 & -(s + 2) \\ 0 & -(s + 2) & s^2 + 2s + 3 \end{bmatrix} \begin{bmatrix} X_1(s) \\ X_2(s) \\ X_3(s) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ F(s) \end{bmatrix}$$

The solution for $X_1(s)$ is $G(s) = \frac{X_1(s)}{F(s)} = \frac{\det(A_1)}{\det(A)} = \frac{\Delta_1}{\Delta}$.

Calculating the Numerator Determinant, Δ_1 :

$$\Delta_1 = \det \begin{bmatrix} 0 & -(2s + 2) & 0 \\ 0 & 2s^2 + 4s + 6 & -(s + 2) \\ F(s) & -(s + 2) & s^2 + 2s + 3 \end{bmatrix} = (s + 2)F(s)$$

Calculating the Denominator Determinant, Δ :

$$\Delta = \det(A) = s^6 + 8s^5 + 25s^4 + 46s^3 + 49s^2 + 26s + 10$$

Constructing the Transfer Function

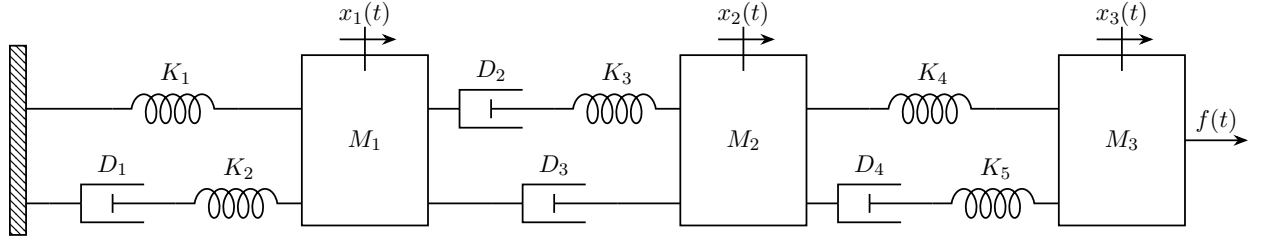
$$G(s) = \frac{X_1(s)}{F(s)} = \frac{\Delta_1/F(s)}{\Delta}$$

Expanding the denominator yields the simplified standard form:

$$G(s) = \frac{s + 2}{s^6 + 8s^5 + 25s^4 + 46s^3 + 49s^2 + 26s + 10}$$

Problem 10

For the translational mechanical system diagram shown below, determine the transfer function $G(s) = X_1(s)/F(s)$.



Given:

$$\begin{aligned} M_1 &= 2 \text{ kg} & K_1 &= 4 \text{ N/m} & D_1 &= 6 \text{ N} \cdot \text{s/m} \\ M_2 &= 1 \text{ kg} & K_2 &= 5 \text{ N/m} & D_2 &= 2 \text{ N} \cdot \text{s/m} \\ M_3 &= 1 \text{ kg} & K_3 &= 3 \text{ N/m} & D_3 &= 1 \text{ N} \cdot \text{s/m} \\ & & K_4 &= 2 \text{ N/m} & D_4 &= 4 \text{ N} \cdot \text{s/m} \\ & & K_5 &= 1 \text{ N/m} \end{aligned}$$

Required:

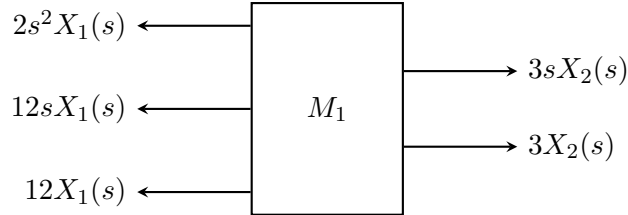
$$G(s) = \frac{X_1(s)}{F(s)}$$

Solution:

Deriving the Equations of Motion

We apply Newton's second law ($\sum F = ma$) to each mass, with the positive direction of motion to the right.

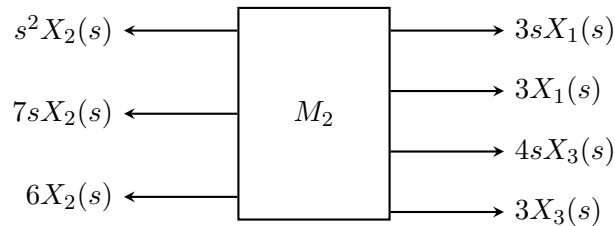
For Mass M_1 :



The resulting equation in the Laplace domain is:

$$(2s^2 + 12s + 12)X_1(s) - (3s + 3)X_2(s) = 0 \quad (1)$$

For Mass M_2 :



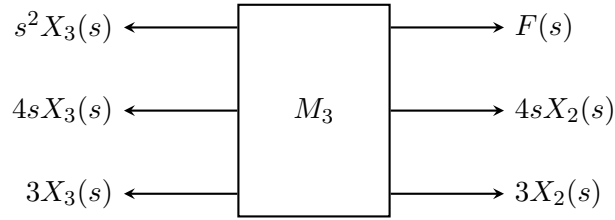
The resulting equation in the Laplace domain is:

$$-(3s + 3)X_1(s) + (s^2 + 7s + 6)X_2(s) - (4s + 3)X_3(s) = 0 \quad (2)$$

For Mass M_3 :

The resulting equation in the Laplace domain is:

$$-(4s + 3)X_2(s) + (s^2 + 4s + 3)X_3(s) = F(s) \quad (3)$$



Solving the System using Cramer's Rule

The system can be expressed in matrix form $A\mathbf{x} = \mathbf{b}$:

$$\begin{bmatrix} 2s^2 + 12s + 12 & -(3s + 3) & 0 \\ -(3s + 3) & s^2 + 7s + 6 & -(4s + 3) \\ 0 & -(4s + 3) & s^2 + 4s + 3 \end{bmatrix} \begin{bmatrix} X_1(s) \\ X_2(s) \\ X_3(s) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ F(s) \end{bmatrix}$$

The solution for $X_1(s)$ is $G(s) = \frac{X_1(s)}{F(s)} = \frac{\det(A_1)}{\det(A)} = \frac{\Delta_1}{\Delta}$.

Calculating the Numerator Determinant, Δ_1 :

$$\Delta_1 = \det \begin{bmatrix} 0 & -(3s + 3) & 0 \\ 0 & s^2 + 7s + 6 & -(4s + 3) \\ F(s) & -(4s + 3) & s^2 + 4s + 3 \end{bmatrix} = (3s + 3)(4s + 3)F(s)$$

Calculating the Denominator Determinant, Δ :

$$\Delta = \det(A) = (2s^2 + 12s + 12) [(s^2 + 7s + 6)(s^2 + 4s + 3) + (3 + 4s)(3 + 3s)]$$

Constructing the Transfer Function

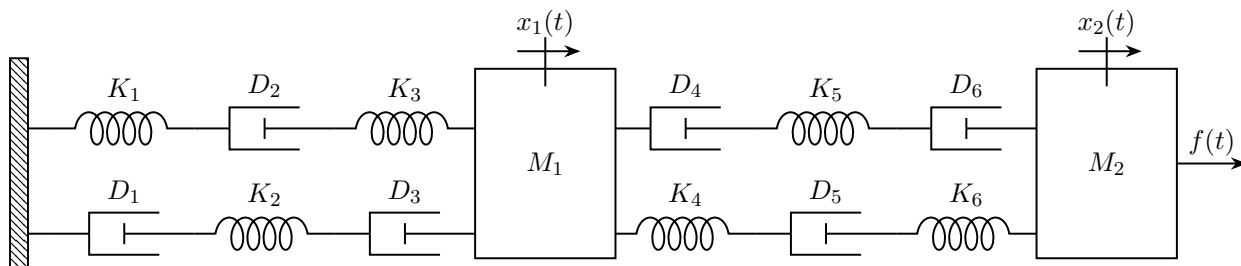
$$G(s) = \frac{X_1(s)}{F(s)} = \frac{\Delta_1/F(s)}{\Delta}$$

Expanding the denominator yields the simplified standard form:

$$G(s) = \frac{(3s + 3)(4s + 3)}{(2s^2 + 12s + 12) [(s^2 + 7s + 6)(s^2 + 4s + 3) + (3 + 4s)(3 + 3s)]}$$

Problem 11

For the translational mechanical system diagram shown below, determine the transfer function $G(s) = X_1(s)/F(s)$.



Given:

$$\begin{aligned}
 M_1 &= 1 \text{ kg} & K_1 &= 1 \text{ N/m} & D_1 &= 1 \text{ N} \cdot \text{s/m} \\
 M_2 &= 1 \text{ kg} & K_2 &= 1 \text{ N/m} & D_2 &= 1 \text{ N} \cdot \text{s/m} \\
 & & K_3 &= 1 \text{ N/m} & D_3 &= 1 \text{ N} \cdot \text{s/m} \\
 & & K_4 &= 1 \text{ N/m} & D_4 &= 1 \text{ N} \cdot \text{s/m} \\
 & & K_5 &= 1 \text{ N/m} & D_5 &= 1 \text{ N} \cdot \text{s/m} \\
 & & K_6 &= 1 \text{ N/m} & D_6 &= 1 \text{ N} \cdot \text{s/m}
 \end{aligned}$$

Required:

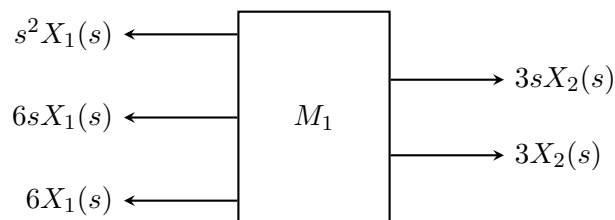
$$G(s) = \frac{X_1(s)}{F(s)}$$

Solution:

Deriving the Equations of Motion

We apply Newton's second law ($\sum F = ma$) to each mass, with the positive direction of motion to the right.

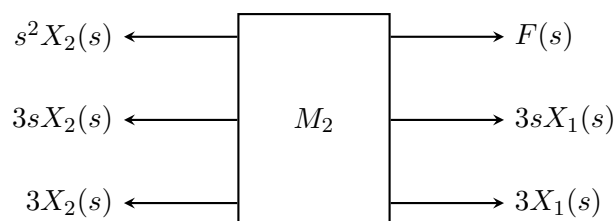
For Mass M_1 :



The resulting equation in the Laplace domain is:

$$(s^2 + 6s + 6)X_1(s) - (3s + 3)X_2(s) = 0 \quad (1)$$

For Mass M_2 :



The resulting equation in the Laplace domain is:

$$-(3s + 3)X_1(s) + (s^2 + 3s + 3)X_2(s) = F(s) \quad (2)$$

Solving the System using Cramer's Rule

The system can be expressed in matrix form $A\mathbf{x} = \mathbf{b}$:

$$\begin{bmatrix} s^2 + 6s + 6 & -(3s + 3) \\ -(3s + 3) & s^2 + 3s + 3 \end{bmatrix} \begin{bmatrix} X_1(s) \\ X_2(s) \end{bmatrix} = \begin{bmatrix} 0 \\ F(s) \end{bmatrix}$$

The solution for $X_1(s)$ is $G(s) = \frac{X_1(s)}{F(s)} = \frac{\det(A_1)}{\det(A)} = \frac{\Delta_1}{\Delta}$.

Calculating the Numerator Determinant, Δ_1 :

$$\Delta_1 = \det \begin{bmatrix} 0 & -(3s + 3) \\ F(s) & s^2 + 3s + 3 \end{bmatrix} = (3s + 3)F(s)$$

Calculating the Denominator Determinant, Δ :

$$\Delta = \det(A) = s^4 + 9s^3 + 18s^2 + 18s + 9$$

Constructing the Transfer Function

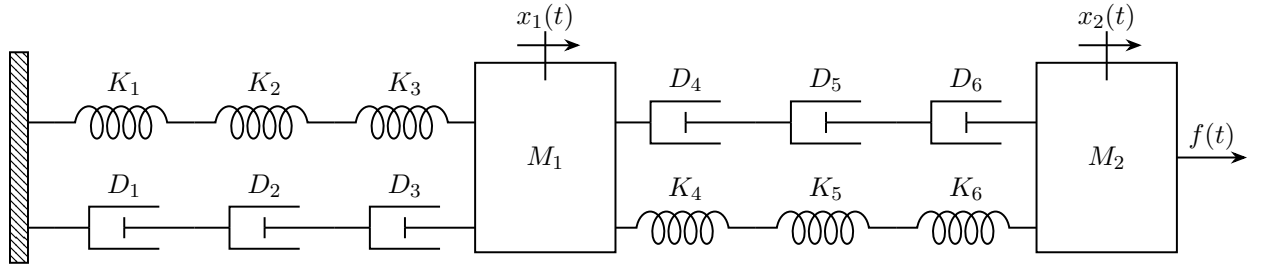
$$G(s) = \frac{X_1(s)}{F(s)} = \frac{\Delta_1/F(s)}{\Delta}$$

Expanding the denominator yields the simplified standard form:

$$G(s) = \frac{3s + 3}{s^4 + 9s^3 + 18s^2 + 18s + 9}$$

Problem 12

For the translational mechanical system diagram shown below, determine the transfer function $G(s) = X_1(s)/F(s)$.



Given:

$$\begin{aligned} M_1 &= 2 \text{ kg} & K_g &= 4 \text{ N/m} & C_g &= 4 \text{ N} \cdot \text{s/m} \\ M_2 &= 1 \text{ kg} & K_b &= 4 \text{ N/m} & C_b &= 3 \text{ N} \cdot \text{s/m} \end{aligned}$$

Required:

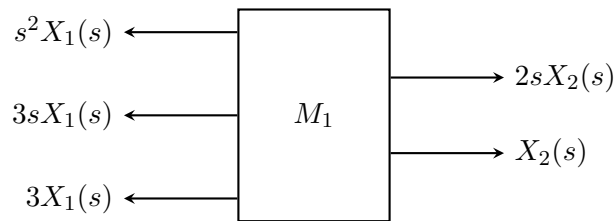
$$G(s) = \frac{X_1(s)}{F(s)}$$

Solution:

Deriving the Equations of Motion

We apply Newton's second law ($\sum F = ma$) to each mass, with the positive direction of motion to the right.

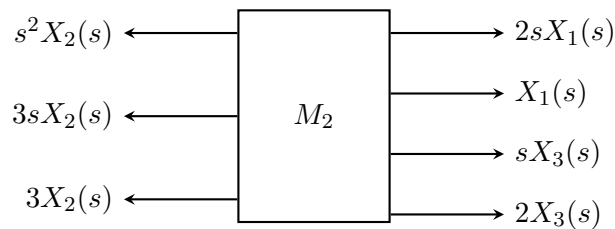
For Mass M_1 :



The resulting equation in the Laplace domain is:

$$(s^2 + 3s + 3)X_1(s) - (2s + 1)X_2(s) = 0 \quad (1)$$

For Mass M_2 :



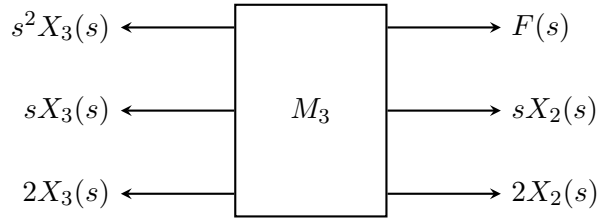
The resulting equation in the Laplace domain is:

$$-(2s + 1)X_1(s) + (s^2 + 3s + 3)X_2(s) - (s + 2)X_3(s) = 0 \quad (2)$$

For Mass M_3 :

The resulting equation in the Laplace domain is:

$$-(s + 2)X_2(s) + (s^2 + s + 2)X_3(s) = F(s) \quad (3)$$



Solving the System using Cramer's Rule

The system can be expressed in matrix form $A\mathbf{x} = \mathbf{b}$:

$$\begin{bmatrix} s^2 + 3s + 3 & -(2s + 1) & 0 \\ -(2s + 1) & s^2 + 3s + 3 & -(s + 2) \\ 0 & -(s + 2) & s^2 + s + 2 \end{bmatrix} \begin{bmatrix} X_1(s) \\ X_2(s) \\ X_3(s) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ F(s) \end{bmatrix}$$

The solution for $X_1(s)$ is $G(s) = \frac{X_1(s)}{F(s)} = \frac{\det(A_1)}{\det(A)} = \frac{\Delta_1}{\Delta}$.

Calculating the Numerator Determinant, Δ_1 :

$$\Delta_1 = \det \begin{bmatrix} 0 & -(2s + 1) & 0 \\ 0 & s^2 + 3s + 3 & -(s + 2) \\ F(s) & -(s + 2) & s^2 + s + 2 \end{bmatrix} = (2s^2 + 5s + 2)F(s)$$

Calculating the Denominator Determinant, Δ :

$$\Delta = \det(A) = s^6 + 7s^5 + 18s^4 + 30s^3 + 25s^2 + 12s + 4$$

Constructing the Transfer Function

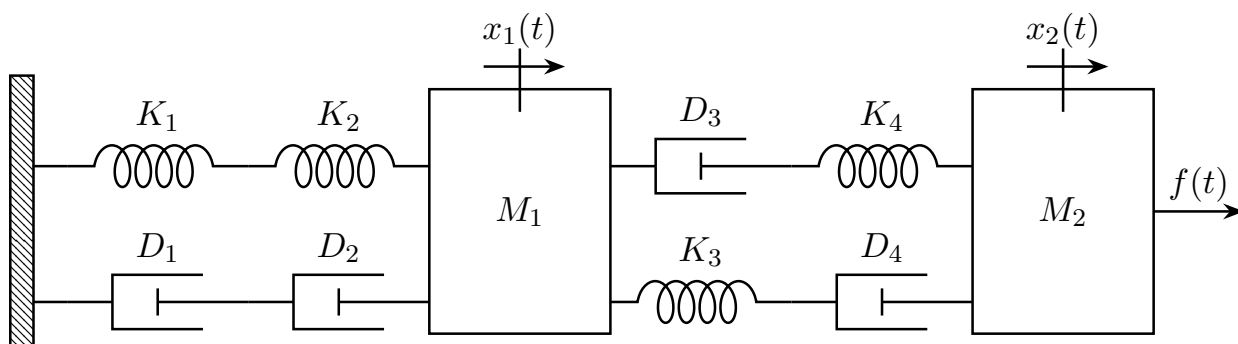
$$G(s) = \frac{X_1(s)}{F(s)} = \frac{\Delta_1/F(s)}{\Delta}$$

Expanding the denominator yields the simplified standard form:

$$G(s) = \frac{2s^2 + 5s + 2}{s^6 + 7s^5 + 18s^4 + 30s^3 + 25s^2 + 12s + 4}$$

Problem 13

For the translational mechanical system diagram shown below, determine the transfer function $G(s) = X_1(s)/F(s)$.



Given:

$$\begin{aligned} M_1 &= 1 \text{ kg} & K_1 &= 1 \text{ N/m} & D_1 &= 1 \text{ N} \cdot \text{s/m} \\ M_2 &= 1 \text{ kg} & K_2 &= 1 \text{ N/m} & D_2 &= 1 \text{ N} \cdot \text{s/m} \\ & & K_3 &= 1 \text{ N/m} & D_3 &= 1 \text{ N} \cdot \text{s/m} \\ & & K_4 &= 1 \text{ N/m} & D_4 &= 1 \text{ N} \cdot \text{s/m} \end{aligned}$$

Required:

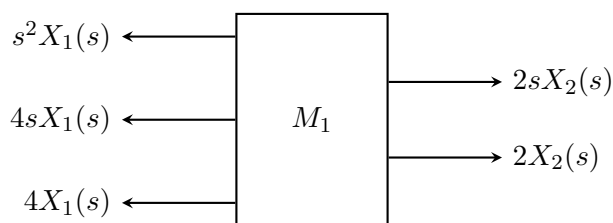
$$G(s) = \frac{X_1(s)}{F(s)}$$

Solution:

Deriving the Equations of Motion

We apply Newton's second law ($\sum F = ma$) to each mass, with the positive direction of motion to the right.

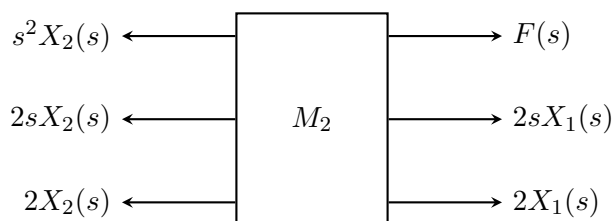
For Mass M_1 :



The resulting equation in the Laplace domain is:

$$(s^2 + 4s + 4)X_1(s) - (2s + 2)X_2(s) = 0 \quad (1)$$

For Mass M_2 :



The resulting equation in the Laplace domain is:

$$-(2s + 2)X_1(s) + (s^2 + 2s + 2)X_2(s) = F(s) \quad (2)$$

Solving the System using Cramer’s Rule

The system can be expressed in matrix form $A\mathbf{x} = \mathbf{b}$:

$$\begin{bmatrix} s^2 + 4s + 4 & -(2s + 2) \\ -(2s + 2) & s^2 + 2s + 2 \end{bmatrix} \begin{bmatrix} X_1(s) \\ X_2(s) \end{bmatrix} = \begin{bmatrix} 0 \\ F(s) \end{bmatrix}$$

The solution for $X_1(s)$ is $G(s) = \frac{X_1(s)}{F(s)} = \frac{\det(A_1)}{\det(A)} = \frac{\Delta_1}{\Delta}$.

Calculating the Numerator Determinant, Δ_1 :

$$\Delta_1 = \det \begin{bmatrix} 0 & -(2s + 2) \\ F(s) & s^2 + 2s + 2 \end{bmatrix} = (2s + 2)F(s)$$

Calculating the Denominator Determinant, Δ :

$$\Delta = \det(A) = s^4 + 6s^3 + 10s^2 + 8s + 4$$

Constructing the Transfer Function

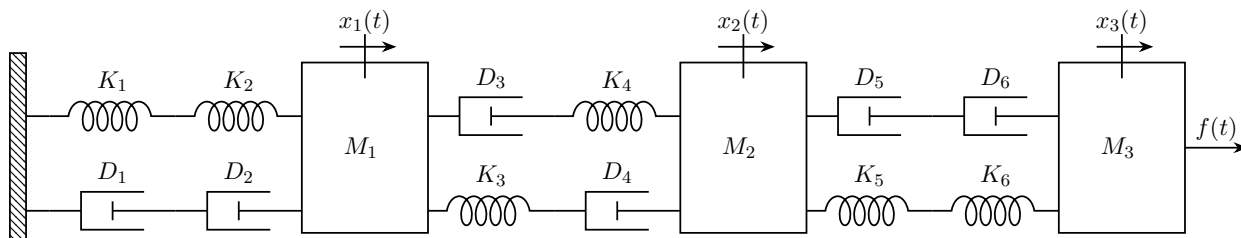
$$G(s) = \frac{X_1(s)}{F(s)} = \frac{\Delta_1/F(s)}{\Delta}$$

Expanding the denominator yields the simplified standard form:

$$G(s) = \frac{2s + 2}{s^4 + 6s^3 + 10s^2 + 8s + 4}$$

Problem 14

For the translational mechanical system diagram shown below, determine the transfer function $G(s) = X_1(s)/F(s)$.



Given:

$$\begin{aligned}
 M_1 &= 1 \text{ kg} & K_1 &= 1 \text{ N/m} & D_1 &= 1 \text{ N} \cdot \text{s/m} \\
 M_2 &= 1 \text{ kg} & K_2 &= 1 \text{ N/m} & D_2 &= 1 \text{ N} \cdot \text{s/m} \\
 M_3 &= 1 \text{ kg} & K_3 &= 1 \text{ N/m} & D_3 &= 1 \text{ N} \cdot \text{s/m} \\
 & & K_4 &= 1 \text{ N/m} & D_4 &= 1 \text{ N} \cdot \text{s/m} \\
 & & K_5 &= 1 \text{ N/m} & D_5 &= 1 \text{ N} \cdot \text{s/m} \\
 & & K_6 &= 1 \text{ N/m} & D_6 &= 1 \text{ N} \cdot \text{s/m}
 \end{aligned}$$

Required:

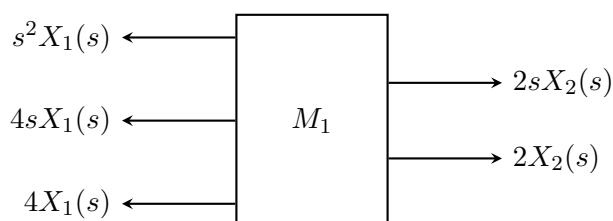
$$G(s) = \frac{X_1(s)}{F(s)}$$

Solution:

Deriving the Equations of Motion

We apply Newton's second law ($\sum F = ma$) to each mass, with the positive direction of motion to the right.

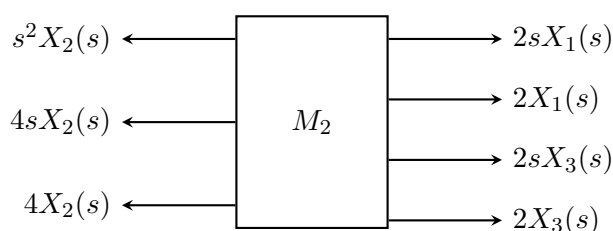
For Mass M_1 :



The resulting equation in the Laplace domain is:

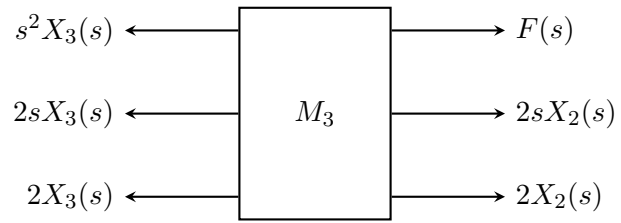
$$(s^2 + 4s + 4)X_1(s) - (2s + 2)X_2(s) = 0 \quad (1)$$

For Mass M_2 :



The resulting equation in the Laplace domain is:

$$-(2s + 2)X_1(s) + (s^2 + 4s + 4)X_2(s) - (2s + 2)X_3(s) = 0 \quad (2)$$



For Mass M_3 :

The resulting equation in the Laplace domain is:

$$-(2s + 2)X_2(s) + (s^2 + 2s + 2)X_3(s) = F(s) \quad (3)$$

Solving the System using Cramer’s Rule

The system can be expressed in matrix form $A\mathbf{x} = \mathbf{b}$:

$$\begin{bmatrix} s^2 + 4s + 4 & -(2s + 2) & 0 \\ -(2s + 2) & s^2 + 4s + 4 & -(2s + 2) \\ 0 & -(2s + 2) & s^2 + 2s + 2 \end{bmatrix} \begin{bmatrix} X_1(s) \\ X_2(s) \\ X_3(s) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ F(s) \end{bmatrix}$$

The solution for $X_1(s)$ is $G(s) = \frac{X_1(s)}{F(s)} = \frac{\det(A_1)}{\det(A)} = \frac{\Delta_1}{\Delta}$.

Calculating the Numerator Determinant, Δ_1 :

$$\Delta_1 = \det \begin{bmatrix} 0 & -(2s + 2) & 0 \\ 0 & s^2 + 4s + 4 & -(2s + 2) \\ F(s) & -(2s + 2) & s^2 + 2s + 2 \end{bmatrix} = (2s + 2)^2 F(s)$$

Calculating the Denominator Determinant, Δ :

$$\Delta = \det(A) = s^6 + 10s^5 + 37s^4 + 64s^3 + 54s^2 + 20s + 4$$

Constructing the Transfer Function

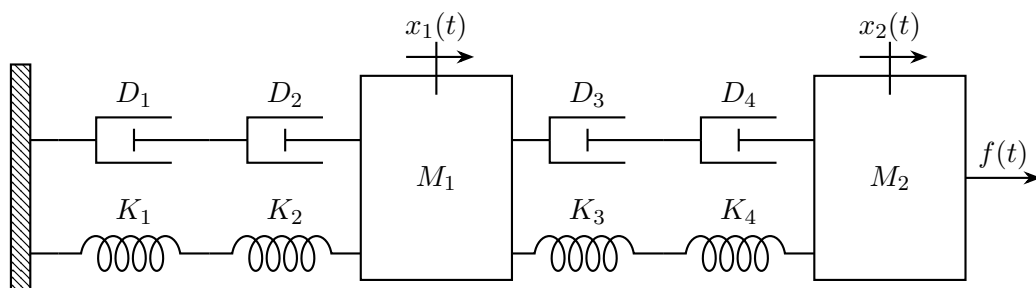
$$G(s) = \frac{X_1(s)}{F(s)} = \frac{\Delta_1/F(s)}{\Delta}$$

Expanding the denominator yields the simplified standard form:

$$G(s) = \frac{(2s + 2)^2}{s^6 + 10s^5 + 37s^4 + 64s^3 + 54s^2 + 20s + 4}$$

Problem 15

For the translational mechanical system diagram shown below, determine the transfer function $G(s) = X_1(s)/F(s)$.



Given:

$$\begin{aligned} M_1 &= 2 \text{ kg} & K_1 &= 3 \text{ N/m} & D_1 &= 1 \text{ N} \cdot \text{s/m} \\ M_2 &= 3 \text{ kg} & K_2 &= 2 \text{ N/m} & D_2 &= 2 \text{ N} \cdot \text{s/m} \\ & & K_3 &= 1 \text{ N/m} & D_3 &= 1 \text{ N} \cdot \text{s/m} \\ & & K_4 &= 2 \text{ N/m} & D_4 &= 2 \text{ N} \cdot \text{s/m} \end{aligned}$$

Required:

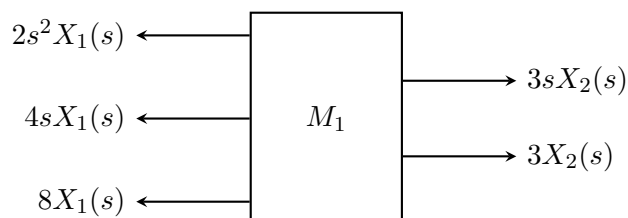
$$G(s) = \frac{X_1(s)}{F(s)}$$

Solution:

Deriving the Equations of Motion

We apply Newton's second law ($\sum F = ma$) to each mass, with the positive direction of motion to the right.

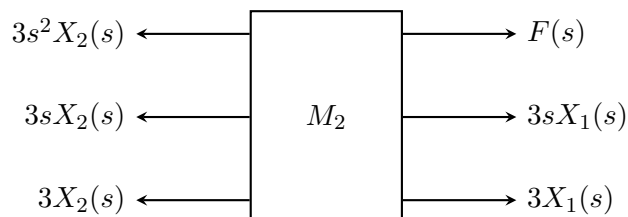
For Mass M_1 :



The resulting equation in the Laplace domain is:

$$(2s^2 + 4s + 8)X_1(s) - (3s + 3)X_2(s) = 0 \quad (1)$$

For Mass M_2 :



The resulting equation in the Laplace domain is:

$$-(3s + 3)X_1(s) + (3s^2 + 3s + 3)X_2(s) = F(s) \quad (2)$$

Solving the System using Cramer's Rule

The system can be expressed in matrix form $A\mathbf{x} = \mathbf{b}$:

$$\begin{bmatrix} 2s^2 + 4s + 8 & -(3s + 3) \\ -(3s + 3) & 3s^2 + 3s + 3 \end{bmatrix} \begin{bmatrix} X_1(s) \\ X_2(s) \end{bmatrix} = \begin{bmatrix} 0 \\ F(s) \end{bmatrix}$$

The solution for $X_1(s)$ is $G(s) = \frac{X_1(s)}{F(s)} = \frac{\det(A_1)}{\det(A)} = \frac{\Delta_1}{\Delta}$.

Calculating the Numerator Determinant, Δ_1 :

$$\Delta_1 = \det \begin{bmatrix} 0 & -(3s + 3) \\ F(s) & 3s^2 + 3s + 3 \end{bmatrix} = (3s + 3)F(s)$$

Calculating the Denominator Determinant, Δ :

$$\Delta = \det(A) = 6s^4 + 18s^3 + 33s^2 + 18s + 15$$

Constructing the Transfer Function

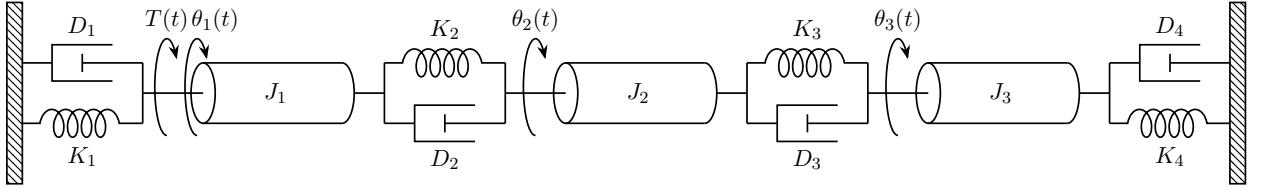
$$G(s) = \frac{X_1(s)}{F(s)} = \frac{\Delta_1/F(s)}{\Delta}$$

Expanding the denominator yields the simplified standard form:

$$G(s) = \frac{3s + 3}{6s^4 + 18s^3 + 33s^2 + 18s + 15}$$

Problem 16

For the rotational mechanical system shown in the diagram below, determine the transfer function $G(s) = \theta_3(s)/T(s)$.



Given:

$$\begin{aligned} J_1 &= 2 \text{ kg} \cdot \text{m}^2 & K_1 &= 4 \text{ N} \cdot \text{m/rad} & D_1 &= 6 \text{ N} \cdot \text{m} \cdot \text{s/rad} \\ J_2 &= 1 \text{ kg} \cdot \text{m}^2 & K_2 &= 3 \text{ N} \cdot \text{m/rad} & D_2 &= 2 \text{ N} \cdot \text{m} \cdot \text{s/rad} \\ J_3 &= 1 \text{ kg} \cdot \text{m}^2 & K_3 &= 2 \text{ N} \cdot \text{m/rad} & D_3 &= 1 \text{ N} \cdot \text{m} \cdot \text{s/rad} \\ & & K_4 &= 1 \text{ N} \cdot \text{m/rad} & D_4 &= 4 \text{ N} \cdot \text{m} \cdot \text{s/rad} \end{aligned}$$

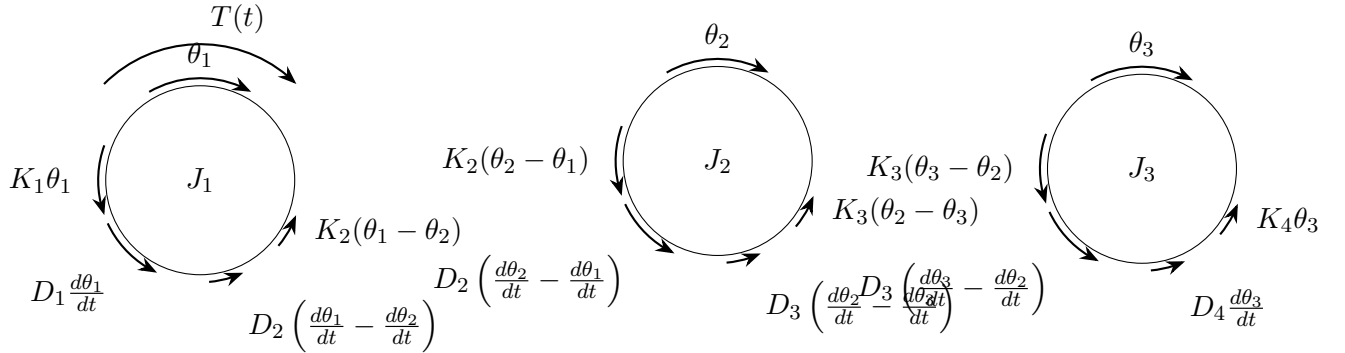
Required:

$$G(s) = \frac{\theta_3(s)}{T(s)}$$

Solution:

Free-Body Diagrams (FBDs)

We assume clockwise rotation is positive.



Equations of Motion

We apply Newton's second law for rotation ($\sum \tau = J \frac{d^2\theta}{dt^2}$) to each mass.

For Inertia 1 (J_1):

$$\begin{aligned} J_1 \frac{d^2\theta_1}{dt^2} + D_1 \frac{d\theta_1}{dt} + K_1\theta_1 + D_2 \left(\frac{d\theta_1}{dt} - \frac{d\theta_2}{dt} \right) + K_2(\theta_1 - \theta_2) &= T(t) \\ J_1 \frac{d^2\theta_1}{dt^2} + (D_1 + D_2) \frac{d\theta_1}{dt} + (K_1 + K_2)\theta_1 - D_2 \frac{d\theta_2}{dt} - K_2\theta_2 &= T(t) \end{aligned}$$

For Inertia 2 (J_2):

$$\begin{aligned} J_2 \frac{d^2\theta_2}{dt^2} + D_2 \left(\frac{d\theta_2}{dt} - \frac{d\theta_1}{dt} \right) + K_2(\theta_2 - \theta_1) + D_3 \left(\frac{d\theta_2}{dt} - \frac{d\theta_3}{dt} \right) + K_3(\theta_2 - \theta_3) &= 0 \\ J_2 \frac{d^2\theta_2}{dt^2} + (D_2 + D_3) \frac{d\theta_2}{dt} + (K_2 + K_3)\theta_2 - D_2 \frac{d\theta_1}{dt} - K_2\theta_1 - D_3 \frac{d\theta_3}{dt} - K_3\theta_3 &= 0 \end{aligned}$$

For Inertia 3 (J_3):

$$\begin{aligned} J_3 \frac{d^2\theta_3}{dt^2} + D_3 \left(\frac{d\theta_3}{dt} - \frac{d\theta_2}{dt} \right) + K_3(\theta_3 - \theta_2) + D_4 \frac{d\theta_3}{dt} + K_4\theta_3 &= 0 \\ J_3 \frac{d^2\theta_3}{dt^2} + (D_3 + D_4) \frac{d\theta_3}{dt} + (K_3 + K_4)\theta_3 - D_3 \frac{d\theta_2}{dt} - K_3\theta_2 &= 0 \end{aligned}$$

Laplace Transform

We take the Laplace transform of the equations, assuming zero initial conditions.

Equation for Inertia 1:

$$[J_1 s^2 + (D_1 + D_2)s + (K_1 + K_2)]\Theta_1(s) - [D_2 s + K_2]\Theta_2(s) = T(s)$$

Equation for Inertia 2:

$$-[D_2 s + K_2]\Theta_1(s) + [J_2 s^2 + (D_2 + D_3)s + (K_2 + K_3)]\Theta_2(s) - [D_3 s + K_3]\Theta_3(s) = 0$$

Equation for Inertia 3:

$$-[D_3 s + K_3]\Theta_2(s) + [J_3 s^2 + (D_3 + D_4)s + (K_3 + K_4)]\Theta_3(s) = 0$$

Substituting the given numerical values:

$$(2s^2 + 8s + 7)\Theta_1(s) - (2s + 3)\Theta_2(s) = T(s) \quad (1)$$

$$-(2s + 3)\Theta_1(s) + (s^2 + 3s + 5)\Theta_2(s) - (s + 2)\Theta_3(s) = 0 \quad (2)$$

$$-(s + 2)\Theta_2(s) + (s^2 + 5s + 3)\Theta_3(s) = 0 \quad (3)$$

Solving for the Transfer Function

The system can be expressed in matrix form $A\Theta = \mathbf{b}$:

$$\begin{bmatrix} 2s^2 + 8s + 7 & -(2s + 3) & 0 \\ -(2s + 3) & s^2 + 3s + 5 & -(s + 2) \\ 0 & -(s + 2) & s^2 + 5s + 3 \end{bmatrix} \begin{bmatrix} \Theta_1(s) \\ \Theta_2(s) \\ \Theta_3(s) \end{bmatrix} = \begin{bmatrix} T(s) \\ 0 \\ 0 \end{bmatrix}$$

Using Cramer's Rule to solve for $\Theta_3(s)$:

$$\Theta_3(s) = \frac{\det(A_3)}{\det(A)}$$

Determinant of A_3 :

$$\det(A_3) = T(s) \begin{vmatrix} -(2s + 3) & s^2 + 3s + 5 \\ 0 & -(s + 2) \end{vmatrix} = T(s)(2s + 3)(s + 2) = T(s)(2s^2 + 7s + 6)$$

Determinant of A :

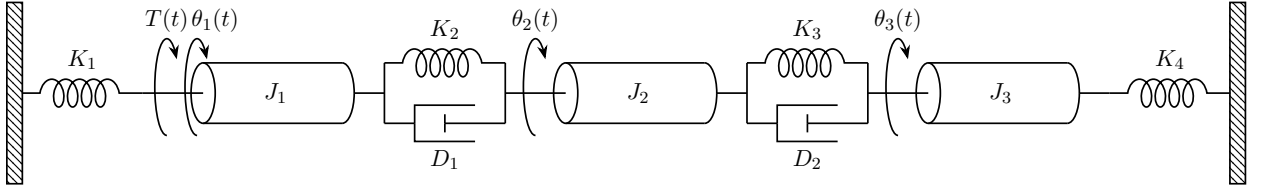
$$\begin{aligned} \det(A) &= (2s^2 + 8s + 7)[(s^2 + 3s + 5)(s^2 + 5s + 3) - (s + 2)^2] - (2s + 3)^2(s^2 + 5s + 3) \\ \det(A) &= 2s^6 + 24s^5 + 123s^4 + 356s^3 + 591s^2 + 499s + 160 \end{aligned}$$

The final transfer function is:

$$\boxed{\frac{\Theta_3(s)}{T(s)} = \frac{2s^2 + 7s + 6}{2s^6 + 24s^5 + 123s^4 + 356s^3 + 591s^2 + 499s + 160}}$$

Problem 17

For the rotational mechanical system shown in the diagram below, determine the transfer function $G(s) = \theta_3(s)/T(s)$.



Given:

$$\begin{aligned} J_1 &= 1 \text{ kg} \cdot \text{m}^2 & K_1 &= 2 \text{ N} \cdot \text{m/rad} & D_1 &= 1 \text{ N} \cdot \text{m} \cdot \text{s/rad} \\ J_2 &= 2 \text{ kg} \cdot \text{m}^2 & K_2 &= 3 \text{ N} \cdot \text{m/rad} & D_2 &= 1 \text{ N} \cdot \text{m} \cdot \text{s/rad} \\ J_3 &= 1 \text{ kg} \cdot \text{m}^2 & K_3 &= 1 \text{ N} \cdot \text{m/rad} \\ & & K_4 &= 1 \text{ N} \cdot \text{m/rad} \end{aligned}$$

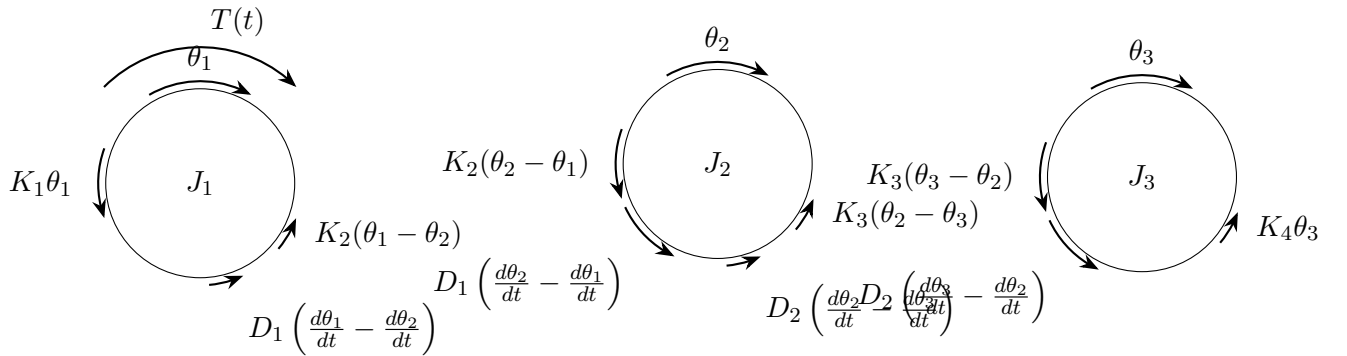
Required:

$$G(s) = \frac{\theta_3(s)}{T(s)}$$

Solution:

Free-Body Diagrams (FBDs)

We assume clockwise rotation is positive.



Equations of Motion

We apply Newton's second law for rotation ($\sum \tau = J \frac{d^2\theta}{dt^2}$) to each mass.

For Inertia 1 (J_1):

$$\begin{aligned} J_1 \frac{d^2\theta_1}{dt^2} + D_1 \left(\frac{d\theta_1}{dt} - \frac{d\theta_2}{dt} \right) + K_1\theta_1 + K_2(\theta_1 - \theta_2) &= T(t) \\ J_1 \frac{d^2\theta_1}{dt^2} + D_1 \frac{d\theta_1}{dt} + (K_1 + K_2)\theta_1 - D_1 \frac{d\theta_2}{dt} - K_2\theta_2 &= T(t) \end{aligned}$$

For Inertia 2 (J_2):

$$\begin{aligned} J_2 \frac{d^2\theta_2}{dt^2} + D_1 \left(\frac{d\theta_2}{dt} - \frac{d\theta_1}{dt} \right) + K_2(\theta_2 - \theta_1) + D_2 \left(\frac{d\theta_2}{dt} - \frac{d\theta_3}{dt} \right) + K_3(\theta_2 - \theta_3) &= 0 \\ J_2 \frac{d^2\theta_2}{dt^2} + (D_1 + D_2) \frac{d\theta_2}{dt} + (K_2 + K_3)\theta_2 - D_1 \frac{d\theta_1}{dt} - K_2\theta_1 - D_2 \frac{d\theta_3}{dt} - K_3\theta_3 &= 0 \end{aligned}$$

For Inertia 3 (J_3):

$$\begin{aligned} J_3 \frac{d^2\theta_3}{dt^2} + D_2 \left(\frac{d\theta_3}{dt} - \frac{d\theta_2}{dt} \right) + K_3(\theta_3 - \theta_2) + K_4\theta_3 &= 0 \\ J_3 \frac{d^2\theta_3}{dt^2} + D_2 \frac{d\theta_3}{dt} + (K_3 + K_4)\theta_3 - D_2 \frac{d\theta_2}{dt} - K_3\theta_2 &= 0 \end{aligned}$$

Laplace Transform

We take the Laplace transform of the equations, assuming zero initial conditions.

Equation for Inertia 1:

$$[J_1 s^2 + D_1 s + (K_1 + K_2)]\Theta_1(s) - [D_1 s + K_2]\Theta_2(s) = T(s)$$

Equation for Inertia 2:

$$-[D_1 s + K_2]\Theta_1(s) + [J_2 s^2 + (D_1 + D_2)s + (K_2 + K_3)]\Theta_2(s) - [D_2 s + K_3]\Theta_3(s) = 0$$

Equation for Inertia 3:

$$-[D_2 s + K_3]\Theta_2(s) + [J_3 s^2 + D_2 s + (K_3 + K_4)]\Theta_3(s) = 0$$

Substituting the given numerical values:

$$(s^2 + s + 5)\Theta_1(s) - (s + 3)\Theta_2(s) = T(s) \quad (1)$$

$$-(s + 3)\Theta_1(s) + (2s^2 + 2s + 4)\Theta_2(s) - (s + 1)\Theta_3(s) = 0 \quad (2)$$

$$-(s + 1)\Theta_2(s) + (s^2 + s + 2)\Theta_3(s) = 0 \quad (3)$$

Solving for the Transfer Function

The system can be expressed in matrix form $A\Theta = \mathbf{b}$:

$$\begin{bmatrix} s^2 + s + 5 & -(s + 3) & 0 \\ -(s + 3) & 2s^2 + 2s + 4 & -(s + 1) \\ 0 & -(s + 1) & s^2 + s + 2 \end{bmatrix} \begin{bmatrix} \Theta_1(s) \\ \Theta_2(s) \\ \Theta_3(s) \end{bmatrix} = \begin{bmatrix} T(s) \\ 0 \\ 0 \end{bmatrix}$$

Using Cramer's Rule to solve for $\Theta_3(s)$:

$$\Theta_3(s) = \frac{\det(A_3)}{\det(A)}$$

Determinant of A_3 :

$$\det(A_3) = T(s) \begin{vmatrix} -(s + 3) & 2s^2 + 2s + 4 \\ 0 & -(s + 1) \end{vmatrix} = T(s)(s + 3)(s + 1) = T(s)(s^2 + 4s + 3)$$

Determinant of A :

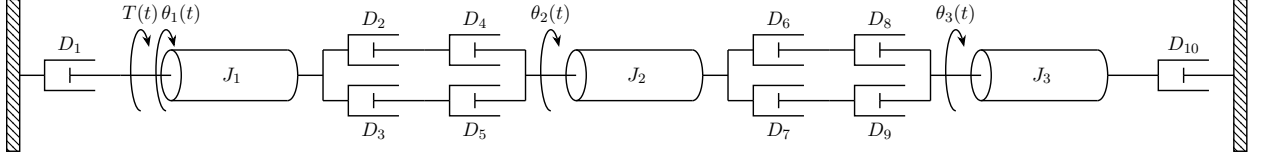
$$\begin{aligned} \det(A) &= (s^2 + s + 5)[(2s^2 + 2s + 4)(s^2 + s + 2) - (s + 1)^2] - (s + 3)^2(s^2 + s + 2) \\ \det(A) &= 2s^6 + 6s^5 + 22s^4 + 28s^3 + 41s^2 + 16s + 17 \end{aligned}$$

The final transfer function is:

$$\boxed{\frac{\Theta_3(s)}{T(s)} = \frac{s^2 + 4s + 3}{2s^6 + 6s^5 + 22s^4 + 28s^3 + 41s^2 + 16s + 17}}$$

Problem 18

For the rotational mechanical system shown in the diagram below, determine the transfer function $G(s) = \theta_3(s)/T(s)$.



Given:

$$\begin{aligned}
 J_1 &= 1 \text{ kg} \cdot \text{m}^2 & K_1 &= 2 \text{ N} \cdot \text{m/rad} & D_1 &= 1 \text{ N} \cdot \text{m} \cdot \text{s/rad} \\
 J_2 &= 1 \text{ kg} \cdot \text{m}^2 & K_2 &= 3 \text{ N} \cdot \text{m/rad} & D_2 &= 1 \text{ N} \cdot \text{m} \cdot \text{s/rad} \\
 J_3 &= 1 \text{ kg} \cdot \text{m}^2 & K_3 &= 1 \text{ N} \cdot \text{m/rad} & D_3 &= 1 \text{ N} \cdot \text{m} \cdot \text{s/rad} \\
 & & K_4 &= 1 \text{ N} \cdot \text{m/rad} & D_4 &= 1 \text{ N} \cdot \text{m} \cdot \text{s/rad} \\
 & & & & D_5 &= 1 \text{ N} \cdot \text{m} \cdot \text{s/rad} \\
 & & & & D_6 &= 1 \text{ N} \cdot \text{m} \cdot \text{s/rad} \\
 & & & & D_7 &= 1 \text{ N} \cdot \text{m} \cdot \text{s/rad} \\
 & & & & D_8 &= 1 \text{ N} \cdot \text{m} \cdot \text{s/rad} \\
 & & & & D_9 &= 1 \text{ N} \cdot \text{m} \cdot \text{s/rad} \\
 & & & & D_{10} &= 1 \text{ N} \cdot \text{m} \cdot \text{s/rad}
 \end{aligned}$$

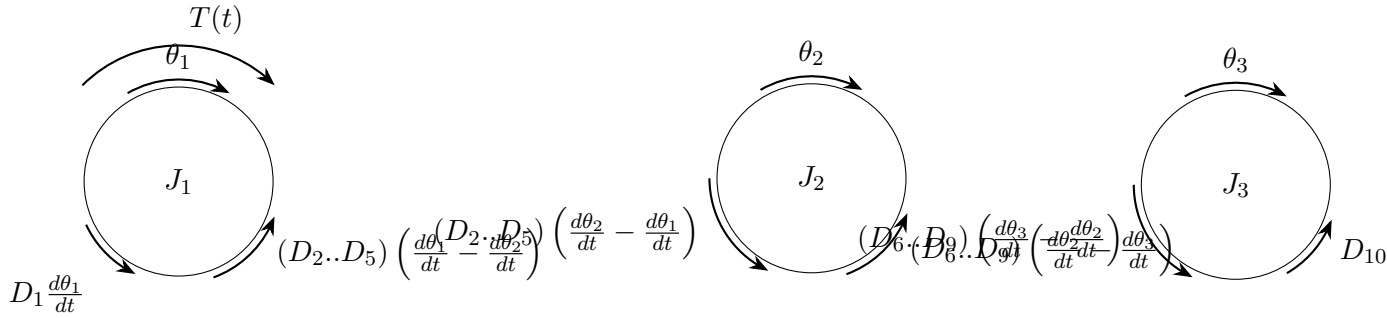
Required:

$$G(s) = \frac{\theta_3(s)}{T(s)}$$

Solution:

Free-Body Diagrams (FBDs)

We assume clockwise rotation is positive.



Equations of Motion

We apply Newton's second law for rotation ($\sum \tau = J \frac{d^2\theta}{dt^2}$) to each mass.

For Inertia 1 (J_1):

$$\begin{aligned}
 J_1 \frac{d^2\theta_1}{dt^2} + D_1 \frac{d\theta_1}{dt} + (D_2 + D_3 + D_4 + D_5) \left(\frac{d\theta_1}{dt} - \frac{d\theta_2}{dt} \right) &= T(t) \\
 J_1 \frac{d^2\theta_1}{dt^2} + (D_1 + D_2 + D_3 + D_4 + D_5) \frac{d\theta_1}{dt} - (D_2 + D_3 + D_4 + D_5) \frac{d\theta_2}{dt} &= T(t)
 \end{aligned}$$

For Inertia 2 (J_2):

$$\begin{aligned}
 J_2 \frac{d^2\theta_2}{dt^2} + (D_2 + D_3 + D_4 + D_5) \left(\frac{d\theta_2}{dt} - \frac{d\theta_1}{dt} \right) + (D_6 + D_7 + D_8 + D_9) \left(\frac{d\theta_2}{dt} - \frac{d\theta_3}{dt} \right) &= 0 \\
 J_2 \frac{d^2\theta_2}{dt^2} + (D_2 + D_3 + D_4 + D_5) \frac{d\theta_2}{dt} - (D_6 + D_7 + D_8 + D_9) \frac{d\theta_3}{dt} &= 0
 \end{aligned}$$

For Inertia 3 (J_3):

$$J_3 \frac{d^2 \theta_3}{dt^2} + (D_6 + D_7 + D_8 + D_9) \left(\frac{d\theta_3}{dt} - \frac{d\theta_2}{dt} \right) + D_{10} \frac{d\theta_3}{dt} = 0$$

$$J_3 \frac{d^2 \theta_3}{dt^2} + (D_6 + D_7 + D_8 + D_9 + D_{10}) \frac{d\theta_3}{dt} - (D_6 + D_7 + D_8 + D_9) \frac{d\theta_2}{dt} = 0$$

Laplace Transform

We take the Laplace transform of the equations, assuming zero initial conditions.

Equation for Inertia 1:

$$[J_1 s^2 + (D_1..D_5)s]\Theta_1(s) - [(D_2..D_5)s]\Theta_2(s) = T(s)$$

Equation for Inertia 2:

$$-[(D_2..D_5)s]\Theta_1(s) + [J_2 s^2 + (D_2..D_9)s]\Theta_2(s) - [(D_6..D_9)s]\Theta_3(s) = 0$$

Equation for Inertia 3:

$$-[(D_6..D_9)s]\Theta_2(s) + [J_3 s^2 + (D_6..D_{10})s]\Theta_3(s) = 0$$

Substituting the given numerical values (all elements equal to 1):

$$(s^2 + 5s)\Theta_1(s) - 4s\Theta_2(s) = T(s) \quad (1)$$

$$-4s\Theta_1(s) + (s^2 + 8s)\Theta_2(s) - 4s\Theta_3(s) = 0 \quad (2)$$

$$-4s\Theta_2(s) + (s^2 + 5s)\Theta_3(s) = 0 \quad (3)$$

Solving for the Transfer Function

The system can be expressed in matrix form $A\Theta = \mathbf{b}$:

$$\begin{bmatrix} s^2 + 5s & -4s & 0 \\ -4s & s^2 + 8s & -4s \\ 0 & -4s & s^2 + 5s \end{bmatrix} \begin{bmatrix} \Theta_1(s) \\ \Theta_2(s) \\ \Theta_3(s) \end{bmatrix} = \begin{bmatrix} T(s) \\ 0 \\ 0 \end{bmatrix}$$

Using Cramer's Rule to solve for $\Theta_3(s)$:

$$\Theta_3(s) = \frac{\det(A_3)}{\det(A)}$$

Determinant of A_3 :

$$\det(A_3) = T(s) \begin{vmatrix} -4s & s^2 + 8s \\ 0 & -4s \end{vmatrix} = T(s)(-4s)(-4s) = 16s^2 T(s)$$

Determinant of A :

$$\det(A) = (s^2 + 5s)[(s^2 + 8s)(s^2 + 5s) - (-4s)^2] - (-4s)^2(s^2 + 5s)$$

$$\det(A) = (s^2 + 5s)(s^4 + 13s^3 + 24s^2) - 16s^4 - 80s^3$$

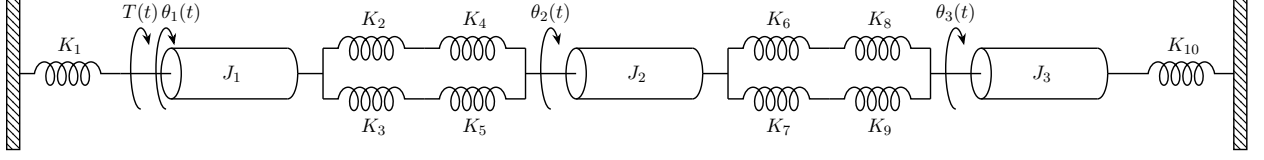
$$\det(A) = s^6 + 18s^5 + 73s^4 + 40s^3$$

The final transfer function is:

$$\boxed{\frac{\Theta_3(s)}{T(s)} = \frac{16s^2}{s^6 + 18s^5 + 73s^4 + 40s^3}}$$

Problem 19

For the rotational mechanical system shown in the diagram below, determine the transfer function $G(s) = \theta_3(s)/T(s)$.



Given:

$$\begin{aligned} J_1 &= 1 \text{ kg} \cdot \text{m}^2 & K_1 &= 1 \text{ N} \cdot \text{m/rad} & K_6 &= 1 \text{ N} \cdot \text{m/rad} \\ J_2 &= 1 \text{ kg} \cdot \text{m}^2 & K_2 &= 1 \text{ N} \cdot \text{m/rad} & K_7 &= 1 \text{ N} \cdot \text{m/rad} \\ J_3 &= 1 \text{ kg} \cdot \text{m}^2 & K_3 &= 1 \text{ N} \cdot \text{m/rad} & K_8 &= 1 \text{ N} \cdot \text{m/rad} \\ & & K_4 &= 1 \text{ N} \cdot \text{m/rad} & K_9 &= 1 \text{ N} \cdot \text{m/rad} \\ & & K_5 &= 1 \text{ N} \cdot \text{m/rad} & K_{10} &= 1 \text{ N} \cdot \text{m/rad} \end{aligned}$$

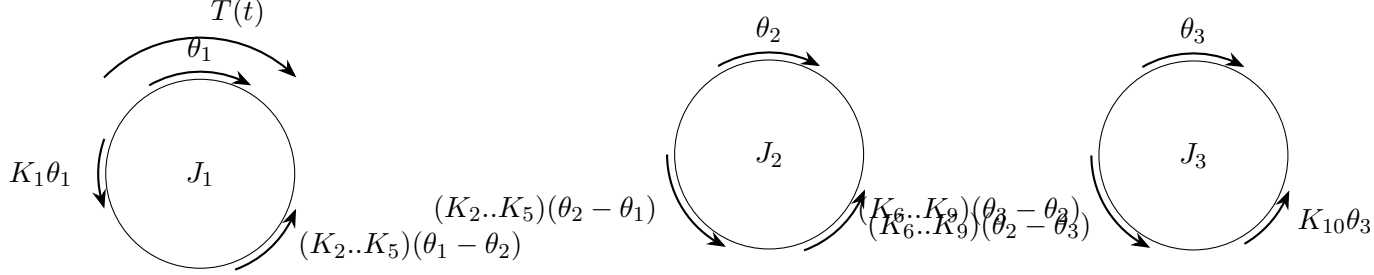
Required:

$$G(s) = \frac{\theta_3(s)}{T(s)}$$

Solution:

Free-Body Diagrams (FBDs)

We assume clockwise rotation is positive.



Equations of Motion

We apply Newton's second law for rotation ($\sum \tau = J \frac{d^2\theta}{dt^2}$) to each mass.

For Inertia 1 (J_1):

$$\begin{aligned} J_1 \frac{d^2\theta_1}{dt^2} + K_1\theta_1 + (K_2 + K_3 + K_4 + K_5)(\theta_1 - \theta_2) &= T(t) \\ J_1 \frac{d^2\theta_1}{dt^2} + (K_1 + K_2 + K_3 + K_4 + K_5)\theta_1 - (K_2 + K_3 + K_4 + K_5)\theta_2 &= T(t) \end{aligned}$$

For Inertia 2 (J_2):

$$\begin{aligned} J_2 \frac{d^2\theta_2}{dt^2} + (K_2 + K_3 + K_4 + K_5)(\theta_2 - \theta_1) + (K_6 + K_7 + K_8 + K_9)(\theta_2 - \theta_3) &= 0 \\ J_2 \frac{d^2\theta_2}{dt^2} + (K_2..K_9)\theta_2 - (K_2 + K_3 + K_4 + K_5)\theta_1 - (K_6 + K_7 + K_8 + K_9)\theta_3 &= 0 \end{aligned}$$

For Inertia 3 (J_3):

$$\begin{aligned} J_3 \frac{d^2\theta_3}{dt^2} + (K_6 + K_7 + K_8 + K_9)(\theta_3 - \theta_2) + K_{10}\theta_3 &= 0 \\ J_3 \frac{d^2\theta_3}{dt^2} + (K_6 + K_7 + K_8 + K_9 + K_{10})\theta_3 - (K_6 + K_7 + K_8 + K_9)\theta_2 &= 0 \end{aligned}$$

Laplace Transform

We take the Laplace transform of the equations, assuming zero initial conditions.

Equation for Inertia 1:

$$[J_1 s^2 + (K_1..K_5)]\Theta_1(s) - [K_2..K_5]\Theta_2(s) = T(s)$$

Equation for Inertia 2:

$$-[K_2..K_5]\Theta_1(s) + [J_2 s^2 + (K_2..K_9)]\Theta_2(s) - [K_6..K_9]\Theta_3(s) = 0$$

Equation for Inertia 3:

$$-[K_6..K_9]\Theta_2(s) + [J_3 s^2 + (K_6..K_{10})]\Theta_3(s) = 0$$

Substituting the given numerical values (all elements equal to 1):

$$(s^2 + 5)\Theta_1(s) - 4\Theta_2(s) = T(s) \quad (1)$$

$$-4\Theta_1(s) + (s^2 + 8)\Theta_2(s) - 4\Theta_3(s) = 0 \quad (2)$$

$$-4\Theta_2(s) + (s^2 + 5)\Theta_3(s) = 0 \quad (3)$$

Solving for the Transfer Function

The system can be expressed in matrix form $A\Theta = \mathbf{b}$:

$$\begin{bmatrix} s^2 + 5 & -4 & 0 \\ -4 & s^2 + 8 & -4 \\ 0 & -4 & s^2 + 5 \end{bmatrix} \begin{bmatrix} \Theta_1(s) \\ \Theta_2(s) \\ \Theta_3(s) \end{bmatrix} = \begin{bmatrix} T(s) \\ 0 \\ 0 \end{bmatrix}$$

Using Cramer's Rule to solve for $\Theta_3(s)$:

$$\Theta_3(s) = \frac{\det(A_3)}{\det(A)}$$

Determinant of A_3 :

$$\det(A_3) = T(s) \begin{vmatrix} -4 & s^2 + 8 \\ 0 & -4 \end{vmatrix} = T(s)(-4)(-4) = 16T(s)$$

Determinant of A :

$$\det(A) = (s^2 + 5)[(s^2 + 8)(s^2 + 5) - (-4)^2] - (-4)^2(s^2 + 5)$$

$$\det(A) = (s^2 + 5)(s^4 + 13s^2 + 24) - 16s^2 - 80$$

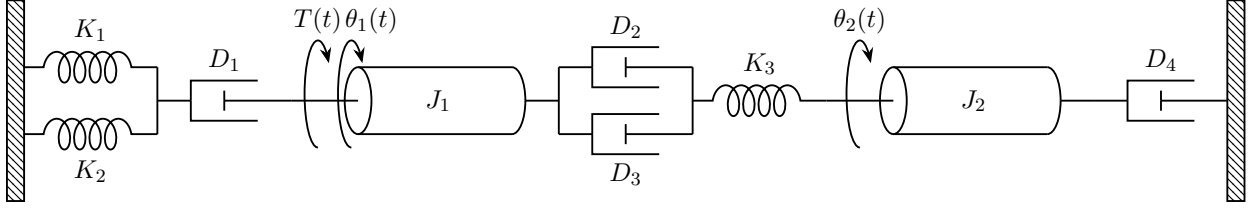
$$\det(A) = s^6 + 18s^4 + 73s^2 + 40$$

The final transfer function is:

$$\boxed{\frac{\Theta_3(s)}{T(s)} = \frac{16}{s^6 + 18s^4 + 73s^2 + 40}}$$

Problem 20

For the rotational mechanical system shown in the diagram below, determine the transfer function $G(s) = \theta_2(s)/T(s)$.



Given:

$$\begin{aligned} J_1 &= 2 \text{ kg} \cdot \text{m}^2 & K_1 &= 4 \text{ N} \cdot \text{m/rad} & D_1 &= 6 \text{ N} \cdot \text{m} \cdot \text{s/rad} \\ J_2 &= 1 \text{ kg} \cdot \text{m}^2 & K_2 &= 5 \text{ N} \cdot \text{m/rad} & D_2 &= 2 \text{ N} \cdot \text{m} \cdot \text{s/rad} \\ J_3 &= 1 \text{ kg} \cdot \text{m}^2 & K_3 &= 3 \text{ N} \cdot \text{m/rad} & D_3 &= 1 \text{ N} \cdot \text{m} \cdot \text{s/rad} \\ & & K_4 &= 2 \text{ N} \cdot \text{m/rad} & D_4 &= 4 \text{ N} \cdot \text{m} \cdot \text{s/rad} \\ & & K_5 &= 1 \text{ N} \cdot \text{m/rad} \end{aligned}$$

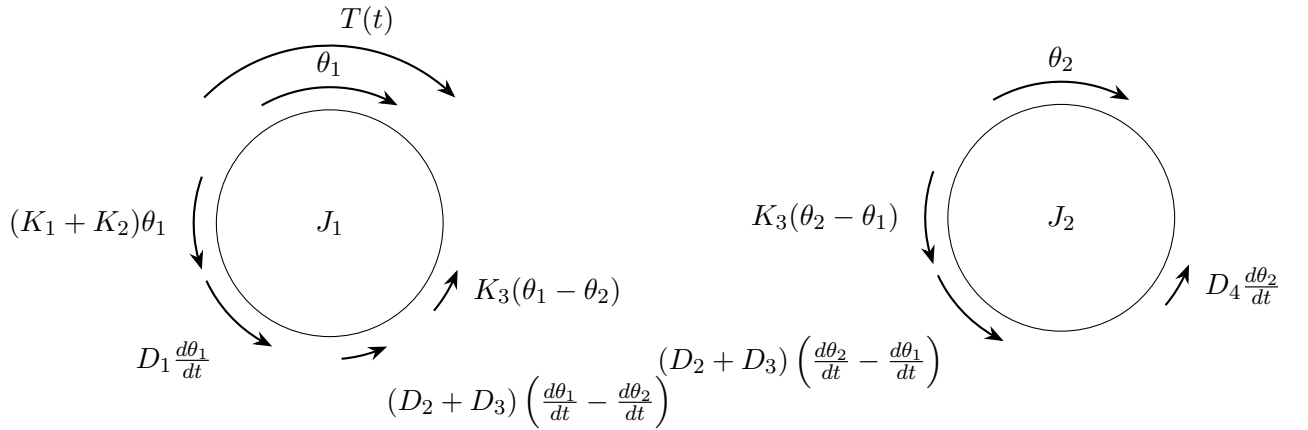
Required:

$$G(s) = \frac{\theta_2(s)}{T(s)}$$

Solution:

Free-Body Diagrams (FBDs)

We assume clockwise rotation is positive.

**Equations of Motion**

We apply Newton's second law for rotation ($\sum \tau = J \frac{d^2\theta}{dt^2}$) to each mass.

For Inertia 1 (J_1):

$$\begin{aligned} J_1 \frac{d^2\theta_1}{dt^2} + D_1 \frac{d\theta_1}{dt} + (K_1 + K_2)\theta_1 + (D_2 + D_3) \left(\frac{d\theta_1}{dt} - \frac{d\theta_2}{dt} \right) + K_3(\theta_1 - \theta_2) &= T(t) \\ J_1 \frac{d^2\theta_1}{dt^2} + (D_1 + D_2 + D_3) \frac{d\theta_1}{dt} + (K_1 + K_2 + K_3)\theta_1 - (D_2 + D_3) \frac{d\theta_2}{dt} - K_3\theta_2 &= T(t) \end{aligned}$$

For Inertia 2 (J_2):

$$\begin{aligned} J_2 \frac{d^2\theta_2}{dt^2} + D_4 \frac{d\theta_2}{dt} + (D_2 + D_3) \left(\frac{d\theta_2}{dt} - \frac{d\theta_1}{dt} \right) + K_3(\theta_2 - \theta_1) &= 0 \\ J_2 \frac{d^2\theta_2}{dt^2} + (D_2 + D_3 + D_4) \frac{d\theta_2}{dt} + K_3\theta_2 - (D_2 + D_3) \frac{d\theta_1}{dt} - K_3\theta_1 &= 0 \end{aligned}$$

Laplace Transform

We take the Laplace transform of the equations, assuming zero initial conditions.

Equation for Inertia 1:

$$[J_1 s^2 + (D_1 + D_2 + D_3)s + (K_1 + K_2 + K_3)]\Theta_1(s) - [(D_2 + D_3)s + K_3]\Theta_2(s) = T(s)$$

Equation for Inertia 2:

$$-[(D_2 + D_3)s + K_3]\Theta_1(s) + [J_2 s^2 + (D_2 + D_3 + D_4)s + K_3]\Theta_2(s) = 0$$

Substituting the given numerical values:

$$(2s^2 + 9s + 12)\Theta_1(s) - (3s + 3)\Theta_2(s) = T(s) \quad (1)$$

$$-(3s + 3)\Theta_1(s) + (s^2 + 7s + 3)\Theta_2(s) = 0 \quad (2)$$

Solving for the Transfer Function

From equation (2), we express $\Theta_1(s)$ in terms of $\Theta_2(s)$:

$$\Theta_1(s) = \frac{s^2 + 7s + 3}{3s + 3}\Theta_2(s)$$

Substitute this into equation (1):

$$(2s^2 + 9s + 12) \left(\frac{s^2 + 7s + 3}{3s + 3}\Theta_2(s) \right) - (3s + 3)\Theta_2(s) = T(s)$$

Factor out $\Theta_2(s)$:

$$\left[\frac{(2s^2 + 9s + 12)(s^2 + 7s + 3) - (3s + 3)^2}{3s + 3} \right] \Theta_2(s) = T(s)$$

Rearrange to find the transfer function $\frac{\Theta_2(s)}{T(s)}$:

$$\frac{\Theta_2(s)}{T(s)} = \frac{3s + 3}{(2s^2 + 9s + 12)(s^2 + 7s + 3) - (3s + 3)^2}$$

Expand and simplify the denominator:

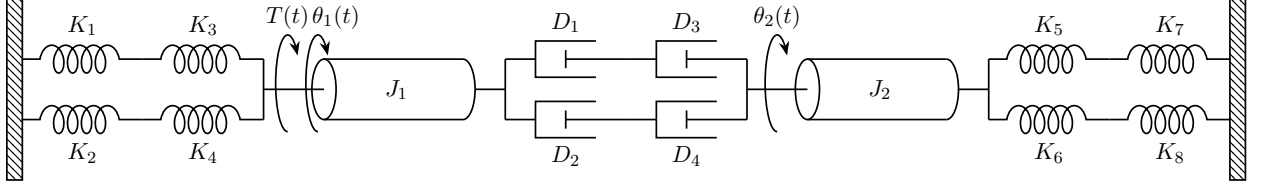
$$\begin{aligned} \text{Denominator} &= (2s^4 + 14s^3 + 6s^2 + 9s^3 + 63s^2 + 27s + 12s^2 + 84s + 36) - (9s^2 + 18s + 9) \\ &= (2s^4 + 23s^3 + 81s^2 + 111s + 36) - (9s^2 + 18s + 9) \\ &= 2s^4 + 23s^3 + 72s^2 + 93s + 27 \end{aligned}$$

The final transfer function is:

$$\boxed{\frac{\Theta_2(s)}{T(s)} = \frac{3s + 3}{2s^4 + 23s^3 + 72s^2 + 93s + 27}}$$

Problem 21

For the rotational mechanical system shown in the diagram below, determine the transfer function $G(s) = \theta_2(s)/T(s)$.



Given:

$$\begin{aligned} J_1 &= 1 \text{ kg} \cdot \text{m}^2 & K_1 &= 1 \text{ N} \cdot \text{m}/\text{rad} & K_5 &= 1 \text{ N} \cdot \text{m}/\text{rad} & D_1 &= 1 \text{ N} \cdot \text{m} \cdot \text{s}/\text{rad} \\ J_2 &= 1 \text{ kg} \cdot \text{m}^2 & K_2 &= 1 \text{ N} \cdot \text{m}/\text{rad} & K_6 &= 1 \text{ N} \cdot \text{m}/\text{rad} & D_2 &= 1 \text{ N} \cdot \text{m} \cdot \text{s}/\text{rad} \\ & & K_3 &= 1 \text{ N} \cdot \text{m}/\text{rad} & K_7 &= 1 \text{ N} \cdot \text{m}/\text{rad} & D_3 &= 1 \text{ N} \cdot \text{m} \cdot \text{s}/\text{rad} \\ & & K_4 &= 1 \text{ N} \cdot \text{m}/\text{rad} & K_8 &= 1 \text{ N} \cdot \text{m}/\text{rad} & D_4 &= 1 \text{ N} \cdot \text{m} \cdot \text{s}/\text{rad} \end{aligned}$$

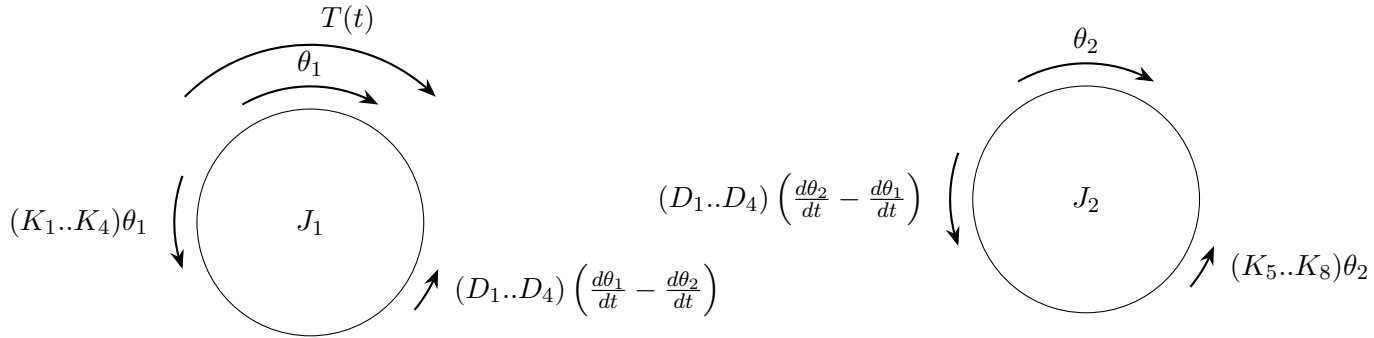
Required:

$$G(s) = \frac{\theta_2(s)}{T(s)}$$

Solution:

Free-Body Diagrams (FBDs)

We assume clockwise rotation is positive.



Equations of Motion

We apply Newton's second law for rotation ($\sum \tau = J \frac{d^2\theta}{dt^2}$) to each mass.

For Inertia 1 (J_1):

$$\begin{aligned} J_1 \frac{d^2\theta_1}{dt^2} + (K_1 + K_2 + K_3 + K_4)\theta_1 + (D_1 + D_2 + D_3 + D_4) \left(\frac{d\theta_1}{dt} - \frac{d\theta_2}{dt} \right) &= T(t) \\ J_1 \frac{d^2\theta_1}{dt^2} + (D_1 + D_2 + D_3 + D_4) \frac{d\theta_1}{dt} + (K_1 + K_2 + K_3 + K_4)\theta_1 - (D_1 + D_2 + D_3 + D_4) \frac{d\theta_2}{dt} &= T(t) \end{aligned}$$

For Inertia 2 (J_2):

$$\begin{aligned} J_2 \frac{d^2\theta_2}{dt^2} + (D_1 + D_2 + D_3 + D_4) \left(\frac{d\theta_2}{dt} - \frac{d\theta_1}{dt} \right) + (K_5 + K_6 + K_7 + K_8)\theta_2 &= 0 \\ J_2 \frac{d^2\theta_2}{dt^2} + (D_1 + D_2 + D_3 + D_4) \frac{d\theta_2}{dt} + (K_5 + K_6 + K_7 + K_8)\theta_2 - (D_1 + D_2 + D_3 + D_4) \frac{d\theta_1}{dt} &= 0 \end{aligned}$$

Laplace Transform

We take the Laplace transform of the equations, assuming zero initial conditions.

Equation for Inertia 1:

$$[J_1 s^2 + (D_1..D_4)s + (K_1..K_4)]\Theta_1(s) - [(D_1..D_4)s]\Theta_2(s) = T(s)$$

Equation for Inertia 2:

$$-[(D_1..D_4)s]\Theta_1(s) + [J_2 s^2 + (D_1..D_4)s + (K_5..K_8)]\Theta_2(s) = 0$$

Substituting the given numerical values:

$$(s^2 + 4s + 4)\Theta_1(s) - 4s\Theta_2(s) = T(s) \quad (1)$$

$$-4s\Theta_1(s) + (s^2 + 4s + 4)\Theta_2(s) = 0 \quad (2)$$

Solving for the Transfer Function

From equation (2), we express $\Theta_1(s)$ in terms of $\Theta_2(s)$:

$$\Theta_1(s) = \frac{s^2 + 4s + 4}{4s}\Theta_2(s)$$

Substitute this into equation (1):

$$(s^2 + 4s + 4) \left(\frac{s^2 + 4s + 4}{4s}\Theta_2(s) \right) - 4s\Theta_2(s) = T(s)$$

Factor out $\Theta_2(s)$:

$$\left[\frac{(s^2 + 4s + 4)^2 - 16s^2}{4s} \right] \Theta_2(s) = T(s)$$

Rearrange to find the transfer function $\frac{\Theta_2(s)}{T(s)}$:

$$\frac{\Theta_2(s)}{T(s)} = \frac{4s}{(s^2 + 4s + 4)^2 - 16s^2}$$

Expand and simplify the denominator:

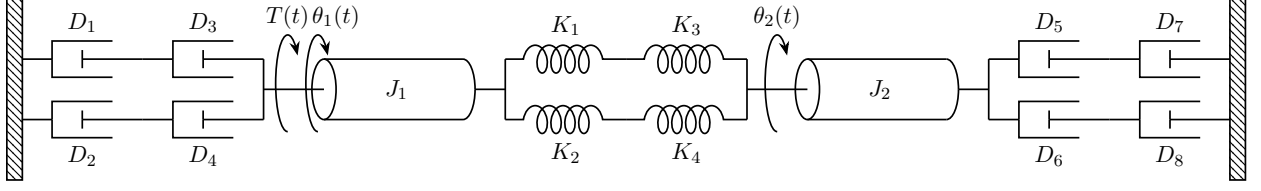
$$\begin{aligned} \text{Denominator} &= (s^4 + 8s^3 + 24s^2 + 32s + 16) - 16s^2 \\ &= s^4 + 8s^3 + 8s^2 + 32s + 16 \end{aligned}$$

The final transfer function is:

$$\boxed{\frac{\Theta_2(s)}{T(s)} = \frac{4s}{s^4 + 8s^3 + 8s^2 + 32s + 16}}$$

Problem 22

For the rotational mechanical system shown in the diagram below, determine the transfer function $G(s) = \theta_2(s)/T(s)$.



Given:

$$\begin{aligned} J_1 &= 1 \text{ kg} \cdot \text{m}^2 & K_1 &= 1 \text{ N} \cdot \text{m}/\text{rad} & D_1 &= 1 \text{ N} \cdot \text{m} \cdot \text{s}/\text{rad} & D_5 &= 1 \text{ N} \cdot \text{m} \cdot \text{s}/\text{rad} \\ J_2 &= 1 \text{ kg} \cdot \text{m}^2 & K_2 &= 1 \text{ N} \cdot \text{m}/\text{rad} & D_2 &= 1 \text{ N} \cdot \text{m} \cdot \text{s}/\text{rad} & D_6 &= 1 \text{ N} \cdot \text{m} \cdot \text{s}/\text{rad} \\ & & K_3 &= 1 \text{ N} \cdot \text{m}/\text{rad} & D_3 &= 1 \text{ N} \cdot \text{m} \cdot \text{s}/\text{rad} & D_7 &= 1 \text{ N} \cdot \text{m} \cdot \text{s}/\text{rad} \\ & & K_4 &= 1 \text{ N} \cdot \text{m}/\text{rad} & D_4 &= 1 \text{ N} \cdot \text{m} \cdot \text{s}/\text{rad} & D_8 &= 1 \text{ N} \cdot \text{m} \cdot \text{s}/\text{rad} \end{aligned}$$

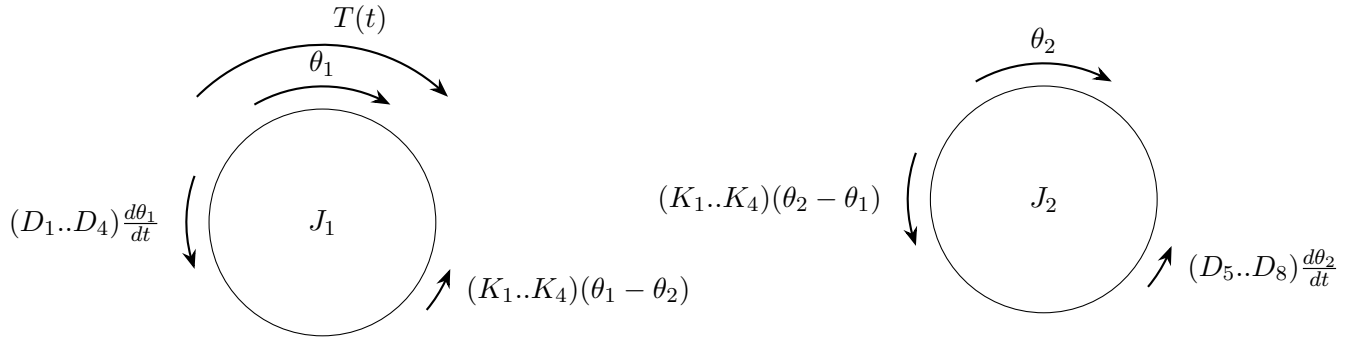
Required:

$$G(s) = \frac{\theta_2(s)}{T(s)}$$

Solution:

Free-Body Diagrams (FBDs)

We assume clockwise rotation is positive.



Equations of Motion

We apply Newton's second law for rotation ($\sum \tau = J \frac{d^2\theta}{dt^2}$) to each mass.

For Inertia 1 (J_1):

$$\begin{aligned} J_1 \frac{d^2\theta_1}{dt^2} + (D_1 + D_2 + D_3 + D_4) \frac{d\theta_1}{dt} + (K_1 + K_2 + K_3 + K_4)(\theta_1 - \theta_2) &= T(t) \\ J_1 \frac{d^2\theta_1}{dt^2} + (D_1 + D_2 + D_3 + D_4) \frac{d\theta_1}{dt} + (K_1 + K_2 + K_3 + K_4)\theta_1 - (K_1 + K_2 + K_3 + K_4)\theta_2 &= T(t) \end{aligned}$$

For Inertia 2 (J_2):

$$\begin{aligned} J_2 \frac{d^2\theta_2}{dt^2} + (D_5 + D_6 + D_7 + D_8) \frac{d\theta_2}{dt} + (K_1 + K_2 + K_3 + K_4)(\theta_2 - \theta_1) &= 0 \\ J_2 \frac{d^2\theta_2}{dt^2} + (D_5 + D_6 + D_7 + D_8) \frac{d\theta_2}{dt} + (K_1 + K_2 + K_3 + K_4)\theta_2 - (K_1 + K_2 + K_3 + K_4)\theta_1 &= 0 \end{aligned}$$

Laplace Transform

We take the Laplace transform of the equations, assuming zero initial conditions.

Equation for Inertia 1:

$$[J_1 s^2 + (D_1..D_4)s + (K_1..K_4)]\Theta_1(s) - [K_1..K_4]\Theta_2(s) = T(s)$$

Equation for Inertia 2:

$$-[K_1..K_4]\Theta_1(s) + [J_2 s^2 + (D_5..D_8)s + (K_1..K_4)]\Theta_2(s) = 0$$

Substituting the given numerical values:

$$(s^2 + 4s + 4)\Theta_1(s) - 4\Theta_2(s) = T(s) \quad (1)$$

$$-4\Theta_1(s) + (s^2 + 4s + 4)\Theta_2(s) = 0 \quad (2)$$

Solving for the Transfer Function

From equation (2), we express $\Theta_1(s)$ in terms of $\Theta_2(s)$:

$$\Theta_1(s) = \frac{s^2 + 4s + 4}{4}\Theta_2(s)$$

Substitute this into equation (1):

$$(s^2 + 4s + 4) \left(\frac{s^2 + 4s + 4}{4}\Theta_2(s) \right) - 4\Theta_2(s) = T(s)$$

Factor out $\Theta_2(s)$:

$$\left[\frac{(s^2 + 4s + 4)^2 - 16}{4} \right] \Theta_2(s) = T(s)$$

Rearrange to find the transfer function $\frac{\Theta_2(s)}{T(s)}$:

$$\frac{\Theta_2(s)}{T(s)} = \frac{4}{(s^2 + 4s + 4)^2 - 16}$$

Expand and simplify the denominator:

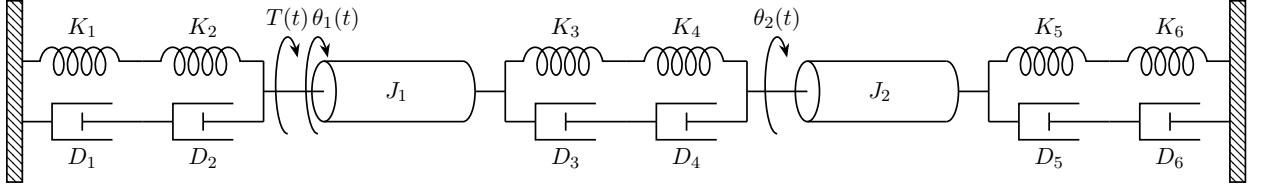
$$\begin{aligned} \text{Denominator} &= (s^4 + 8s^3 + 24s^2 + 32s + 16) - 16 \\ &= s^4 + 8s^3 + 24s^2 + 32s \end{aligned}$$

The final transfer function is:

$$\boxed{\frac{\Theta_2(s)}{T(s)} = \frac{4}{s^4 + 8s^3 + 24s^2 + 32s}}$$

Problem 23

For the rotational mechanical system shown in the diagram below, determine the transfer function $G(s) = \theta_2(s)/T(s)$.



Given:

$$\begin{aligned} J_1 &= 1 \text{ kg} \cdot \text{m}^2 & K_1 &= 1 \text{ N} \cdot \text{m}/\text{rad} & D_1 &= 1 \text{ N} \cdot \text{m} \cdot \text{s}/\text{rad} \\ J_2 &= 1 \text{ kg} \cdot \text{m}^2 & K_2 &= 1 \text{ N} \cdot \text{m}/\text{rad} & D_2 &= 1 \text{ N} \cdot \text{m} \cdot \text{s}/\text{rad} \\ & & K_3 &= 1 \text{ N} \cdot \text{m}/\text{rad} & D_3 &= 1 \text{ N} \cdot \text{m} \cdot \text{s}/\text{rad} \\ & & K_4 &= 1 \text{ N} \cdot \text{m}/\text{rad} & D_4 &= 1 \text{ N} \cdot \text{m} \cdot \text{s}/\text{rad} \\ & & K_5 &= 1 \text{ N} \cdot \text{m}/\text{rad} & D_5 &= 1 \text{ N} \cdot \text{m} \cdot \text{s}/\text{rad} \\ & & K_6 &= 1 \text{ N} \cdot \text{m}/\text{rad} & D_6 &= 1 \text{ N} \cdot \text{m} \cdot \text{s}/\text{rad} \end{aligned}$$

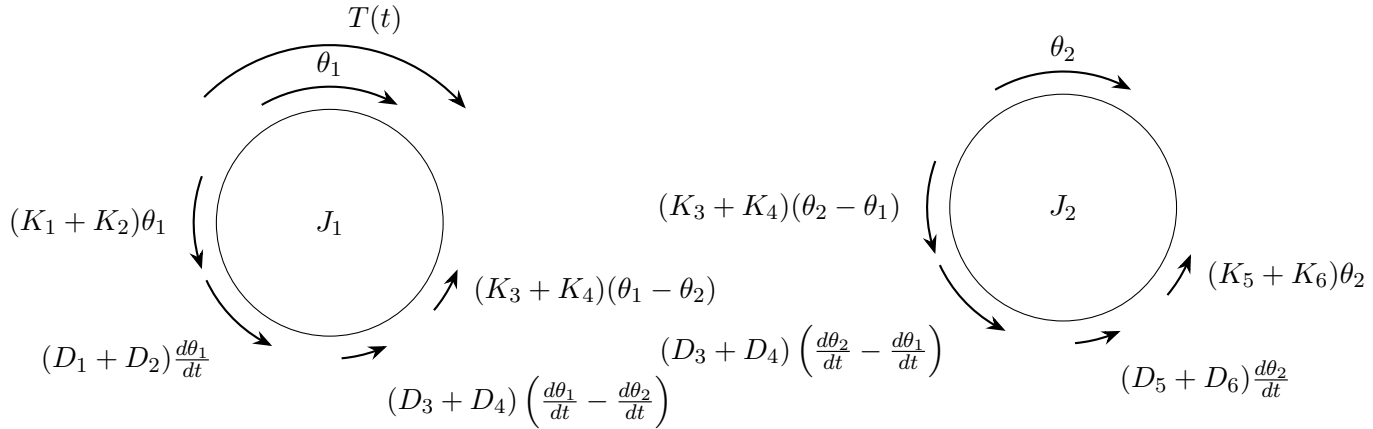
Required:

$$G(s) = \frac{\theta_2(s)}{T(s)}$$

Solution:

Free-Body Diagrams (FBDs)

We assume clockwise rotation is positive.



Equations of Motion

We apply Newton's second law for rotation ($\sum \tau = J \frac{d^2\theta}{dt^2}$) to each mass.

For Inertia 1 (J_1):

$$\begin{aligned} J_1 \frac{d^2\theta_1}{dt^2} + (D_1 + D_2) \frac{d\theta_1}{dt} + (K_1 + K_2)\theta_1 + (D_3 + D_4) \left(\frac{d\theta_1}{dt} - \frac{d\theta_2}{dt} \right) + (K_3 + K_4)(\theta_1 - \theta_2) &= T(t) \\ J_1 \frac{d^2\theta_1}{dt^2} + (D_1 + D_2 + D_3 + D_4) \frac{d\theta_1}{dt} + (K_1 + K_2 + K_3 + K_4)\theta_1 - (D_3 + D_4) \frac{d\theta_2}{dt} - (K_3 + K_4)\theta_2 &= T(t) \end{aligned}$$

For Inertia 2 (J_2):

$$\begin{aligned} J_2 \frac{d^2\theta_2}{dt^2} + (D_5 + D_6) \frac{d\theta_2}{dt} + (K_5 + K_6)\theta_2 + (D_3 + D_4) \left(\frac{d\theta_2}{dt} - \frac{d\theta_1}{dt} \right) + (K_3 + K_4)(\theta_2 - \theta_1) &= 0 \\ J_2 \frac{d^2\theta_2}{dt^2} + (D_3 + D_4 + D_5 + D_6) \frac{d\theta_2}{dt} + (K_3 + K_4 + K_5 + K_6)\theta_2 - (D_3 + D_4) \frac{d\theta_1}{dt} - (K_3 + K_4)\theta_1 &= 0 \end{aligned}$$

Laplace Transform

We take the Laplace transform of the equations, assuming zero initial conditions.

Equation for Inertia 1:

$$[J_1 s^2 + (D_1 + D_2 + D_3 + D_4)s + (K_1 + K_2 + K_3 + K_4)]\Theta_1(s) - [(D_3 + D_4)s + (K_3 + K_4)]\Theta_2(s) = T(s)$$

Equation for Inertia 2:

$$-[(D_3 + D_4)s + (K_3 + K_4)]\Theta_1(s) + [J_2 s^2 + (D_3 + D_4 + D_5 + D_6)s + (K_3 + K_4 + K_5 + K_6)]\Theta_2(s) = 0$$

Substituting the given numerical values:

$$(s^2 + 4s + 4)\Theta_1(s) - (2s + 2)\Theta_2(s) = T(s) \quad (1)$$

$$-(2s + 2)\Theta_1(s) + (s^2 + 4s + 4)\Theta_2(s) = 0 \quad (2)$$

Solving for the Transfer Function

From equation (2), we express $\Theta_1(s)$ in terms of $\Theta_2(s)$:

$$\Theta_1(s) = \frac{s^2 + 4s + 4}{2s + 2}\Theta_2(s)$$

Substitute this into equation (1):

$$(s^2 + 4s + 4) \left(\frac{s^2 + 4s + 4}{2s + 2}\Theta_2(s) \right) - (2s + 2)\Theta_2(s) = T(s)$$

Factor out $\Theta_2(s)$:

$$\left[\frac{(s^2 + 4s + 4)^2 - (2s + 2)^2}{2s + 2} \right] \Theta_2(s) = T(s)$$

Rearrange to find the transfer function $\frac{\Theta_2(s)}{T(s)}$:

$$\frac{\Theta_2(s)}{T(s)} = \frac{2s + 2}{(s^2 + 4s + 4)^2 - (2s + 2)^2}$$

Expand and simplify the denominator:

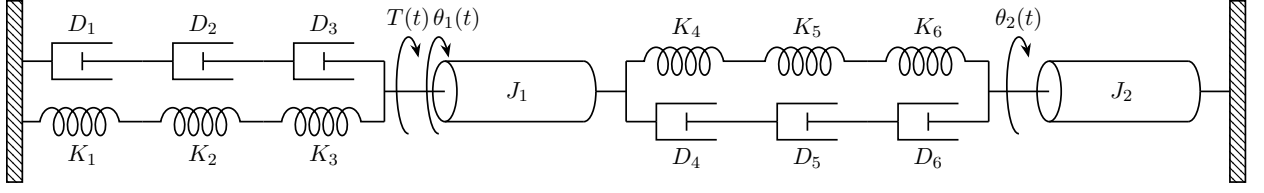
$$\begin{aligned} \text{Denominator} &= (s^4 + 8s^3 + 24s^2 + 32s + 16) - (4s^2 + 8s + 4) \\ &= s^4 + 8s^3 + 20s^2 + 24s + 12 \end{aligned}$$

The final transfer function is:

$$\boxed{\frac{\Theta_2(s)}{T(s)} = \frac{2s + 2}{s^4 + 8s^3 + 20s^2 + 24s + 12}}$$

Problem 24

For the rotational mechanical system shown in the diagram below, determine the transfer function $G(s) = \theta_2(s)/T(s)$.



Given:

$$\begin{aligned} J_1 &= 1 \text{ kg} \cdot \text{m}^2 & K_1 &= 1 \text{ N} \cdot \text{m}/\text{rad} & D_1 &= 1 \text{ N} \cdot \text{m} \cdot \text{s}/\text{rad} \\ J_2 &= 1 \text{ kg} \cdot \text{m}^2 & K_2 &= 1 \text{ N} \cdot \text{m}/\text{rad} & D_2 &= 1 \text{ N} \cdot \text{m} \cdot \text{s}/\text{rad} \\ & & K_3 &= 1 \text{ N} \cdot \text{m}/\text{rad} & D_3 &= 1 \text{ N} \cdot \text{m} \cdot \text{s}/\text{rad} \\ & & K_4 &= 1 \text{ N} \cdot \text{m}/\text{rad} & D_4 &= 1 \text{ N} \cdot \text{m} \cdot \text{s}/\text{rad} \\ & & K_5 &= 1 \text{ N} \cdot \text{m}/\text{rad} & D_5 &= 1 \text{ N} \cdot \text{m} \cdot \text{s}/\text{rad} \\ & & K_6 &= 1 \text{ N} \cdot \text{m}/\text{rad} & D_6 &= 1 \text{ N} \cdot \text{m} \cdot \text{s}/\text{rad} \end{aligned}$$

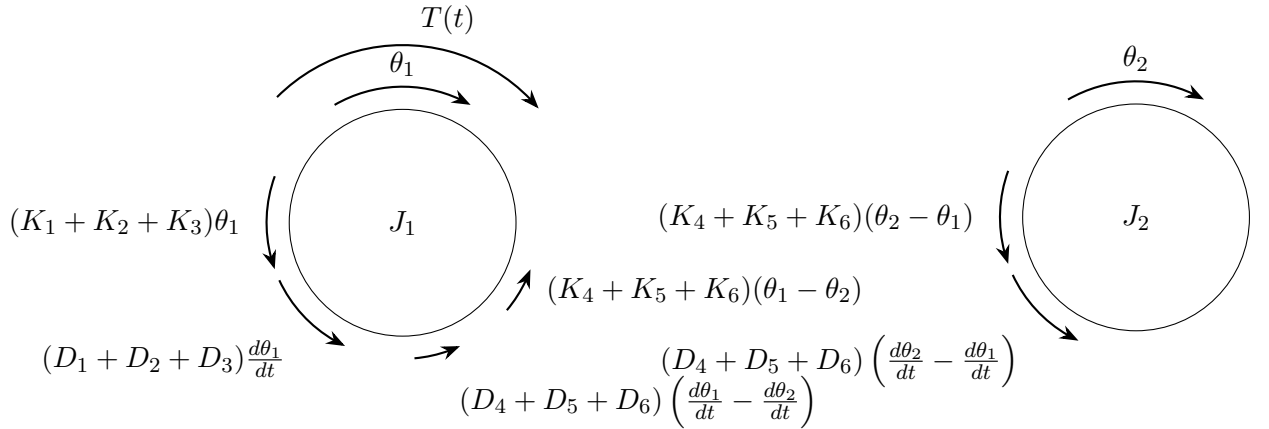
Required:

$$G(s) = \frac{\theta_2(s)}{T(s)}$$

Solution:

Free-Body Diagrams (FBDs)

We assume clockwise rotation is positive.



Equations of Motion

We apply Newton's second law for rotation ($\sum \tau = J \frac{d^2\theta}{dt^2}$) to each mass.

For Inertia 1 (J_1):

$$\begin{aligned} J_1 \frac{d^2\theta_1}{dt^2} + (D_1 + D_2 + D_3) \frac{d\theta_1}{dt} + (K_1 + K_2 + K_3)\theta_1 + (D_4 + D_5 + D_6) \left(\frac{d\theta_1}{dt} - \frac{d\theta_2}{dt} \right) + (K_4 + K_5 + K_6)(\theta_1 - \theta_2) &= T(t) \\ J_1 \frac{d^2\theta_1}{dt^2} + (D_1 + D_2 + D_3 + D_4 + D_5 + D_6) \frac{d\theta_1}{dt} + (K_1 + K_2 + K_3 + K_4 + K_5 + K_6)\theta_1 - (D_4 + D_5 + D_6) \frac{d\theta_2}{dt} - (K_4 + K_5 + K_6)\theta_2 &= T(t) \end{aligned}$$

For Inertia 2 (J_2):

$$\begin{aligned} J_2 \frac{d^2\theta_2}{dt^2} + (D_4 + D_5 + D_6) \left(\frac{d\theta_2}{dt} - \frac{d\theta_1}{dt} \right) + (K_4 + K_5 + K_6)(\theta_2 - \theta_1) &= 0 \\ J_2 \frac{d^2\theta_2}{dt^2} + (D_4 + D_5 + D_6) \frac{d\theta_2}{dt} + (K_4 + K_5 + K_6)\theta_2 - (D_4 + D_5 + D_6) \frac{d\theta_1}{dt} - (K_4 + K_5 + K_6)\theta_1 &= 0 \end{aligned}$$

Laplace Transform

We take the Laplace transform of the equations, assuming zero initial conditions.

Equation for Inertia 1:

$$[J_1 s^2 + (D_1..D_6)s + (K_1..K_6)]\Theta_1(s) - [(D_4 + D_5 + D_6)s + (K_4 + K_5 + K_6)]\Theta_2(s) = T(s)$$

Equation for Inertia 2:

$$-[(D_4 + D_5 + D_6)s + (K_4 + K_5 + K_6)]\Theta_1(s) + [J_2 s^2 + (D_4 + D_5 + D_6)s + (K_4 + K_5 + K_6)]\Theta_2(s) = 0$$

Substituting the given numerical values:

$$(s^2 + 6s + 6)\Theta_1(s) - (3s + 3)\Theta_2(s) = T(s) \quad (1)$$

$$-(3s + 3)\Theta_1(s) + (s^2 + 3s + 3)\Theta_2(s) = 0 \quad (2)$$

Solving for the Transfer Function

From equation (2), we express $\Theta_1(s)$ in terms of $\Theta_2(s)$:

$$\Theta_1(s) = \frac{s^2 + 3s + 3}{3s + 3}\Theta_2(s)$$

Substitute this into equation (1):

$$(s^2 + 6s + 6) \left(\frac{s^2 + 3s + 3}{3s + 3}\Theta_2(s) \right) - (3s + 3)\Theta_2(s) = T(s)$$

Factor out $\Theta_2(s)$:

$$\left[\frac{(s^2 + 6s + 6)(s^2 + 3s + 3) - (3s + 3)^2}{3s + 3} \right] \Theta_2(s) = T(s)$$

Rearrange to find the transfer function $\frac{\Theta_2(s)}{T(s)}$:

$$\frac{\Theta_2(s)}{T(s)} = \frac{3s + 3}{(s^2 + 6s + 6)(s^2 + 3s + 3) - (3s + 3)^2}$$

Expand and simplify the denominator:

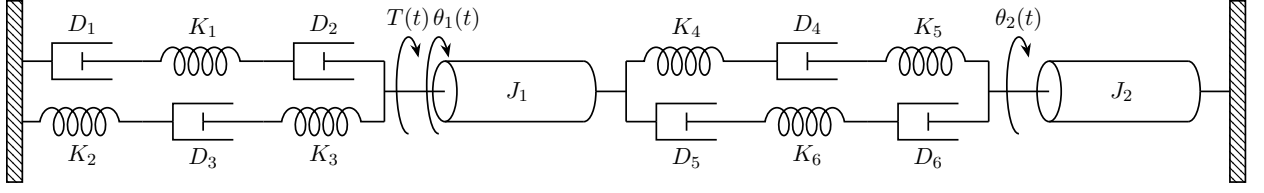
$$\begin{aligned} \text{Denominator} &= (s^4 + 9s^3 + 27s^2 + 36s + 18) - (9s^2 + 18s + 9) \\ &= s^4 + 9s^3 + 18s^2 + 18s + 9 \end{aligned}$$

The final transfer function is:

$$\boxed{\frac{\Theta_2(s)}{T(s)} = \frac{3s + 3}{s^4 + 9s^3 + 18s^2 + 18s + 9}}$$

Problem 25

For the rotational mechanical system shown in the diagram below, determine the transfer function $G(s) = \theta_2(s)/T(s)$.



Given:

$$\begin{aligned}
 J_1 &= 1 \text{ kg} \cdot \text{m}^2 & K_1 &= 1 \text{ N} \cdot \text{m}/\text{rad} & D_1 &= 1 \text{ N} \cdot \text{m} \cdot \text{s}/\text{rad} \\
 J_2 &= 1 \text{ kg} \cdot \text{m}^2 & K_2 &= 2 \text{ N} \cdot \text{m}/\text{rad} & D_2 &= 1 \text{ N} \cdot \text{m} \cdot \text{s}/\text{rad} \\
 & & K_3 &= 1 \text{ N} \cdot \text{m}/\text{rad} & D_3 &= 1 \text{ N} \cdot \text{m} \cdot \text{s}/\text{rad} \\
 & & K_4 &= 1 \text{ N} \cdot \text{m}/\text{rad} & D_4 &= 1 \text{ N} \cdot \text{m} \cdot \text{s}/\text{rad} \\
 & & K_5 &= 1 \text{ N} \cdot \text{m}/\text{rad} & D_5 &= 1 \text{ N} \cdot \text{m} \cdot \text{s}/\text{rad} \\
 & & K_6 &= 1 \text{ N} \cdot \text{m}/\text{rad} & D_6 &= 1 \text{ N} \cdot \text{m} \cdot \text{s}/\text{rad}
 \end{aligned}$$

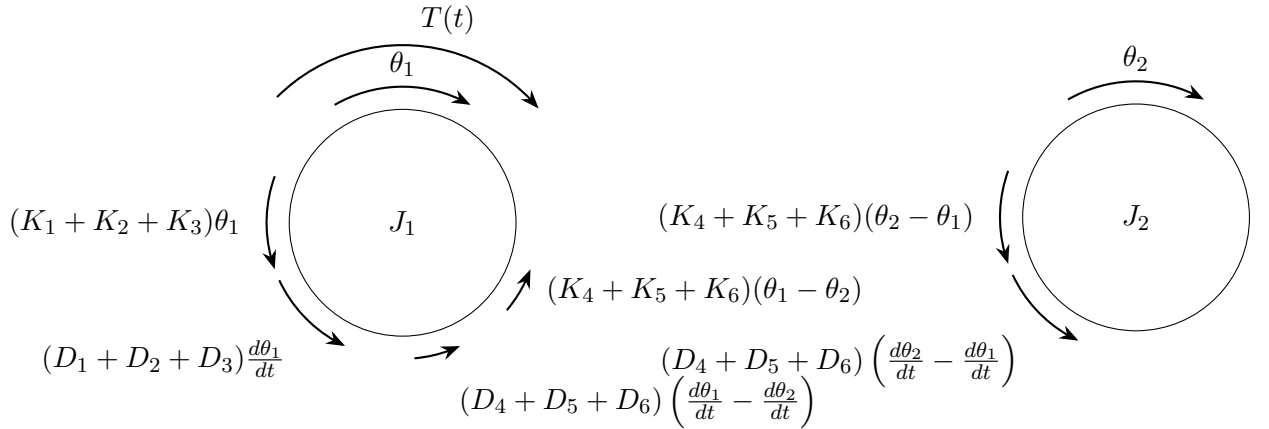
Required:

$$G(s) = \frac{\theta_2(s)}{T(s)}$$

Solution:

Free-Body Diagrams (FBDs)

We assume clockwise rotation is positive.



Equations of Motion

We apply Newton's second law for rotation ($\sum \tau = J \frac{d^2\theta}{dt^2}$) to each mass.

For Inertia 1 (J_1):

$$\begin{aligned}
 J_1 \frac{d^2\theta_1}{dt^2} + (D_1 + D_2 + D_3) \frac{d\theta_1}{dt} + (K_1 + K_2 + K_3)\theta_1 + (D_4 + D_5 + D_6) \left(\frac{d\theta_1}{dt} - \frac{d\theta_2}{dt} \right) + (K_4 + K_5 + K_6)(\theta_1 - \theta_2) &= T(t) \\
 J_1 \frac{d^2\theta_1}{dt^2} + (D_1 + D_2 + D_3 + D_4 + D_5 + D_6) \frac{d\theta_1}{dt} + (K_1 + K_2 + K_3 + K_4 + K_5 + K_6)\theta_1 - (D_4 + D_5 + D_6) \frac{d\theta_2}{dt} - (K_4 + K_5 + K_6)\theta_2 &= T(t)
 \end{aligned}$$

For Inertia 2 (J_2):

$$\begin{aligned}
 J_2 \frac{d^2\theta_2}{dt^2} + (D_4 + D_5 + D_6) \left(\frac{d\theta_2}{dt} - \frac{d\theta_1}{dt} \right) + (K_4 + K_5 + K_6)(\theta_2 - \theta_1) &= 0 \\
 J_2 \frac{d^2\theta_2}{dt^2} + (D_4 + D_5 + D_6) \frac{d\theta_2}{dt} + (K_4 + K_5 + K_6)\theta_2 - (D_4 + D_5 + D_6) \frac{d\theta_1}{dt} - (K_4 + K_5 + K_6)\theta_1 &= 0
 \end{aligned}$$

Laplace Transform

We take the Laplace transform of the equations, assuming zero initial conditions.

Equation for Inertia 1:

$$[J_1 s^2 + (D_1..D_6)s + (K_1..K_6)]\Theta_1(s) - [(D_4 + D_5 + D_6)s + (K_4 + K_5 + K_6)]\Theta_2(s) = T(s)$$

Equation for Inertia 2:

$$-[(D_4 + D_5 + D_6)s + (K_4 + K_5 + K_6)]\Theta_1(s) + [J_2 s^2 + (D_4 + D_5 + D_6)s + (K_4 + K_5 + K_6)]\Theta_2(s) = 0$$

Substituting the given numerical values:

$$(s^2 + 6s + 7)\Theta_1(s) - (3s + 3)\Theta_2(s) = T(s) \quad (1)$$

$$-(3s + 3)\Theta_1(s) + (s^2 + 3s + 3)\Theta_2(s) = 0 \quad (2)$$

Solving for the Transfer Function

From equation (2), we express $\Theta_1(s)$ in terms of $\Theta_2(s)$:

$$\Theta_1(s) = \frac{s^2 + 3s + 3}{3s + 3}\Theta_2(s)$$

Substitute this into equation (1):

$$(s^2 + 6s + 7) \left(\frac{s^2 + 3s + 3}{3s + 3}\Theta_2(s) \right) - (3s + 3)\Theta_2(s) = T(s)$$

Factor out $\Theta_2(s)$:

$$\left[\frac{(s^2 + 6s + 7)(s^2 + 3s + 3) - (3s + 3)^2}{3s + 3} \right] \Theta_2(s) = T(s)$$

Rearrange to find the transfer function $\frac{\Theta_2(s)}{T(s)}$:

$$\frac{\Theta_2(s)}{T(s)} = \frac{3s + 3}{(s^2 + 6s + 7)(s^2 + 3s + 3) - (3s + 3)^2}$$

Expand and simplify the denominator:

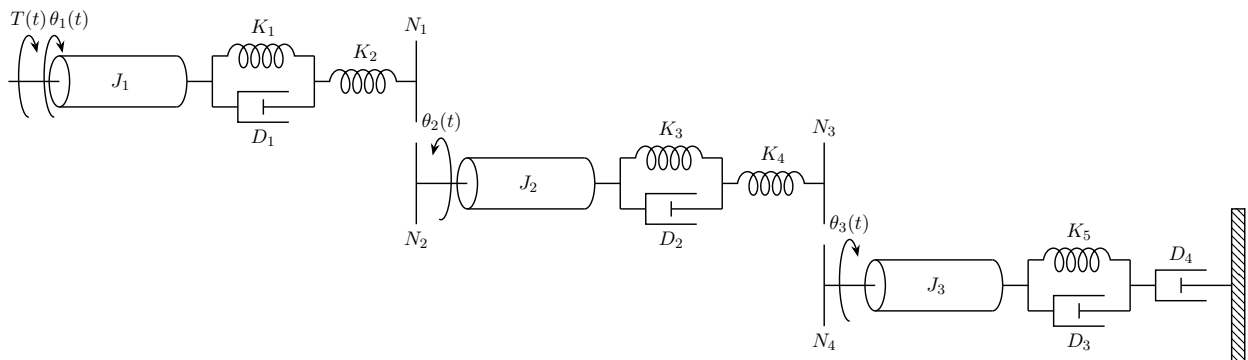
$$\begin{aligned} \text{Denominator} &= (s^4 + 9s^3 + 28s^2 + 39s + 21) - (9s^2 + 18s + 9) \\ &= s^4 + 9s^3 + 19s^2 + 21s + 12 \end{aligned}$$

The final transfer function is:

$$\boxed{\frac{\Theta_2(s)}{T(s)} = \frac{3s + 3}{s^4 + 9s^3 + 19s^2 + 21s + 12}}$$

Problem 26

For the rotational mechanical system with gears shown in the diagram below, determine the transfer function $G(s) = \theta_1(s)/T(s)$.



Given:

$$\begin{aligned}
 J_1 &= 4 \text{ kg} \cdot \text{m}^2 & K_1 &= 3 \text{ N} \cdot \text{m/rad} & D_1 &= 2 \text{ N} \cdot \text{m} \cdot \text{s/rad} & N_1 &= 20 \\
 J_2 &= 8 \text{ kg} \cdot \text{m}^2 & K_2 &= 5 \text{ N} \cdot \text{m/rad} & D_2 &= 4 \text{ N} \cdot \text{m} \cdot \text{s/rad} & N_2 &= 40 \\
 J_3 &= 36 \text{ kg} \cdot \text{m}^2 & K_3 &= 2 \text{ N} \cdot \text{m/rad} & D_3 &= 6 \text{ N} \cdot \text{m} \cdot \text{s/rad} & N_3 &= 60 \\
 & & K_4 &= 6 \text{ N} \cdot \text{m/rad} & D_4 &= 2 \text{ N} \cdot \text{m} \cdot \text{s/rad} & N_4 &= 30 \\
 & & K_5 &= 4 \text{ N} \cdot \text{m/rad} & & & &
 \end{aligned}$$

Required:

$$G(s) = \frac{\theta_1(s)}{T(s)}$$

Solution:

Reflected parameters to shaft of Inertia 1:

$$J'_2 = J_2 \left(\frac{N_1}{N_2} \right)^2$$

$$J'_2 = 8 \cdot \left(\frac{20}{40} \right)^2$$

$$J'_2 = 8 \cdot \frac{1}{4}$$

$$J'_2 = 2$$

$$J'_3 = J_3 \left(\frac{N_1 N_3}{N_2 N_4} \right)^2$$

$$J'_3 = 36 \cdot \left(\frac{20 \cdot 60}{40 \cdot 30} \right)^2$$

$$J'_3 = 36 \cdot 1$$

$$J'_3 = 36$$

$$J_1 = \text{final inertia} = 4$$

$$J_{\text{eq}} = J_1 + J'_2 + J'_3$$

$$J_{\text{eq}} = 4 + 2 + 36$$

$$J_{\text{eq}} = 42$$

Reflected spring constants:

$$K'_3 = K_3 \left(\frac{N_1}{N_2} \right)^2 = 2 \cdot \frac{1}{4} = 0.5$$

$$K'_4 = K_4 \left(\frac{N_1}{N_2} \right)^2 = 6 \cdot \frac{1}{4} = 1.5$$

$$K'_5 = K_5 \left(\frac{N_1 N_3}{N_2 N_4} \right)^2 = 4 \cdot 1 = 4$$

$$K_1 = 3$$

$$K_2 = 5$$

$$K_{\text{eq}} = K_1 + K_2 + K'_3 + K'_4 + K'_5$$

$$K_{\text{eq}} = 3 + 5 + 0.5 + 1.5 + 4$$

$$K_{\text{eq}} = 14$$

Reflected damping constants:

$$D'_2 = D_2 \left(\frac{N_1}{N_2} \right)^2$$

$$D'_2 = 4 \cdot \frac{1}{4}$$

$$D'_2 = 1$$

$$D'_3 = D_3 \left(\frac{N_1 N_3}{N_2 N_4} \right)^2$$

$$D'_3 = 6 \cdot 1$$

$$D'_3 = 6$$

$$D'_4 = D_4 \left(\frac{N_1 N_3}{N_2 N_4} \right)^2$$

$$D'_4 = 2 \cdot 1$$

$$D'_4 = 2$$

$$D_1 = 2$$

$$D_{\text{eq}} = D_1 + D'_2 + D'_3 + D'_4$$

$$D_{\text{eq}} = 2 + 1 + 6 + 2$$

$$D_{\text{eq}} = 11$$

Transfer function:

$$T(s) = J_{\text{eq}} s^2 \theta_1(s) + D_{\text{eq}} s \theta_1(s) + K_{\text{eq}} \theta_1(s)$$

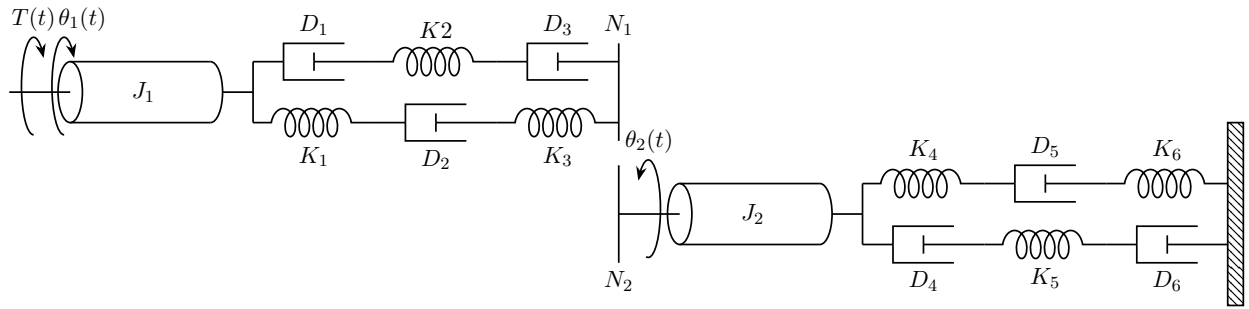
$$\frac{\theta_1(s)}{T(s)} = \frac{1}{J_{\text{eq}} s^2 + D_{\text{eq}} s + K_{\text{eq}}}$$

$$\frac{\theta_1(s)}{T(s)} = \frac{1}{42s^2 + 11s + 14}$$

$$\boxed{\frac{\theta_1(s)}{T(s)} = \frac{1}{42s^2 + 11s + 14}}$$

Problem 27

For the rotational mechanical system with gears shown in the diagram below, determine the transfer function $G(s) = \theta_1(s)/T(s)$.



Given:

$$\begin{aligned} J_1 &= 5 \text{ kg} \cdot \text{m}^2 & K_1 &= 2 \text{ N} \cdot \text{m/rad} & D_1 &= 4 \text{ N} \cdot \text{m} \cdot \text{s/rad} & N_1 &= 20 \\ J_2 &= 12 \text{ kg} \cdot \text{m}^2 & K_2 &= 2 \text{ N} \cdot \text{m/rad} & D_2 &= 4 \text{ N} \cdot \text{m} \cdot \text{s/rad} & N_2 &= 40 \\ & & K_3 &= 4 \text{ N} \cdot \text{m/rad} & D_3 &= 8 \text{ N} \cdot \text{m} \cdot \text{s/rad} & & \\ & & K_4 &= 16 \text{ N} \cdot \text{m/rad} & D_4 &= 16 \text{ N} \cdot \text{m} \cdot \text{s/rad} & & \\ & & K_5 &= 32 \text{ N} \cdot \text{m/rad} & D_5 &= 16 \text{ N} \cdot \text{m} \cdot \text{s/rad} & & \\ & & K_6 &= 32 \text{ N} \cdot \text{m/rad} & D_6 &= 48 \text{ N} \cdot \text{m} \cdot \text{s/rad} & & \end{aligned}$$

Required:

$$G(s) = \frac{\theta_1(s)}{T(s)}$$

Solution:

Reflected parameters to shaft of Inertia 1:

$$J'_2 = J_2 \left(\frac{N_1}{N_2} \right)^2$$

$$J'_2 = 12 \cdot \left(\frac{20}{40} \right)^2$$

$$J'_2 = 12 \cdot \frac{1}{4}$$

$$J'_2 = 3$$

$$J_1 = \text{final inertia} = 5$$

$$J_{\text{eq}} = J_1 + J'_2$$

$$J_{\text{eq}} = 5 + 3$$

$$J_{\text{eq}} = 8$$

Reflected spring constants:

$$K'_4 = K_4 \left(\frac{N_1}{N_2} \right)^2 = 16 \cdot \frac{1}{4} = 4$$

$$K'_5 = K_5 \left(\frac{N_1}{N_2} \right)^2 = 32 \cdot \frac{1}{4} = 8$$

$$K'_6 = K_6 \left(\frac{N_1}{N_2} \right)^2 = 32 \cdot \frac{1}{4} = 8$$

$$K_1 = 2$$

$$K_2 = 2$$

$$K_3 = 4$$

$$K_{\text{eq}} = K_1 + K_2 + K_3 + K'_4 + K'_5 + K'_6$$

$$K_{\text{eq}} = 2 + 2 + 4 + 4 + 8 + 8$$

$$K_{\text{eq}} = 28$$

Reflected damping constants:

$$D'_4 = D_4 \left(\frac{N_1}{N_2} \right)^2$$

$$D'_4 = 16 \cdot \frac{1}{4}$$

$$D'_4 = 4$$

$$D'_5 = D_5 \left(\frac{N_1}{N_2} \right)^2$$

$$D'_5 = 16 \cdot \frac{1}{4}$$

$$D'_5 = 4$$

$$D'_6 = D_6 \left(\frac{N_1}{N_2} \right)^2$$

$$D'_6 = 48 \cdot \frac{1}{4}$$

$$D'_6 = 12$$

$$D_1 = 4$$

$$D_2 = 4$$

$$D_3 = 8$$

$$D_{\text{eq}} = D_1 + D_2 + D_3 + D'_4 + D'_5 + D'_6$$

$$D_{\text{eq}} = 4 + 4 + 8 + 4 + 4 + 12$$

$$D_{\text{eq}} = 36$$

Transfer function:

$$T(s) = J_{\text{eq}} s^2 \theta_1(s) + D_{\text{eq}} s \theta_1(s) + K_{\text{eq}} \theta_1(s)$$

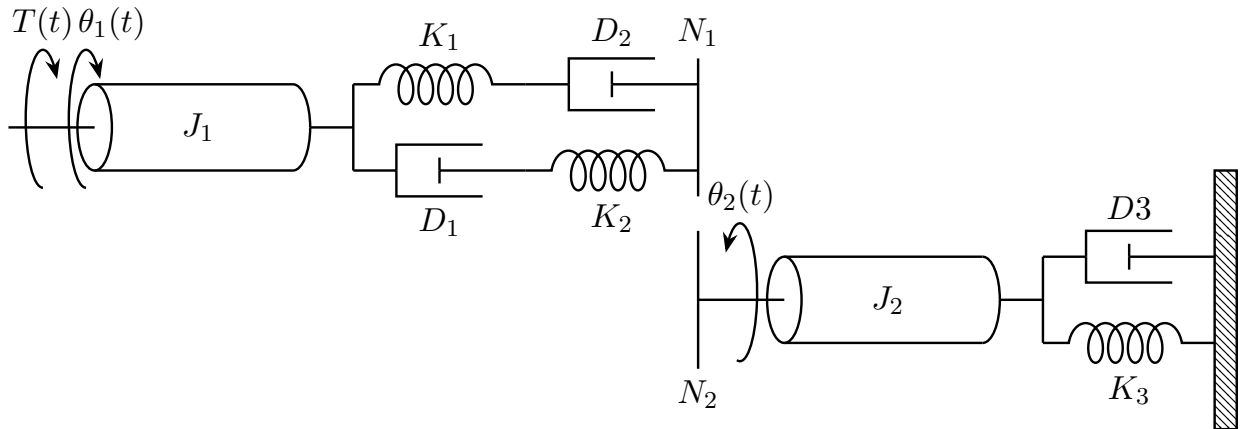
$$\frac{\theta_1(s)}{T(s)} = \frac{1}{J_{\text{eq}} s^2 + D_{\text{eq}} s + K_{\text{eq}}}$$

$$\frac{\theta_1(s)}{T(s)} = \frac{1}{8s^2 + 36s + 28}$$

$$\boxed{\frac{\theta_1(s)}{T(s)} = \frac{1}{8s^2 + 36s + 28}}$$

Problem 28

For the rotational mechanical system with gears shown in the diagram below, determine the transfer function $G(s) = \theta_1(s)/T(s)$.



Given:

$$\begin{aligned} J_1 &= 5 \text{ kg} \cdot \text{m}^2 & K_1 &= 2 \text{ N} \cdot \text{m/rad} & D_1 &= 4 \text{ N} \cdot \text{m} \cdot \text{s/rad} & N_1 &= 20 \\ J_2 &= 12 \text{ kg} \cdot \text{m}^2 & K_2 &= 3 \text{ N} \cdot \text{m/rad} & D_2 &= 2 \text{ N} \cdot \text{m} \cdot \text{s/rad} & N_2 &= 40 \\ & & K_3 &= 20 \text{ N} \cdot \text{m/rad} & D_3 &= 24 \text{ N} \cdot \text{m} \cdot \text{s/rad} \end{aligned}$$

Required:

$$G(s) = \frac{\theta_1(s)}{T(s)}$$

Solution:

Reflected parameters to shaft of Inertia 1:

$$J'_2 = J_2 \left(\frac{N_1}{N_2} \right)^2$$

$$J'_2 = 12 \cdot \left(\frac{20}{40} \right)^2$$

$$J'_2 = 12 \cdot \frac{1}{4}$$

$$J'_2 = 3$$

$$J_1 = \text{final inertia} = 5$$

$$J_{\text{eq}} = J_1 + J'_2$$

$$J_{\text{eq}} = 5 + 3$$

$$J_{\text{eq}} = 8$$

Reflected spring constants:

$$K'_3 = K_3 \left(\frac{N_1}{N_2} \right)^2 = 20 \cdot \frac{1}{4} = 5$$

$$K_1 = 2$$

$$K_2 = 3$$

$$K_{\text{eq}} = K_1 + K_2 + K'_3$$

$$K_{\text{eq}} = 2 + 3 + 5$$

$$K_{\text{eq}} = 10$$

Reflected damping constants:

$$D'_3 = D_3 \left(\frac{N_1}{N_2} \right)^2$$

$$D'_3 = 24 \cdot \frac{1}{4}$$

$$D'_3 = 6$$

$$D_1 = 4$$

$$D_2 = 2$$

$$D_{\text{eq}} = D_1 + D_2 + D'_3$$

$$D_{\text{eq}} = 4 + 2 + 6$$

$$D_{\text{eq}} = 12$$

Transfer function:

$$T(s) = J_{\text{eq}} s^2 \theta_1(s) + D_{\text{eq}} s \theta_1(s) + K_{\text{eq}} \theta_1(s)$$

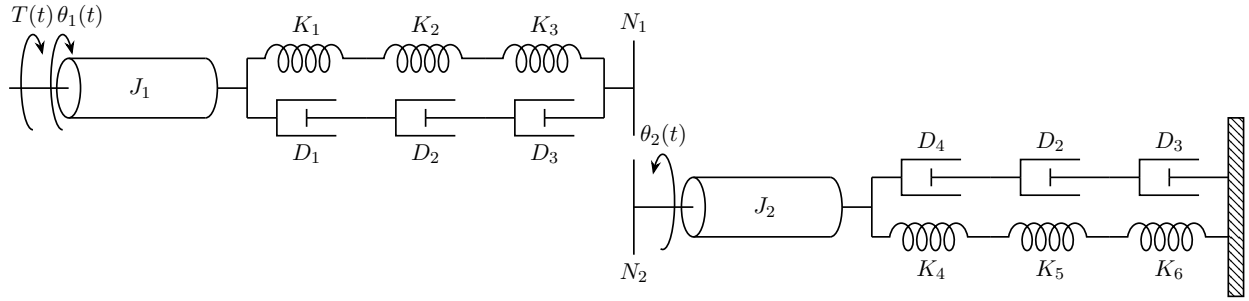
$$\frac{\theta_1(s)}{T(s)} = \frac{1}{J_{\text{eq}} s^2 + D_{\text{eq}} s + K_{\text{eq}}}$$

$$\frac{\theta_1(s)}{T(s)} = \frac{1}{8s^2 + 12s + 10}$$

$$\boxed{\frac{\theta_1(s)}{T(s)} = \frac{1}{8s^2 + 12s + 10}}$$

Problem 29

For the rotational mechanical system with gears shown in the diagram below, determine the transfer function $G(s) = \theta_2(s)/T(s)$.



Given:

$$\begin{aligned} J_1 &= 5 \text{ kg} \cdot \text{m}^2 & K_1 &= 1 \text{ N} \cdot \text{m}/\text{rad} & D_1 &= 2 \text{ N} \cdot \text{m} \cdot \text{s}/\text{rad} & N_1 &= 20 \\ J_2 &= 12 \text{ kg} \cdot \text{m}^2 & K_2 &= 1 \text{ N} \cdot \text{m}/\text{rad} & D_2 &= 2 \text{ N} \cdot \text{m} \cdot \text{s}/\text{rad} & N_2 &= 40 \\ & & K_3 &= 2 \text{ N} \cdot \text{m}/\text{rad} & D_3 &= 4 \text{ N} \cdot \text{m} \cdot \text{s}/\text{rad} & & \\ & & K_4 &= 4 \text{ N} \cdot \text{m}/\text{rad} & D_4 &= 4 \text{ N} \cdot \text{m} \cdot \text{s}/\text{rad} & & \\ & & K_5 &= 8 \text{ N} \cdot \text{m}/\text{rad} & D_5 &= 8 \text{ N} \cdot \text{m} \cdot \text{s}/\text{rad} & & \\ & & K_6 &= 12 \text{ N} \cdot \text{m}/\text{rad} & D_6 &= 12 \text{ N} \cdot \text{m} \cdot \text{s}/\text{rad} & & \end{aligned}$$

Required:

$$G(s) = \frac{\theta_2(s)}{T(s)}$$

Solution:

Reflected parameters to shaft of Inertia 1:

$$J'_2 = J_2 \left(\frac{N_1}{N_2} \right)^2$$

$$J'_2 = 12 \cdot \left(\frac{20}{40} \right)^2$$

$$J'_2 = 12 \cdot \frac{1}{4}$$

$$J'_2 = 3$$

$$J_1 = \text{final inertia} = 5$$

$$J_{\text{eq}} = J_1 + J'_2$$

$$J_{\text{eq}} = 5 + 3$$

$$J_{\text{eq}} = 8$$

Reflected spring constants:

$$K'_4 = K_4 \left(\frac{N_1}{N_2} \right)^2 = 4 \cdot \frac{1}{4} = 1$$

$$K'_5 = K_5 \left(\frac{N_1}{N_2} \right)^2 = 8 \cdot \frac{1}{4} = 2$$

$$K'_6 = K_6 \left(\frac{N_1}{N_2} \right)^2 = 12 \cdot \frac{1}{4} = 3$$

$$K_1 = 1$$

$$K_2 = 1$$

$$K_3 = 2$$

$$K_{\text{eq}} = K_1 + K_2 + K_3 + K'_4 + K'_5 + K'_6$$

$$K_{\text{eq}} = 1 + 1 + 2 + 1 + 2 + 3$$

$$K_{\text{eq}} = 10$$

Reflected damping constants:

$$D'_4 = D_4 \left(\frac{N_1}{N_2} \right)^2$$

$$D'_4 = 4 \cdot \frac{1}{4}$$

$$D'_4 = 1$$

$$D'_5 = D_5 \left(\frac{N_1}{N_2} \right)^2$$

$$D'_5 = 8 \cdot \frac{1}{4}$$

$$D'_5 = 2$$

$$D'_6 = D_6 \left(\frac{N_1}{N_2} \right)^2$$

$$D'_6 = 12 \cdot \frac{1}{4}$$

$$D'_6 = 3$$

$$D_1 = 2$$

$$D_2 = 2$$

$$D_3 = 4$$

$$D_{\text{eq}} = D_1 + D_2 + D_3 + D'_4 + D'_5 + D'_6$$

$$D_{\text{eq}} = 2 + 2 + 4 + 1 + 2 + 3$$

$$D_{\text{eq}} = 14$$

Transfer function:

$$T(s) = J_{\text{eq}} s^2 \theta_1(s) + D_{\text{eq}} s \theta_1(s) + K_{\text{eq}} \theta_1(s)$$

$$\frac{\theta_1(s)}{T(s)} = \frac{1}{J_{\text{eq}} s^2 + D_{\text{eq}} s + K_{\text{eq}}}$$

$$\frac{\theta_1(s)}{T(s)} = \frac{1}{8s^2 + 14s + 10}$$

$$\theta_1(s) = \left(\frac{N_2}{N_1} \right) \theta_2(s) = \left(\frac{40}{20} \right) \theta_2(s) = 2\theta_2(s)$$

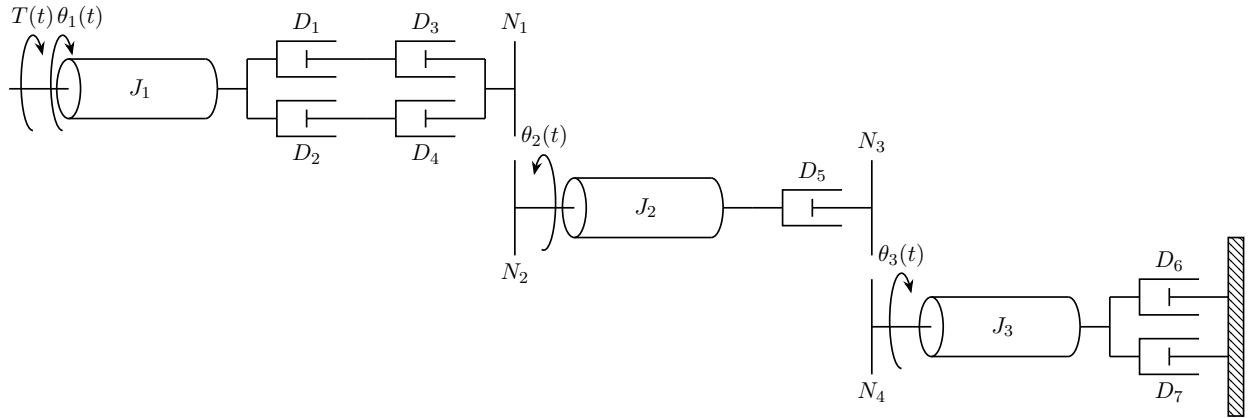
$$\frac{2\theta_2(s)}{T(s)} = \frac{1}{8s^2 + 14s + 10}$$

$$\frac{\theta_2(s)}{T(s)} = \frac{1}{16s^2 + 28s + 20}$$

$\frac{\theta_2(s)}{T(s)} = \frac{1}{16s^2 + 28s + 20}$

Problem 30

For the rotational mechanical system with gears shown in the diagram below, determine the transfer function $G(s) = \theta_1(s)/T(s)$.



Given:

$$\begin{aligned}
 J_1 &= 1 \text{ kg} \cdot \text{m}^2 & D_1 &= 1 \text{ N} \cdot \text{m} \cdot \text{s}/\text{rad} & N_1 &= 20 \\
 J_2 &= 1 \text{ kg} \cdot \text{m}^2 & D_2 &= 1 \text{ N} \cdot \text{m} \cdot \text{s}/\text{rad} & N_2 &= 10 \\
 J_3 &= 1 \text{ kg} \cdot \text{m}^2 & D_3 &= 1 \text{ N} \cdot \text{m} \cdot \text{s}/\text{rad} & N_3 &= 30 \\
 & & D_4 &= 1 \text{ N} \cdot \text{m} \cdot \text{s}/\text{rad} & N_4 &= 15 \\
 & & D_5 &= 1 \text{ N} \cdot \text{m} \cdot \text{s}/\text{rad} \\
 & & D_6 &= 1 \text{ N} \cdot \text{m} \cdot \text{s}/\text{rad} \\
 & & D_7 &= 1 \text{ N} \cdot \text{m} \cdot \text{s}/\text{rad}
 \end{aligned}$$

Required:

$$G(s) = \frac{\theta_1(s)}{T(s)}$$

Solution:

Reflected parameters to shaft of Inertia 1:

$$J'_2 = J_2 \left(\frac{N_1}{N_2} \right)^2$$

$$J'_2 = 1 \cdot \left(\frac{20}{10} \right)^2$$

$$J'_2 = 1 \cdot 4$$

$$J'_2 = 4$$

$$J'_3 = J_3 \left(\frac{N_1 N_3}{N_2 N_4} \right)^2$$

$$J'_3 = 1 \cdot \left(\frac{20 \cdot 30}{10 \cdot 15} \right)^2$$

$$J'_3 = 1 \cdot 16$$

$$J'_3 = 16$$

$$J_1 = \text{final inertia} = 1$$

$$J_{\text{eq}} = J_1 + J'_2 + J'_3$$

$$J_{\text{eq}} = 1 + 4 + 16$$

$$J_{\text{eq}} = 21$$

Reflected damping constants:

$$D'_5 = D_5 \left(\frac{N_1}{N_2} \right)^2$$

$$D'_5 = 1 \cdot 4$$

$$D'_5 = 4$$

$$D'_6 = D_6 \left(\frac{N_1 N_3}{N_2 N_4} \right)^2$$

$$D'_6 = 1 \cdot 16$$

$$D'_6 = 16$$

$$D'_7 = D_7 \left(\frac{N_1 N_3}{N_2 N_4} \right)^2$$

$$D'_7 = 1 \cdot 16$$

$$D'_7 = 16$$

$$D_1 = 1$$

$$D_2 = 1$$

$$D_3 = 1$$

$$D_4 = 1$$

$$D_{\text{eq}} = D_1 + D_2 + D_3 + D_4 + D'_5 + D'_6 + D'_7$$

$$D_{\text{eq}} = 1 + 1 + 1 + 1 + 4 + 16 + 16$$

$$D_{\text{eq}} = 40$$

Transfer function:

$$T(s) = J_{\text{eq}} s^2 \theta_1(s) + D_{\text{eq}} s \theta_1(s)$$

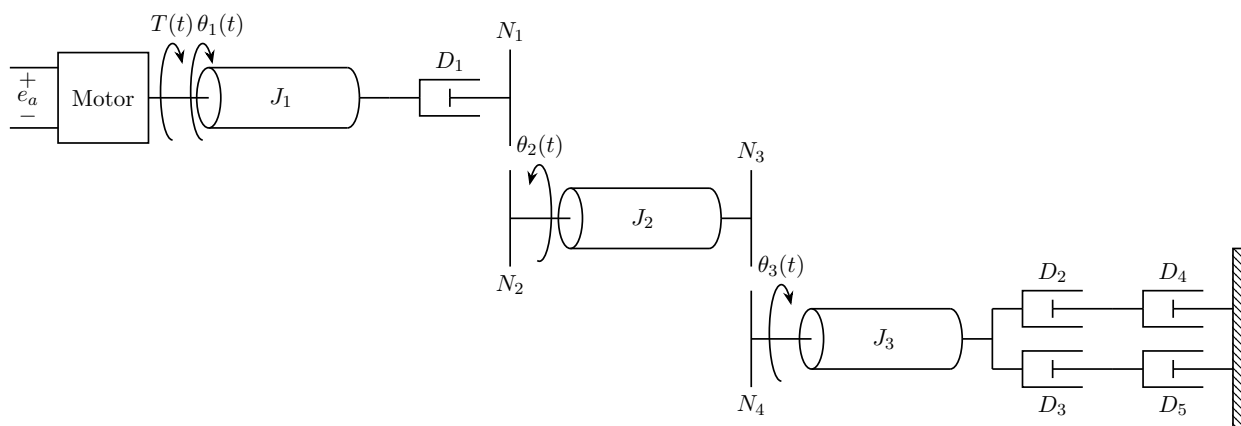
$$\frac{\theta_1(s)}{T(s)} = \frac{1}{J_{\text{eq}} s^2 + D_{\text{eq}} s}$$

$$\frac{\theta_1(s)}{T(s)} = \frac{1}{21s^2 + 40s}$$

$\frac{\theta_1(s)}{T(s)} = \frac{1}{21s^2 + 40s}$
--

Problem 31

A DC motor drives a rotational mechanical system as shown in the figure. Assume that J_1 and D_1 correspond to the motor-side inertia J_a and damping D_a , respectively. Determine the electromechanical transfer function $\frac{\theta_m(s)}{E_a(s)}$.



Given:

$$\begin{aligned}
 J_a &= 4 \text{ kg} \cdot \text{m}^2 & D_a &= 2 \text{ N} \cdot \text{m} \cdot \text{s/rad} & N_1 &= 20 & e_a &= 12 \text{ V} \\
 J_2 &= 16 \text{ kg} \cdot \text{m}^2 & D_2 &= 2 \text{ N} \cdot \text{m} \cdot \text{s/rad} & N_2 &= 40 & T_{\text{stall}} &= 6 \text{ N} \cdot \text{m} \\
 J_3 &= 36 \text{ kg} \cdot \text{m}^2 & D_3 &= 4 \text{ N} \cdot \text{m} \cdot \text{s/rad} & N_3 &= 60 & \omega_{\text{no-load}} &= 3 \text{ rad/s} \\
 & & D_4 &= 6 \text{ N} \cdot \text{m} \cdot \text{s/rad} & N_4 &= 30 & & \\
 & & D_5 &= 8 \text{ N} \cdot \text{m} \cdot \text{s/rad} & & & &
 \end{aligned}$$

Required:

$$G(s) = \frac{\theta_m(s)}{E_a(s)}$$

Solution:

Gear (kinematic) ratio:

$$r \equiv \frac{\theta_m}{\theta_3} = \frac{N_1 N_3}{N_2 N_4}.$$

With $N_1 = 20$, $N_2 = 40$, $N_3 = 60$, $N_4 = 30$ we get

$$r = \frac{20 \cdot 60}{40 \cdot 30} = 1,$$

so the motor and load rotate at the same speed:

$$\omega_m = \omega_3, \quad \theta_m = \theta_3.$$

Reflection factor for inertias and viscous damping at first gear stage is $\left(\frac{N_1}{N_2}\right)^2$.

Equivalent inertia reflected to motor shaft:

$$J_m = J_a + \left(\frac{N_1}{N_2}\right)^2 (J_2) + \left(\frac{N_1 N_3}{N_2 N_4}\right)^2 (J_3)$$

$$J_m = 4 + \left(\frac{1}{2}\right)^2 (16) + (1)^2 (36)$$

$$J_m = 4 + 4 + 36 = 44 \text{ kg} \cdot \text{m}^2,$$

Equivalent damping reflected to motor shaft:

$$D_m = D_a + \left(\frac{N_1 N_3}{N_2 N_4} \right)^2 (D_2 + D_3 + D_4 + D_5)$$

$$D_m = 2 + (1)^2(2 + 4 + 6 + 8)$$

$$D_m = 2 + 20 = 22 \text{ N} \cdot \text{m} \cdot \text{s} / \text{rad}.$$

Motor electrical constants from specifications:

$$\frac{K_t}{R_a} = \frac{T_{\text{stall}}}{e_a} = \frac{6}{12} = \frac{1}{2}$$

$$K_b = \frac{e_a}{\omega_{\text{no-load}}} = \frac{12}{3} = 4.$$

Transfer function (Laplace domain):

$$\frac{\theta_m(s)}{E_a(s)} = \frac{\frac{K_t}{R_a J_m}}{s \left[s + \frac{1}{J_m} \left(D_m + \frac{K_t K_b}{R_a} \right) \right]}$$

Substituting values:

$$\frac{\theta_m(s)}{E_a(s)} = \frac{\frac{1/2}{44}}{s \left[s + \frac{1}{44} \left(22 + \frac{1}{2} \cdot 4 \right) \right]}$$

$$\frac{\theta_m(s)}{E_a(s)} = \frac{\frac{1}{88}}{s \left[s + \frac{24}{44} \right]}$$

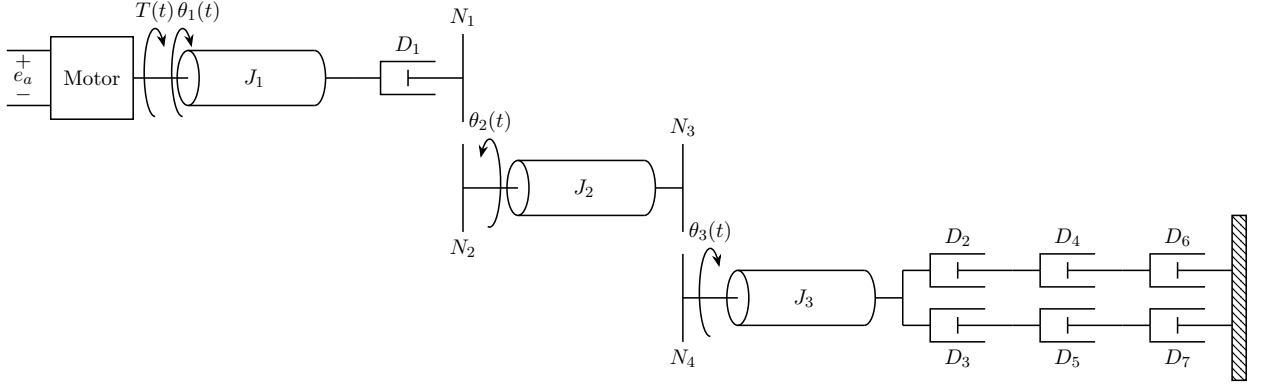
$$\frac{\theta_m(s)}{E_a(s)} = \frac{\frac{1}{88}}{s \left(s + \frac{6}{11} \right)}$$

Multiplying numerator and denominator by 88:

$$\boxed{\frac{\theta_m(s)}{E_a(s)} = \frac{1}{88s^2 + 48s}}$$

Problem 32

A DC motor drives a rotational mechanical system as shown in the figure. Assume that J_1 and D_1 correspond to the motor-side inertia J_a and damping D_a , respectively. Determine the electromechanical transfer function $\frac{\theta_m(s)}{E_a(s)}$.



Given:

$$\begin{aligned}
 J_a &= 5 \text{ kg} \cdot \text{m}^2 & D_a &= 2 \text{ N} \cdot \text{m} \cdot \text{s}/\text{rad} & N_1 &= 20 & e_a &= 20 \text{ V} \\
 J_2 &= 20 \text{ kg} \cdot \text{m}^2 & D_2 &= 1 \text{ N} \cdot \text{m} \cdot \text{s}/\text{rad} & N_2 &= 40 & T_{\text{stall}} &= 10 \text{ N} \cdot \text{m} \\
 J_3 &= 10 \text{ kg} \cdot \text{m}^2 & D_3 &= 1 \text{ N} \cdot \text{m} \cdot \text{s}/\text{rad} & N_3 &= 60 & \omega_{\text{no-load}} &= 5 \text{ rad/s} \\
 & & D_4 &= 1 \text{ N} \cdot \text{m} \cdot \text{s}/\text{rad} & N_4 &= 30 & & \\
 & & D_5 &= 1 \text{ N} \cdot \text{m} \cdot \text{s}/\text{rad} & & & & \\
 & & D_6 &= 1 \text{ N} \cdot \text{m} \cdot \text{s}/\text{rad} & & & & \\
 & & D_7 &= 1 \text{ N} \cdot \text{m} \cdot \text{s}/\text{rad} & & & &
 \end{aligned}$$

Required:

$$G(s) = \frac{\theta_m(s)}{E_a(s)}$$

Solution:

Gear (kinematic) ratio:

$$r \equiv \frac{\theta_m}{\theta_3} = \frac{N_1 N_3}{N_2 N_4}.$$

With $N_1 = 20$, $N_2 = 40$, $N_3 = 60$, $N_4 = 30$ we get

$$r = \frac{20 \cdot 60}{40 \cdot 30} = 1,$$

so the motor and load rotate at the same speed:

$$\omega_m = \omega_3, \quad \theta_m = \theta_3.$$

Reflection factor for inertias and viscous damping at first gear stage is $\left(\frac{N_1}{N_2}\right)^2$.

Equivalent inertia reflected to motor shaft:

$$J_m = J_a + \left(\frac{N_1}{N_2}\right)^2 (J_2) + \left(\frac{N_1 N_3}{N_2 N_4}\right)^2 (J_3)$$

$$J_m = 5 + \left(\frac{1}{2}\right)^2 (20) + (1)^2 (10)$$

$$J_m = 5 + 5 + 10 = 20 \text{ kg} \cdot \text{m}^2,$$

Equivalent damping reflected to motor shaft:

$$D_m = D_a + \left(\frac{N_1 N_3}{N_2 N_4} \right)^2 (D_2 + D_3 + D_4 + D_5 + D_6 + D_7)$$

$$D_m = 2 + (1)^2(1 + 1 + 1 + 1 + 1 + 1)$$

$$D_m = 2 + 6 = 8 \text{ N} \cdot \text{m} \cdot \text{s/rad}.$$

Motor electrical constants from specifications:

$$\frac{K_t}{R_a} = \frac{T_{\text{stall}}}{e_a} = \frac{10}{20} = \frac{1}{2}$$

$$K_b = \frac{e_a}{\omega_{\text{no-load}}} = \frac{20}{5} = 4.$$

Transfer function (Laplace domain):

$$\frac{\theta_m(s)}{E_a(s)} = \frac{\frac{K_t}{R_a J_m}}{s \left[s + \frac{1}{J_m} \left(D_m + \frac{K_t K_b}{R_a} \right) \right]}$$

Substituting values:

$$\frac{\theta_m(s)}{E_a(s)} = \frac{\frac{1/2}{20}}{s \left[s + \frac{1}{20} \left(8 + \frac{1}{2} \cdot 4 \right) \right]}$$

$$\frac{\theta_m(s)}{E_a(s)} = \frac{\frac{1}{40}}{s \left[s + \frac{10}{20} \right]}$$

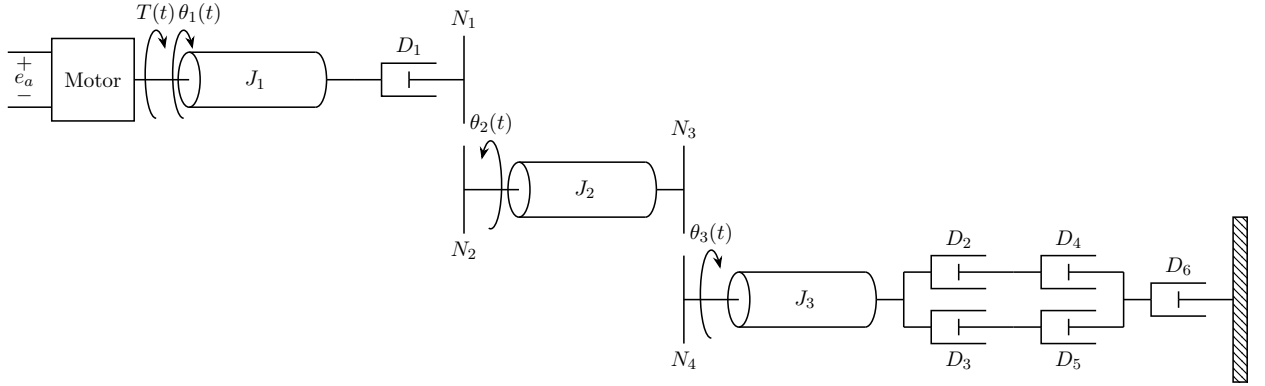
$$\frac{\theta_m(s)}{E_a(s)} = \frac{\frac{1}{40}}{s \left(s + \frac{1}{2} \right)}$$

Multiplying numerator and denominator by 40:

$$\boxed{\frac{\theta_m(s)}{E_a(s)} = \frac{1}{40s^2 + 20s}}$$

Problem 33

A DC motor drives a rotational mechanical system as shown in the figure. Assume that J_1 and D_1 correspond to the motor-side inertia J_a and damping D_a , respectively. Determine the electromechanical transfer function $\frac{\theta_m(s)}{E_a(s)}$.



Given:

$$\begin{aligned}
 J_a &= 2 \text{ kg} \cdot \text{m}^2 & D_a &= 1 \text{ N} \cdot \text{m} \cdot \text{s}/\text{rad} & N_1 &= 20 & e_a &= 20 \text{ V} \\
 J_2 &= 8 \text{ kg} \cdot \text{m}^2 & D_2 &= 1 \text{ N} \cdot \text{m} \cdot \text{s}/\text{rad} & N_2 &= 40 & T_{\text{stall}} &= 10 \text{ N} \cdot \text{m} \\
 J_3 &= 10 \text{ kg} \cdot \text{m}^2 & D_3 &= 1 \text{ N} \cdot \text{m} \cdot \text{s}/\text{rad} & N_3 &= 60 & \omega_{\text{no-load}} &= 4 \text{ rad/s} \\
 & & D_4 &= 1 \text{ N} \cdot \text{m} \cdot \text{s}/\text{rad} & N_4 &= 30 & & \\
 & & D_5 &= 1 \text{ N} \cdot \text{m} \cdot \text{s}/\text{rad} & & & & \\
 & & D_6 &= 1 \text{ N} \cdot \text{m} \cdot \text{s}/\text{rad} & & & &
 \end{aligned}$$

Required:

$$G(s) = \frac{\theta_m(s)}{E_a(s)}$$

Solution:

Gear (kinematic) ratio:

$$r \equiv \frac{\theta_m}{\theta_3} = \frac{N_1 N_3}{N_2 N_4}.$$

With $N_1 = 20$, $N_2 = 40$, $N_3 = 60$, $N_4 = 30$ we get

$$r = \frac{20 \cdot 60}{40 \cdot 30} = 1,$$

so the motor and load rotate at the same speed:

$$\omega_m = \omega_3, \quad \theta_m = \theta_3.$$

Reflection factor for inertias and viscous damping at first gear stage is $\left(\frac{N_1}{N_2}\right)^2$.

Equivalent inertia reflected to motor shaft:

$$J_m = J_a + \left(\frac{N_1}{N_2}\right)^2 (J_2) + \left(\frac{N_1 N_3}{N_2 N_4}\right)^2 (J_3)$$

$$J_m = 2 + \left(\frac{1}{2}\right)^2 (8) + (1)^2 (10)$$

$$J_m = 2 + 2 + 10 = 14 \text{ kg} \cdot \text{m}^2,$$

Equivalent damping reflected to motor shaft:

$$D_m = D_a + \left(\frac{N_1 N_3}{N_2 N_4} \right)^2 (D_2 + D_3 + D_4 + D_5 + D_6)$$

$$D_m = 1 + (1)^2(1 + 1 + 1 + 1 + 1)$$

$$D_m = 1 + 5 = 6 \text{ N} \cdot \text{m} \cdot \text{s/rad}.$$

Motor electrical constants from specifications:

$$\frac{K_t}{R_a} = \frac{T_{\text{stall}}}{e_a} = \frac{10}{20} = \frac{1}{2}$$

$$K_b = \frac{e_a}{\omega_{\text{no-load}}} = \frac{20}{4} = 5.$$

Transfer function (Laplace domain):

$$\frac{\theta_m(s)}{E_a(s)} = \frac{\frac{K_t}{R_a J_m}}{s \left[s + \frac{1}{J_m} \left(D_m + \frac{K_t K_b}{R_a} \right) \right]}$$

Substituting values:

$$\frac{\theta_m(s)}{E_a(s)} = \frac{\frac{1/2}{14}}{s \left[s + \frac{1}{14} \left(6 + \frac{1}{2} \cdot 5 \right) \right]}$$

$$\frac{\theta_m(s)}{E_a(s)} = \frac{\frac{1}{28}}{s \left[s + \frac{1}{14} \left(\frac{12+5}{2} \right) \right]}$$

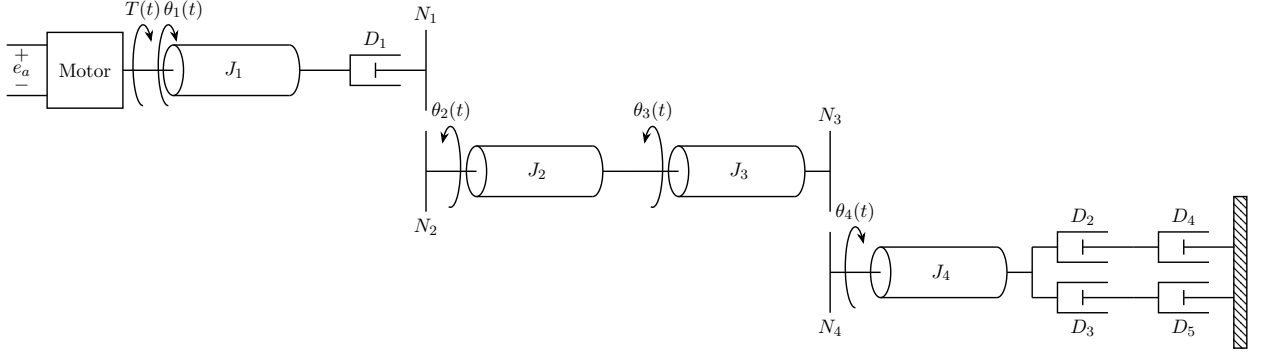
$$\frac{\theta_m(s)}{E_a(s)} = \frac{\frac{1}{28}}{s \left(s + \frac{17}{28} \right)}$$

Multiplying numerator and denominator by 28:

$$\boxed{\frac{\theta_m(s)}{E_a(s)} = \frac{1}{28s^2 + 17s}}$$

Problem 34

A DC motor drives a rotational mechanical system as shown in the figure. Assume that J_1 and D_1 correspond to the motor-side inertia J_a and damping D_a , respectively. Determine the electromechanical transfer function $\frac{\theta_m(s)}{E_a(s)}$.



Given:

$$\begin{aligned}
 J_a &= 5 \text{ kg} \cdot \text{m}^2 & D_a &= 2 \text{ N} \cdot \text{m} \cdot \text{s/rad} & N_1 &= 20 & e_a &= 20 \text{ V} \\
 J_2 &= 12 \text{ kg} \cdot \text{m}^2 & D_2 &= 1 \text{ N} \cdot \text{m} \cdot \text{s/rad} & N_2 &= 40 & T_{\text{stall}} &= 10 \text{ N} \cdot \text{m} \\
 J_3 &= 8 \text{ kg} \cdot \text{m}^2 & D_3 &= 1 \text{ N} \cdot \text{m} \cdot \text{s/rad} & N_3 &= 60 & \omega_{\text{no-load}} &= 4 \text{ rad/s} \\
 J_4 &= 16 \text{ kg} \cdot \text{m}^2 & D_4 &= 1 \text{ N} \cdot \text{m} \cdot \text{s/rad} & N_4 &= 30 & & \\
 & & D_5 &= 1 \text{ N} \cdot \text{m} \cdot \text{s/rad} & & & &
 \end{aligned}$$

Required:

$$G(s) = \frac{\theta_m(s)}{E_a(s)}$$

Solution:

Gear (kinematic) ratio:

$$r \equiv \frac{\theta_m}{\theta_4} = \frac{N_1 N_3}{N_2 N_4}.$$

With $N_1 = 20$, $N_2 = 40$, $N_3 = 60$, $N_4 = 30$ we get

$$r = \frac{20 \cdot 60}{40 \cdot 30} = 1,$$

so the motor and load rotate at the same speed:

$$\omega_m = \omega_4, \quad \theta_m = \theta_4.$$

Reflection factor for inertias and viscous damping at first gear stage is $\left(\frac{N_1}{N_2}\right)^2$.

Equivalent inertia reflected to motor shaft:

$$J_m = J_a + \left(\frac{N_1}{N_2}\right)^2 (J_2 + J_3) + \left(\frac{N_1 N_3}{N_2 N_4}\right)^2 (J_4)$$

$$J_m = 5 + \left(\frac{1}{2}\right)^2 (12 + 8) + (1)^2 (16)$$

$$J_m = 5 + 5 + 16 = 26 \text{ kg} \cdot \text{m}^2,$$

Equivalent damping reflected to motor shaft:

$$D_m = D_a + \left(\frac{N_1 N_3}{N_2 N_4}\right)^2 (D_2 + D_3 + D_4 + D_5)$$

$$D_m = 2 + (1)^2(1 + 1 + 1 + 1)$$

$$D_m = 2 + 4 = 6 \text{ N} \cdot \text{m} - \text{s/rad}.$$

Motor electrical constants from specifications:

$$\frac{K_t}{R_a} = \frac{T_{\text{stall}}}{e_a} = \frac{10}{20} = \frac{1}{2}$$

$$K_b = \frac{e_a}{\omega_{\text{no-load}}} = \frac{20}{4} = 5.$$

Transfer function (Laplace domain):

$$\frac{\theta_m(s)}{E_a(s)} = \frac{\frac{K_t}{R_a J_m}}{s \left[s + \frac{1}{J_m} \left(D_m + \frac{K_t K_b}{R_a} \right) \right]}$$

Substituting values:

$$\frac{\theta_m(s)}{E_a(s)} = \frac{\frac{1/2}{26}}{s \left[s + \frac{1}{26} \left(6 + \frac{1}{2} \cdot 5 \right) \right]}$$

$$\frac{\theta_m(s)}{E_a(s)} = \frac{\frac{1}{52}}{s \left[s + \frac{1}{26} \left(\frac{12+5}{2} \right) \right]}$$

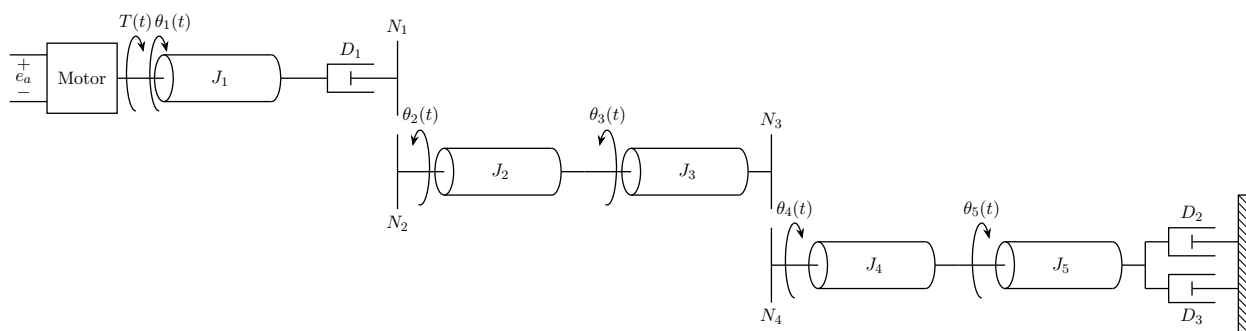
$$\frac{\theta_m(s)}{E_a(s)} = \frac{\frac{1}{52}}{s \left(s + \frac{17}{52} \right)}$$

Multiplying numerator and denominator by 52:

$$\frac{\theta_m(s)}{E_a(s)} = \frac{1}{52s^2 + 17s}$$

Problem 35

A DC motor drives a rotational mechanical system as shown in the figure. Assume that J_1 and D_1 correspond to the motor-side inertia J_a and damping D_a , respectively. Determine the electromechanical transfer function $\frac{\theta_m(s)}{E_a(s)}$.



Given:

$$\begin{aligned}
 J_a &= 5 \text{ kg} \cdot \text{m}^2 & D_a &= 2 \text{ N} \cdot \text{m} \cdot \text{s/rad} & N_1 &= 20 & e_a &= 20 \text{ V} \\
 J_2 &= 8 \text{ kg} \cdot \text{m}^2 & D_2 &= 4 \text{ N} \cdot \text{m} \cdot \text{s/rad} & N_2 &= 40 & T_{\text{stall}} &= 10 \text{ N} \cdot \text{m} \\
 J_3 &= 12 \text{ kg} \cdot \text{m}^2 & D_3 &= 4 \text{ N} \cdot \text{m} \cdot \text{s/rad} & N_3 &= 20 & \omega_{\text{no-load}} &= 5 \text{ rad/s} \\
 J_4 &= 10 \text{ kg} \cdot \text{m}^2 & & & N_4 &= 40 & & \\
 J_5 &= 30 \text{ kg} \cdot \text{m}^2 & & & & & &
 \end{aligned}$$

Required:

$$G(s) = \frac{\theta_m(s)}{E_a(s)}$$

Solution:

Gear (kinematic) ratio:

$$r \equiv \frac{\theta_m}{\theta_5} = \frac{N_1 N_3}{N_2 N_4}.$$

With $N_1 = 20$, $N_2 = 40$, $N_3 = 20$, $N_4 = 40$ we get

$$r = \frac{20 \cdot 20}{40 \cdot 40} = \frac{1}{4},$$

so the load rotates 4 times slower than the motor:

$$\omega_5 = \frac{1}{4} \omega_m, \quad \theta_5 = \frac{1}{4} \theta_m.$$

Reflection factor for inertias and viscous damping is $1/r^2 = 16$.

Equivalent inertia reflected to motor shaft:

$$J_m = J_a + \left(\frac{N_1}{N_2}\right)^2 (J_2 + J_3) + \left(\frac{N_1 N_3}{N_2 N_4}\right)^2 (J_4 + J_5)$$

$$J_m = 5 + \left(\frac{1}{2}\right)^2 (8 + 12) + \left(\frac{1}{4}\right)^2 (10 + 30)$$

$$J_m = 5 + 5 + \frac{5}{2} = \frac{25}{2} \text{ kg} \cdot \text{m}^2,$$

Equivalent damping reflected to motor shaft:

$$D_m = D_a + \left(\frac{N_1 N_3}{N_2 N_4}\right)^2 (D_2 + D_3)$$

$$D_m = 2 + \left(\frac{1}{4}\right)^2 (4 + 4)$$

$$D_m = 2 + \frac{1}{2} = \frac{5}{2} \text{ N} \cdot \text{m} \cdot \text{s} / \text{rad}.$$

Motor electrical constants from specifications:

$$\frac{K_t}{R_a} = \frac{T_{\text{stall}}}{e_a} = \frac{10}{20} = \frac{1}{2}$$

$$K_b = \frac{e_a}{\omega_{\text{no-load}}} = \frac{20}{5} = 4.$$

Transfer function (Laplace domain):

$$\frac{\theta_m(s)}{E_a(s)} = \frac{\frac{K_t}{R_a J_m}}{s \left[s + \frac{1}{J_m} \left(D_m + \frac{K_t K_b}{R_a} \right) \right]}$$

Substituting values:

$$\frac{\theta_m(s)}{E_a(s)} = \frac{\frac{1/2}{25/2}}{s \left[s + \frac{1}{25/2} \left(\frac{5}{2} + \frac{1}{2} \cdot 4 \right) \right]}$$

$$\frac{\theta_m(s)}{E_a(s)} = \frac{\frac{1}{25}}{s \left[s + \frac{2}{25} \left(\frac{5+4}{2} \right) \right]}$$

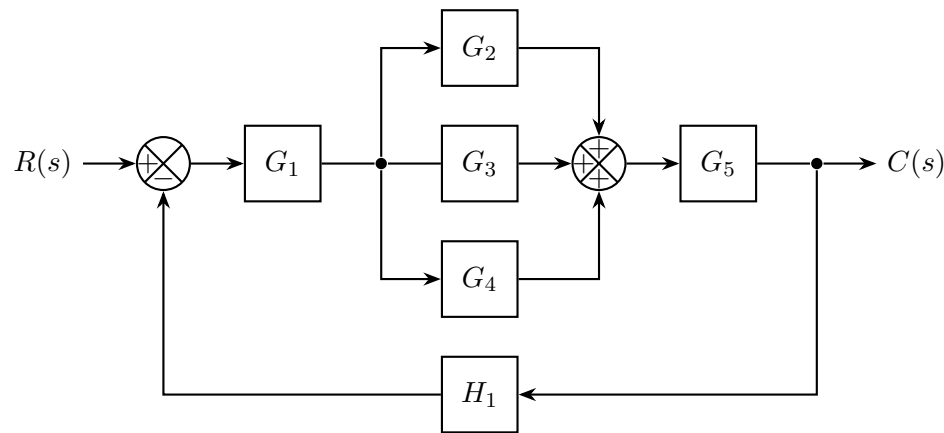
$$\frac{\theta_m(s)}{E_a(s)} = \frac{\frac{1}{25}}{s \left(s + \frac{9}{25} \right)}$$

Multiplying numerator and denominator by 25:

$$\boxed{\frac{\theta_m(s)}{E_a(s)} = \frac{1}{25s^2 + 9s}}$$

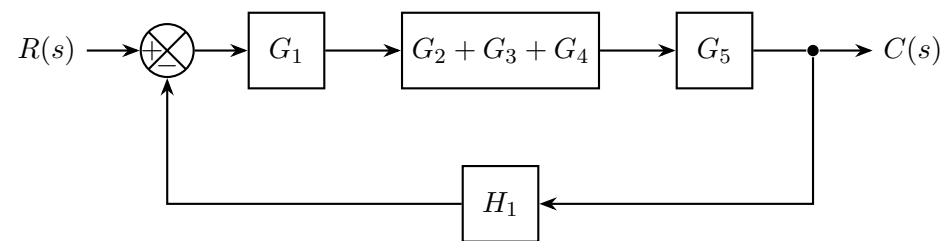
Problem 36

Reduce the given block diagram to its canonical form.

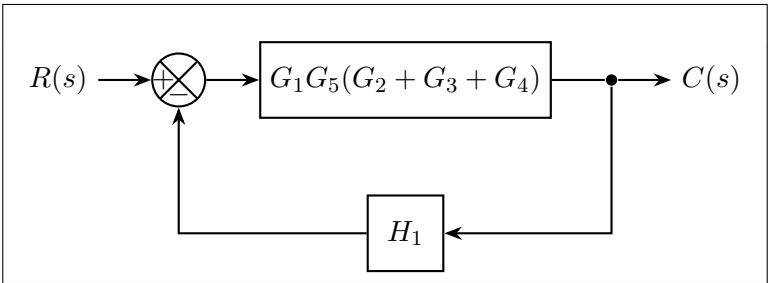


Solution:

G_2 , G_3 , and G_4 are in parallel forming $G_2 + G_3 + G_4$.

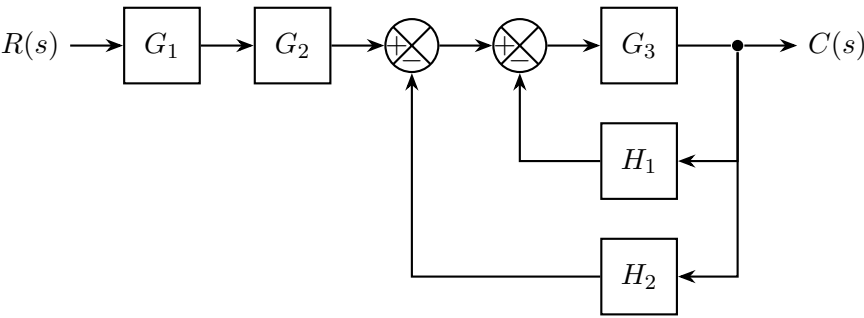


Combining the parallel blocks with G_1 and G_5 in cascade forms $G_1 G_5 (G_2 + G_3 + G_4)$. This is then placed in a feedback loop with H_1 .



Problem 37

Reduce the given block diagram to its canonical form.



Given:

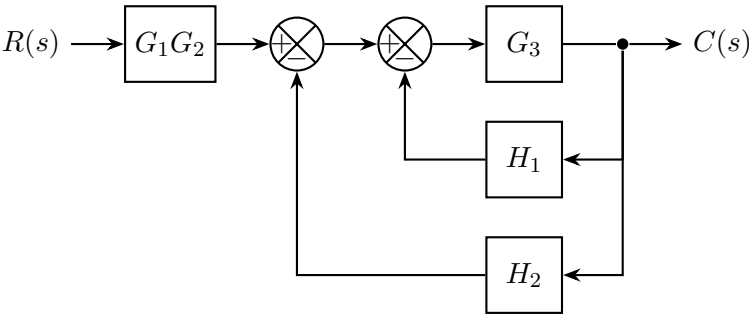
As per drawing

Required:

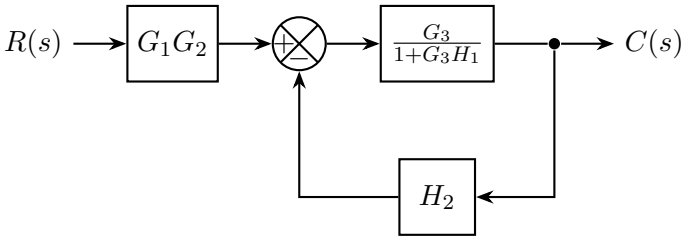
Canonical form, transfer function $\frac{C(s)}{R(s)}$.

Solution:

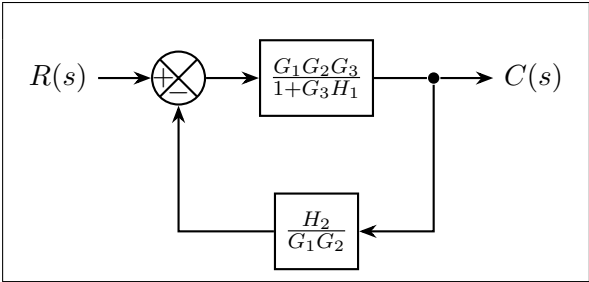
G_1 and G_2 are in cascade forming G_1G_2 .



G_3 and H_1 are in a feedback loop forming $\frac{G_3}{1+G_3H_1}$.

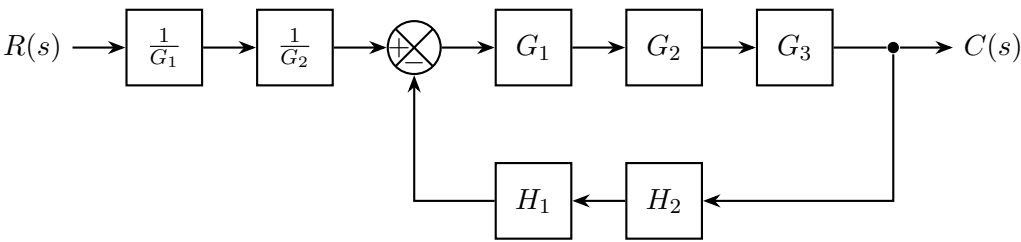


G_1G_2 and $\frac{G_3}{1+G_3H_1}$ are in cascade forming $\frac{G_1G_2G_3}{1+G_3H_1}$. Moving the summing point of H_2 ahead of G_1G_2 scales the feedback by $\frac{1}{G_1G_2}$, forming $\frac{H_2}{G_1G_2}$.



Problem 38

Reduce the given block diagram to its canonical form.



Given:

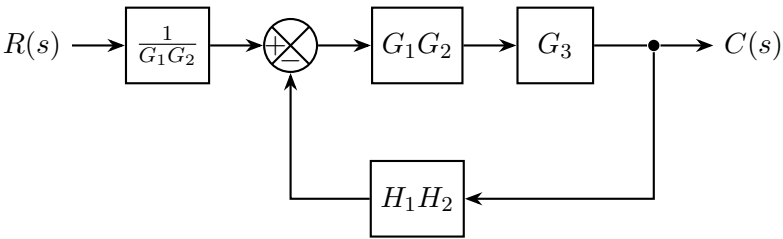
As per drawing

Required:

Canonical form, transfer function $\frac{C(s)}{R(s)}$.

Solution:

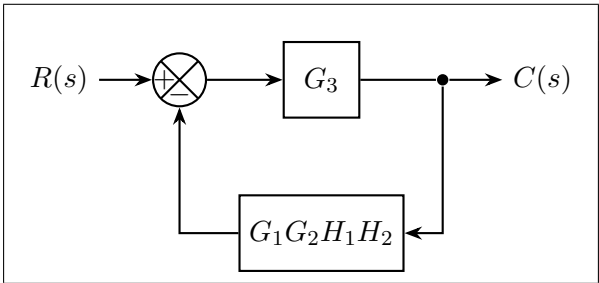
$\frac{1}{G_1}$ and $\frac{1}{G_2}$ are in cascade forming $\frac{1}{G_1G_2}$. G_1 and G_2 are in cascade forming G_1G_2 . H_2 and H_1 are in cascade forming H_1H_2 .



G_1 , $G_2 + G_3 + G_4$, and G_5 are in cascade forming $G_1G_5(G_2 + G_3 + G_4)$.

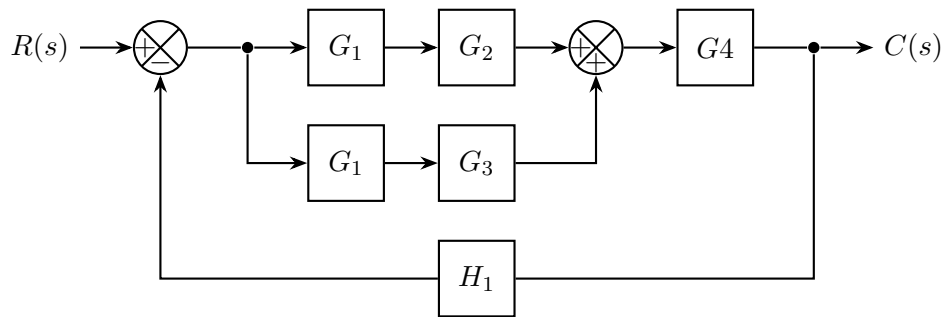
$G_1 + G_2 + G_3 + G_4$ and G_5 are in parallel forming $G_1 + G_2 + G_3 + G_4 + G_5$.

G_6 , G_7 , and the central block are in parallel forming the overall transfer function.



Problem 39

Reduce the given block diagram to its canonical form.



Given:

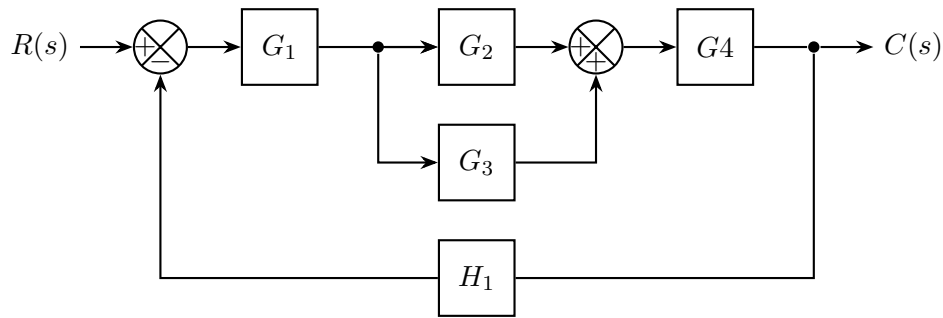
As per drawing

Required:

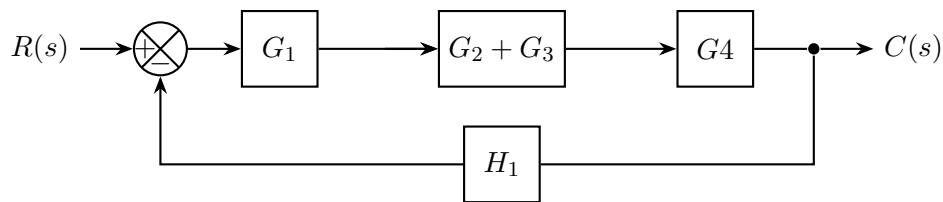
Canonical form, transfer function $\frac{C(s)}{R(s)}$.

Solution:

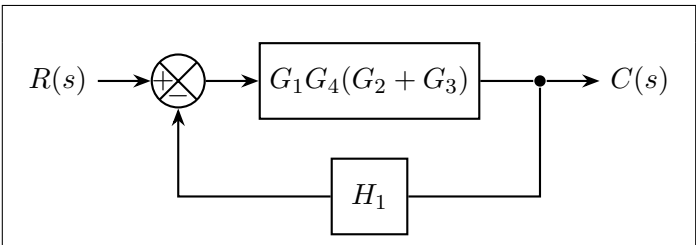
Moving G_1 ahead of the split point.



G_2 and G_3 are in parallel forming $G_2 + G_3$.



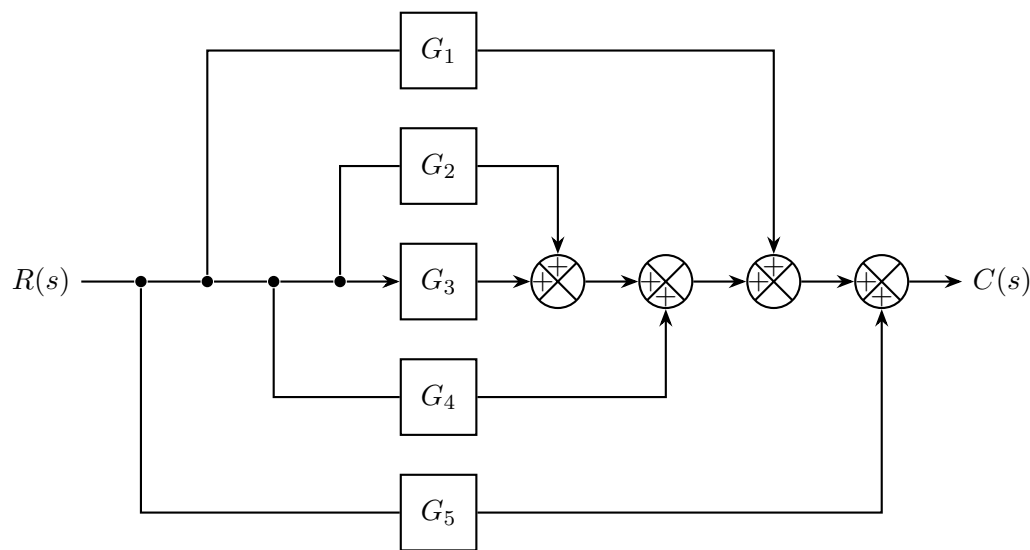
G_1 , $G_2 + G_3$, and G_4 are in cascade forming $G_1G_4(G_2 + G_3)$.



Transfer Function:

$$\frac{C(s)}{R(s)} = \frac{G_1G_4(G_2 + G_3)}{1 + G_1G_4H_1(G_2 + G_3)}$$

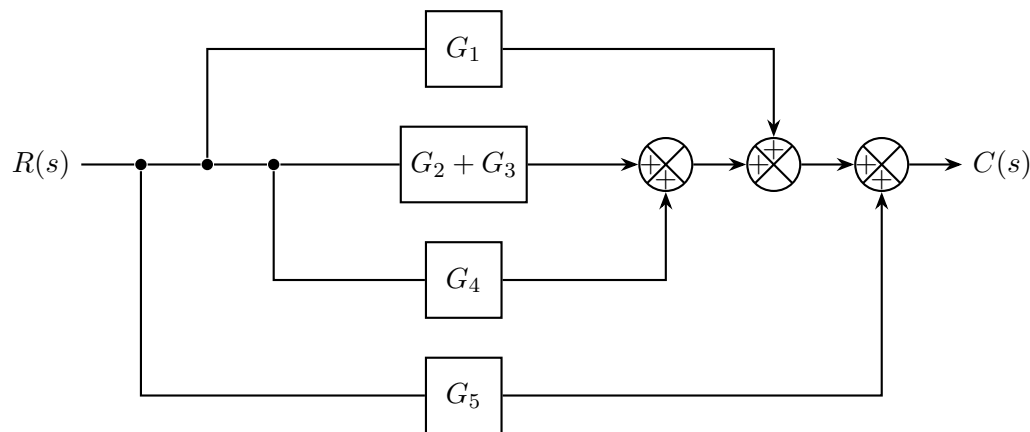
Problem 40
 Reduce the given block diagram to its canonical form.



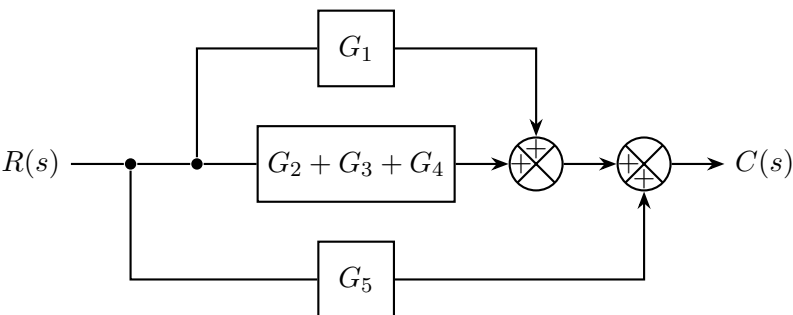
Given:
 As per drawing

Required:
 Canonical form, transfer function $\frac{C(s)}{R(s)}$.

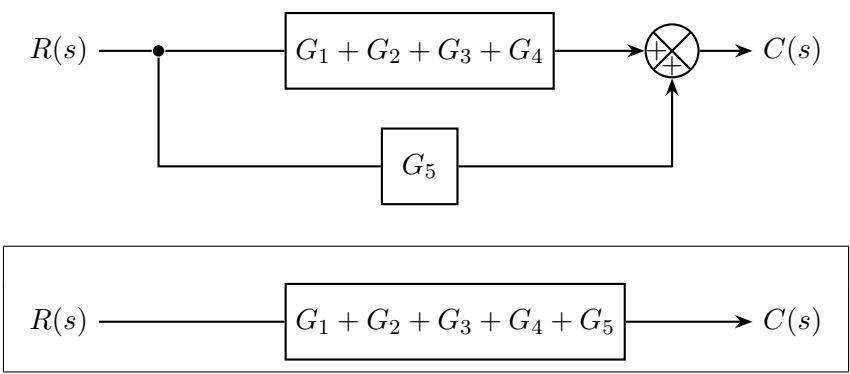
Solution:
 G_2 and G_3 are in parallel forming $G_2 + G_3$.



G_4 and G_5 are in parallel forming $G_4 + G_5$. We also combine the parallel $G_2 + G_3$ with the additional parallel structure.

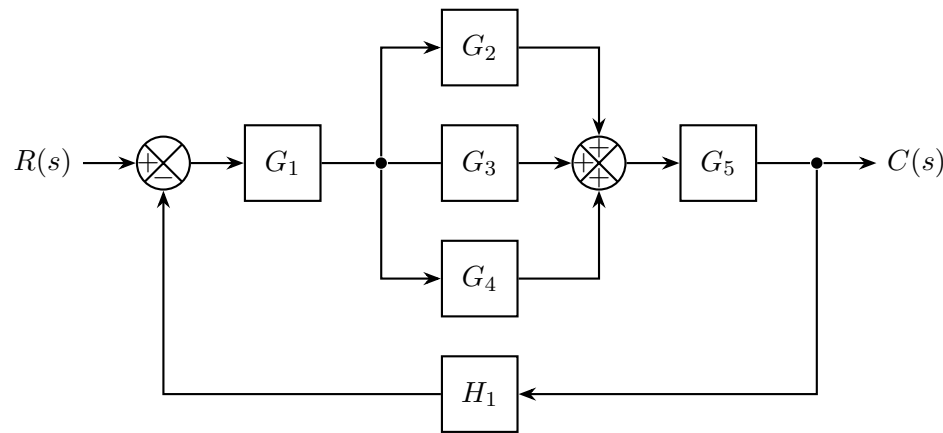


$G_2 + G_3 + G_4$ and G_1 are in parallel forming $G_1 + G_2 + G_3 + G_4$.



Problem 41

Reduce the given block diagram to its canonical form.



Given:

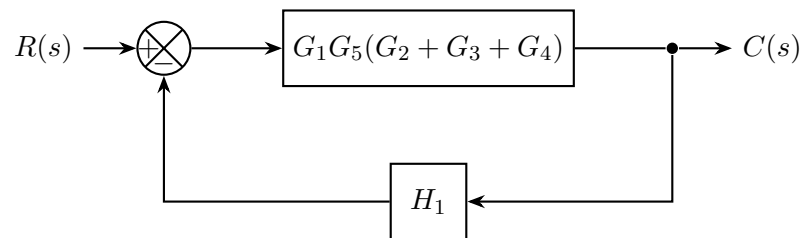
As per drawing

Required:

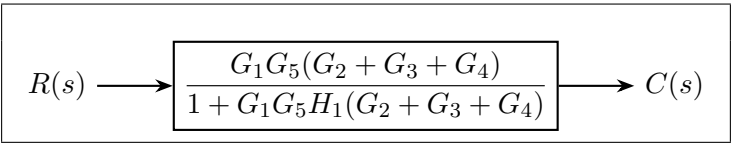
Canonical form, transfer function $\frac{C(s)}{R(s)}$.

Solution:

G_2 , G_3 , and G_4 are in parallel forming $G_2 + G_3 + G_4$. This is in cascade with G_1 and G_5 , forming $G_1G_5(G_2 + G_3 + G_4)$.

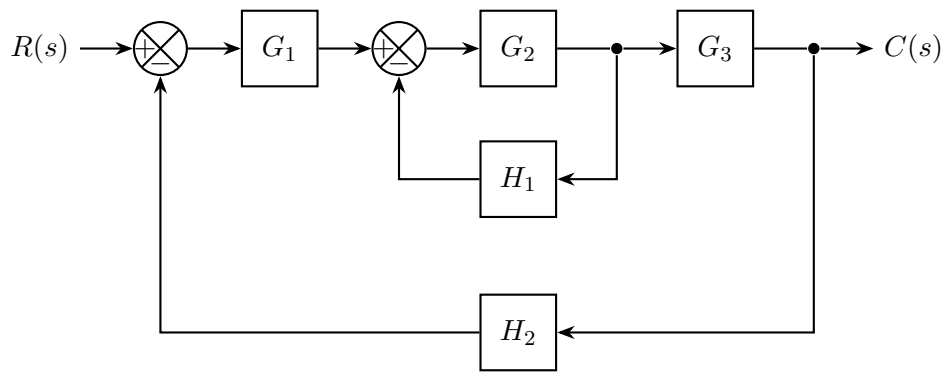


The forward path and feedback loop H_1 are combined forming the closed-loop transfer function.



Problem 42

Reduce the given block diagram to its canonical form.



Given:

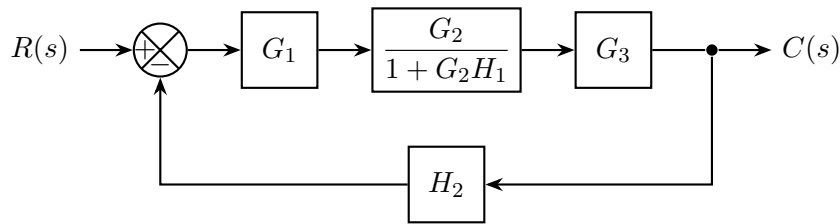
As per drawing

Required:

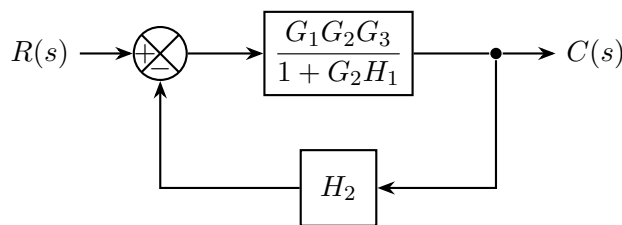
Canonical form, transfer function $\frac{C(s)}{R(s)}$.

Solution:

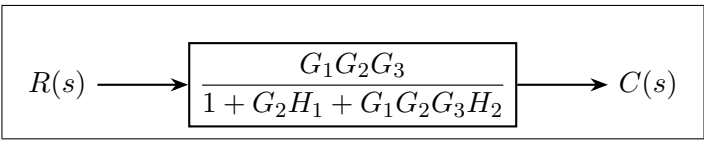
G_2 and H_1 form a feedback loop resulting in $\frac{G_2}{1+G_2H_1}$.



G_1 , $\frac{G_2}{1+G_2H_1}$, and G_3 are in cascade forming $\frac{G_1G_2G_3}{1+G_2H_1}$.

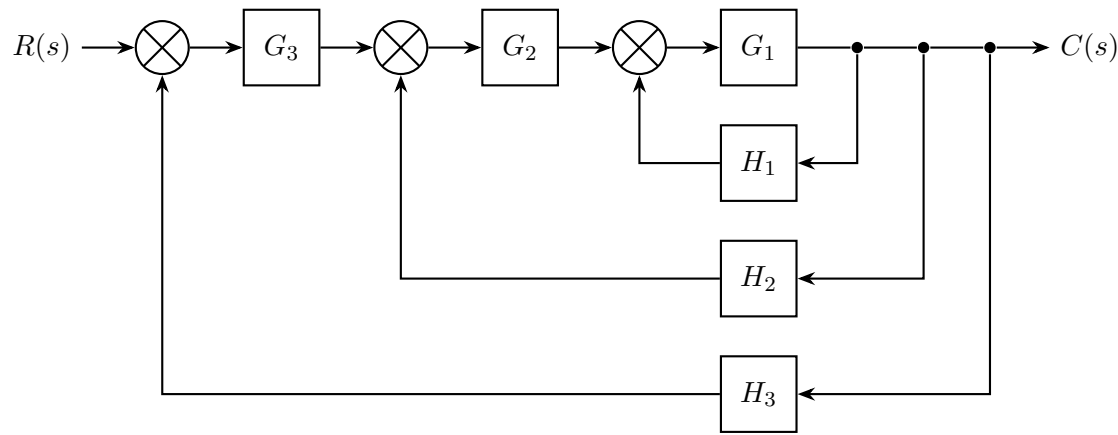


The forward path and feedback loop H_2 are combined forming the closed-loop transfer function.



Problem 43

Reduce the given block diagram to its canonical form.



Given:

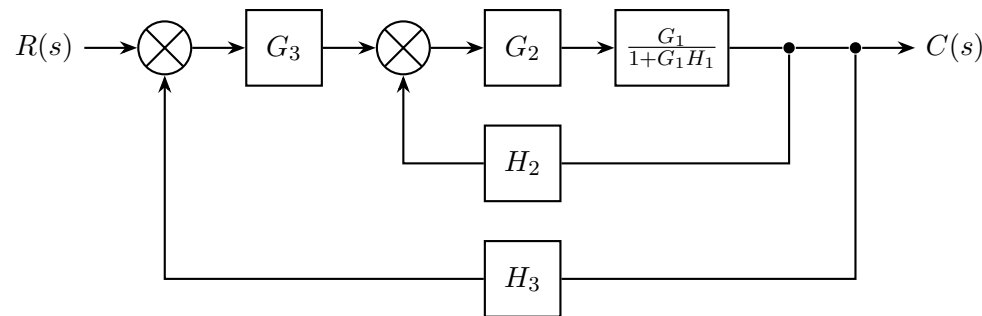
As per drawing

Required:

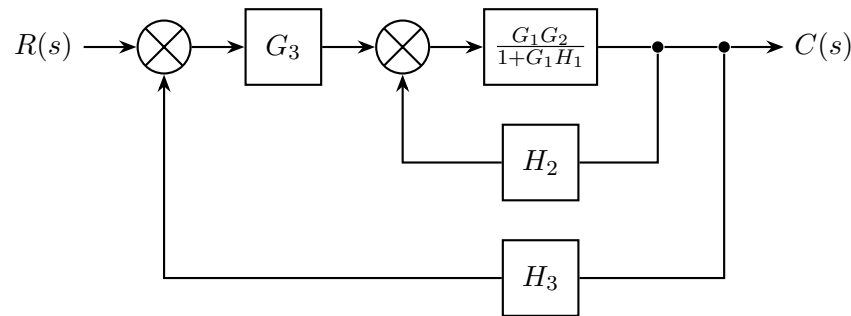
Canonical form, transfer function $\frac{C(s)}{R(s)}$.

Solution:

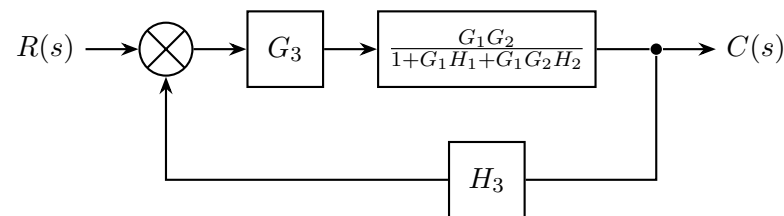
G_1 and H_1 form a feedback loop resulting in $\frac{G_1}{1+G_1H_1}$.



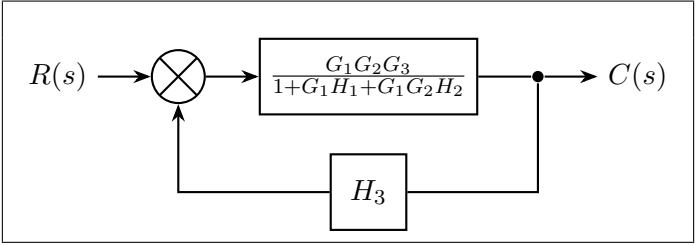
G_2 and $\frac{G_1}{1+G_1H_1}$ are in cascade forming $\frac{G_1G_2}{1+G_1H_1}$.



The forward block $\frac{G_1G_2}{1+G_1H_1}$ and feedback loop H_2 form a closed loop resulting in $\frac{G_1G_2}{1+G_1H_1+G_1G_2H_2}$.

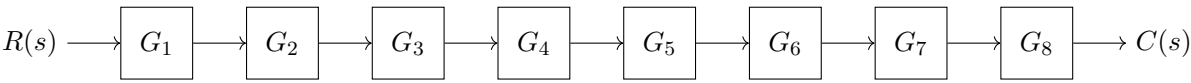


Moving the G_1G_2 block out of the forward path cancels with $\frac{1}{G_1G_2}$ and multiplies the feedback path by G_1G_2 , forming $G_1G_2H_1H_2$.



Problem 44

Reduce the given block diagram to its canonical form.



Given:

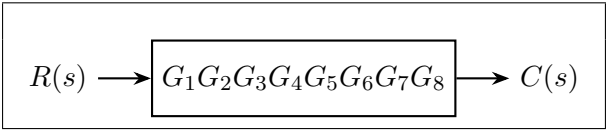
As per drawing

Required:

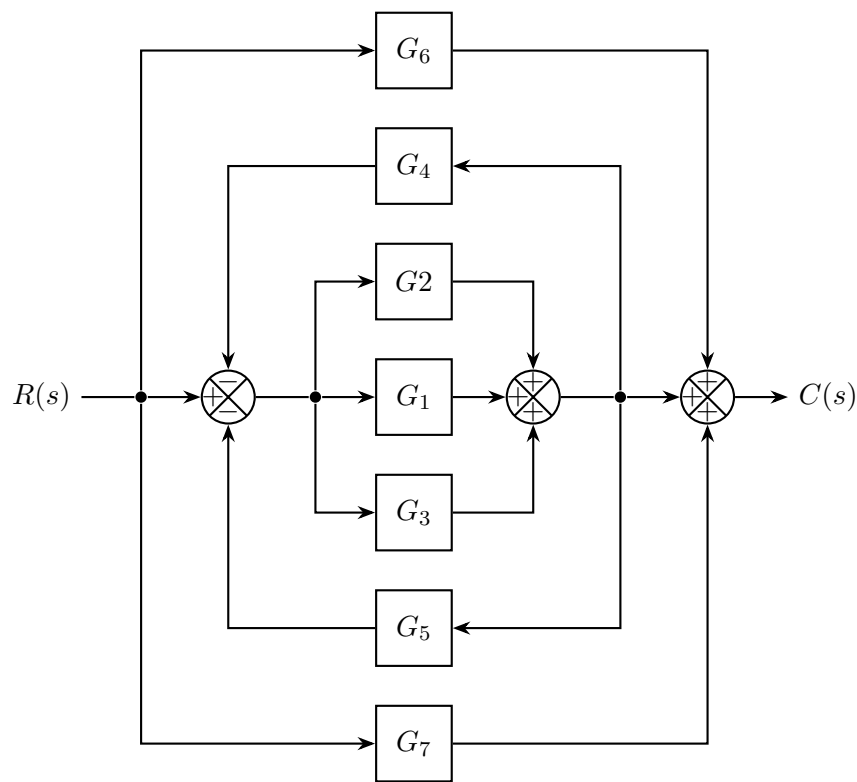
Canonical form, transfer function $\frac{C(s)}{R(s)}$.

Solution:

G_1 through G_8 are in cascade forming $G_1G_2G_3G_4G_5G_6G_7G_8$.



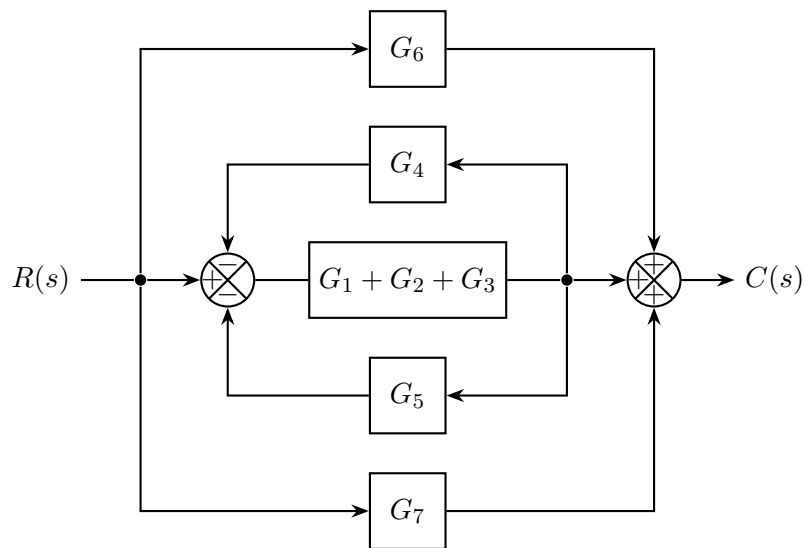
Problem 45
 Reduce the given block diagram to its canonical form.



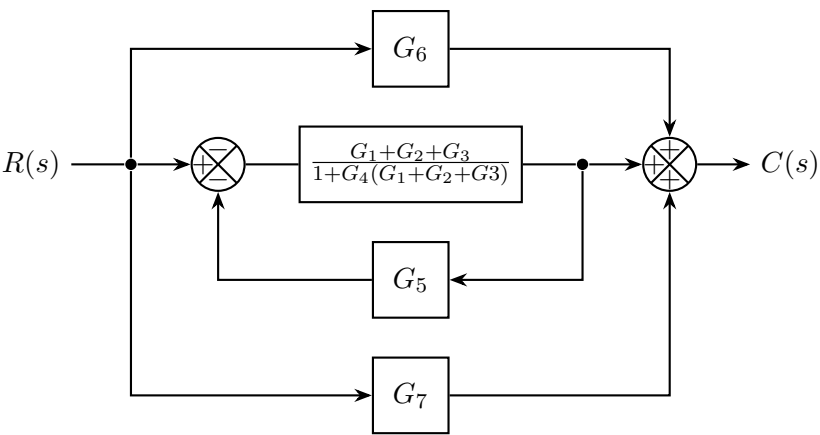
Given:
 As per drawing

Required:
 Canonical form, transfer function $\frac{C(s)}{R(s)}$.

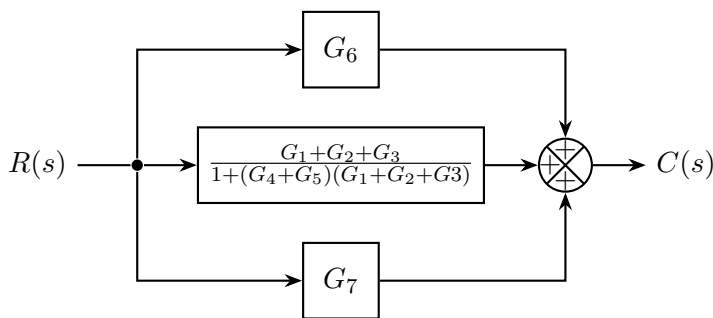
Solution:
 $G_1, G_2,$ and G_3 are in parallel forming $G_1 + G_2 + G_3$.



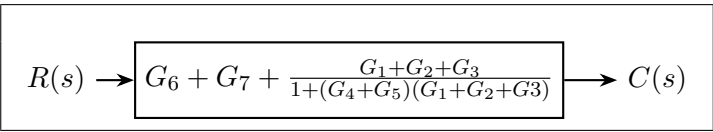
$G_2 + G_3$ and G_4 are in parallel forming $G_2 + G_3 + G_4$.



G_5 is in parallel with the overall structure, forming additional feedback. Combining all parallel and series elements.

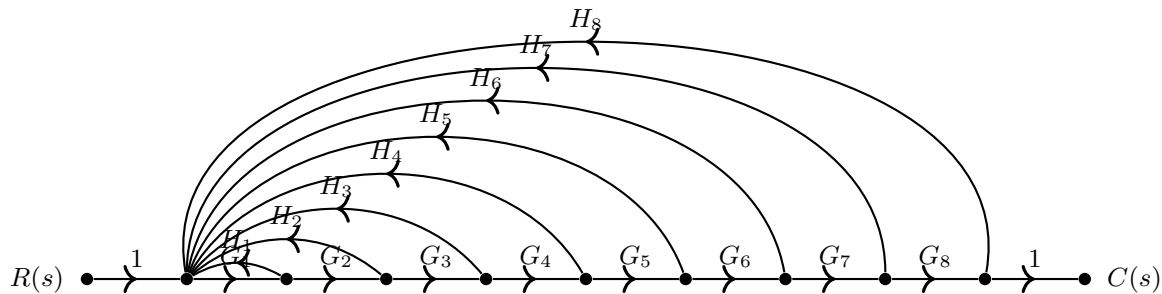


The cascaded parallel blocks form the final transfer function structure, accounting for all G_i and feedback elements.



Problem 46

Determine the transfer function $C(s)/R(s)$ for the signal flow graph shown in the figure using Mason's Gain Formula.



Given:

$$\begin{aligned} G_1 &= 1 & G_5 &= 1 & H_1 &= 1 & H_5 &= 1 \\ G_2 &= 1 & G_6 &= 1 & H_2 &= 1 & H_6 &= 1 \\ G_3 &= 1 & G_7 &= 1 & H_3 &= 1 & H_7 &= 1 \\ G_4 &= 1 & G_8 &= 1 & H_4 &= 1 & H_8 &= 1 \end{aligned}$$

Required:

$$\text{Transfer function } \frac{C(s)}{R(s)}.$$

Solution:

Using Mason's Rule:

$$\frac{C(s)}{R(s)} = \frac{\sum_{k=1}^n T_k \Delta_k}{\Delta}$$

Forward Paths:

$$T_1 = G_1 G_2 G_3 G_4 G_5 G_6 G_7 G_8$$

Feedback Loops:

$$\begin{aligned} L_1 &= G_1 H_1 \\ L_2 &= G_1 G_2 H_2 \\ L_3 &= G_1 G_2 G_3 H_3 \\ L_4 &= G_1 G_2 G_3 G_4 H_4 \\ L_5 &= G_1 G_2 G_3 G_4 G_5 H_5 \\ L_6 &= G_1 G_2 G_3 G_4 G_5 G_6 H_6 \\ L_7 &= G_1 G_2 G_3 G_4 G_5 G_6 G_7 H_7 \\ L_8 &= G_1 G_2 G_3 G_4 G_5 G_6 G_7 G_8 H_8 \end{aligned}$$

Non-touching Loop Pairs:

None

Delta:

$$\begin{aligned} \Delta &= 1 - (L_1 + L_2 + L_3 + L_4 + L_5 + L_6 + L_7 + L_8) \\ &= 1 - (G_1 H_1 + G_1 G_2 H_2 + G_1 G_2 G_3 H_3 + G_1 G_2 G_3 G_4 H_4 + G_1 G_2 G_3 G_4 G_5 H_5 \\ &\quad + G_1 G_2 G_3 G_4 G_5 G_6 H_6 + G_1 G_2 G_3 G_4 G_5 G_6 G_7 H_7 + G_1 G_2 G_3 G_4 G_5 G_6 G_7 G_8 H_8) \end{aligned}$$

Delta for Individual Forward Paths:

$$\Delta_1 = 1$$

Transfer Function:

$$\frac{C(s)}{R(s)} = \frac{T_1 \Delta_1}{\Delta}$$

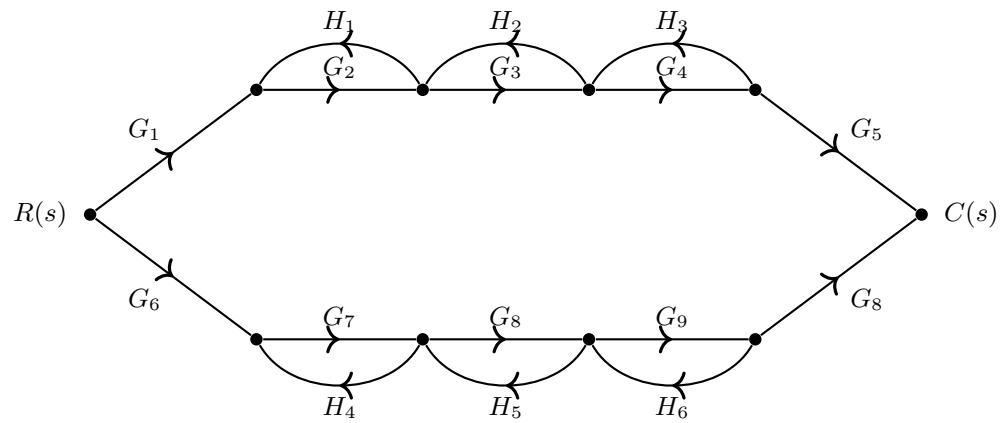
$$\frac{C(s)}{R(s)} = \frac{G_1 G_2 G_3 G_4 G_5 G_6 G_7 G_8}{1 - (G_1 H_1 + G_1 G_2 H_2 + G_1 G_2 G_3 H_3 + G_1 G_2 G_3 G_4 H_4 + G_1 G_2 G_3 G_4 G_5 H_5 + G_1 G_2 G_3 G_4 G_5 G_6 H_6 + G_1 G_2 G_3 G_4 G_5 G_6 G_7 H_7 + G_1 G_2 G_3 G_4 G_5 G_6 G_7 G_8 H_8)}$$

Substituting the given values:

$$\frac{C(s)}{R(s)} = \frac{1}{1 - 8} = -\frac{1}{7}$$

Problem 47

Determine the transfer function $C(s)/R(s)$ for the signal flow graph shown in the figure using Mason's Gain Formula.



Given:

$$\begin{aligned} G_1 &= 1 & G_6 &= 1 & H_1 &= 1 & H_6 &= 1 \\ G_2 &= 1 & G_7 &= 1 & H_2 &= 1 & & \\ G_3 &= 1 & G_8 &= 1 & H_3 &= 1 & & \\ G_4 &= 1 & G_9 &= 1 & H_4 &= 1 & & \\ G_5 &= 1 & & & H_5 &= 1 & & \end{aligned}$$

Required:

Transfer function $\frac{C(s)}{R(s)}$.

Solution:

Using Mason's Rule:

$$\frac{C(s)}{R(s)} = \frac{\sum_{k=1}^n T_k \Delta_k}{\Delta}$$

Forward Paths:

$$\begin{aligned} T_1 &= G_1 G_2 G_3 G_4 G_5 \\ T_2 &= G_6 G_7 G_8 G_9 \end{aligned}$$

Feedback Loops:

$$\begin{aligned} L_1 &= G_2 H_1 \\ L_2 &= G_3 H_2 \\ L_3 &= G_4 H_3 \\ L_4 &= G_7 H_4 \\ L_5 &= G_8 H_5 \\ L_6 &= G_9 H_6 \end{aligned}$$

Non-touching Loop Pairs:

$$\begin{aligned}
L_1L_3 &= G_2G_4H_1H_3 \\
L_4L_6 &= G_7G_9H_4H_6 \\
L_1L_4 &= G_2G_7H_1H_4 \\
L_1L_5 &= G_2G_8H_1H_5 \\
L_1L_6 &= G_2G_9H_1H_6 \\
L_2L_4 &= G_3G_7H_2H_4 \\
L_2L_5 &= G_3G_8H_2H_5 \\
L_2L_6 &= G_3G_9H_2H_6 \\
L_3L_4 &= G_4G_7H_3H_4 \\
L_3L_5 &= G_4G_8H_3H_5 \\
L_3L_6 &= G_4G_9H_3H_6
\end{aligned}$$

Three Non-touching Loops:

$$\begin{aligned}
L_1L_3L_4 &= G_2G_4G_7H_1H_3H_4 \\
L_1L_3L_5 &= G_2G_4G_8H_1H_3H_5 \\
L_1L_3L_6 &= G_2G_4G_9H_1H_3H_6 \\
L_1L_4L_6 &= G_2G_7G_9H_1H_4H_6 \\
L_2L_4L_6 &= G_3G_7G_9H_2H_4H_6 \\
L_3L_4L_6 &= G_4G_7G_9H_3H_4H_6
\end{aligned}$$

Four Non-touching Loops:

$$L_1L_3L_4L_6 = G_2G_4G_7G_9H_1H_3H_4H_6$$

Delta:

$$\begin{aligned}
\Delta &= 1 - (L_1 + L_2 + L_3 + L_4 + L_5 + L_6) \\
&\quad + (L_1L_3 + L_4L_6 + L_1L_4 + L_1L_5 + L_1L_6 + L_2L_4 + L_2L_5 + L_2L_6 + L_3L_4 + L_3L_5 + L_3L_6) \\
&\quad - (L_1L_3L_4 + L_1L_3L_5 + L_1L_3L_6 + L_1L_4L_6 + L_2L_4L_6 + L_3L_4L_6) \\
&\quad + (L_1L_3L_4L_6)
\end{aligned}$$

Delta for Individual Forward Paths:

$$\begin{aligned}
\Delta_1 &= 1 - (L_4 + L_5 + L_6) + L_4L_6 \\
\Delta_2 &= 1 - (L_1 + L_2 + L_3) + L_1L_3
\end{aligned}$$

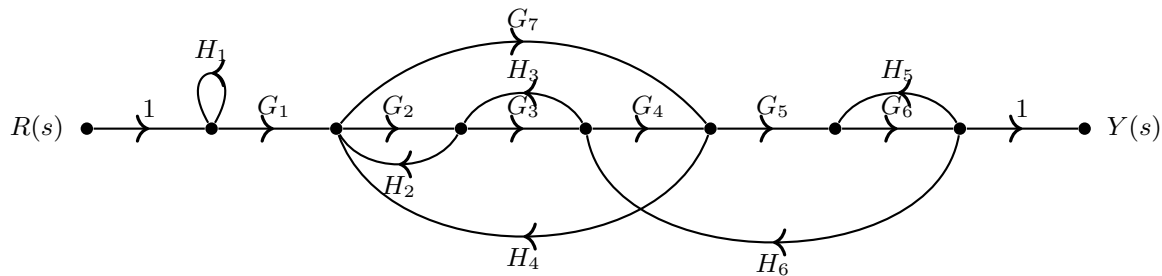
Transfer Function:

$$\frac{C(s)}{R(s)} = \frac{T_1\Delta_1 + T_2\Delta_2}{\Delta}$$

$ \begin{aligned} &G_1G_2G_3G_4G_5(1 - G_7H_4 - G_8H_5 - G_9H_6 + G_7G_9H_4H_6) \\ &+ G_6G_7G_8G_9G_8(1 - G_2H_1 - G_3H_2 - G_4H_3 + G_2G_4H_1H_3) \end{aligned} $ $ \frac{C(s)}{R(s)} = \frac{\text{above}}{1 - (G_2H_1 + G_3H_2 + G_4H_3 + G_7H_4 + G_8H_5 + G_9H_6)} $ $ \begin{aligned} &+ (G_2G_4H_1H_3 + G_7G_9H_4H_6 + G_2G_7H_1H_4 + G_2G_8H_1H_5 + G_2G_9H_1H_6 \\ &\quad + G_3G_7H_2H_4 + G_3G_8H_2H_5 + G_3G_9H_2H_6 + G_4G_7H_3H_4 + G_4G_8H_3H_5 + G_4G_9H_3H_6) \\ &- (G_2G_4G_7H_1H_3H_4 + G_2G_4G_8H_1H_3H_5 + G_2G_4G_9H_1H_3H_6 \\ &\quad + G_2G_7G_9H_1H_4H_6 + G_3G_7G_9H_2H_4H_6 + G_4G_7G_9H_3H_4H_6) \\ &+ (G_2G_4G_7G_9H_1H_3H_4H_6) \end{aligned} $ Substituting the given values: $ \frac{C(s)}{R(s)} = \frac{1(1 - 3 + 1) + 1(1 - 3 + 1)}{1 - 6 + 11 - 6 + 1} = \frac{-1 - 1}{1} = -2 $

Problem 48

Determine the transfer function $Y(s)/R(s)$ for the signal flow graph shown in the figure using Mason's Gain Formula.



Given:

$$\begin{aligned} G_1 &= 2 & G_5 &= 1 & H_2 &= 1 & H_6 &= 1 \\ G_2 &= 1 & G_6 &= 1 & H_3 &= 1 & & \\ G_3 &= 1 & G_7 &= 2 & H_4 &= 1 & & \\ G_4 &= 1 & H_1 &= 1 & H_5 &= 1 & & \end{aligned}$$

Required:

Transfer function $\frac{Y(s)}{R(s)}$.

Solution:

Using Mason's Rule:

$$\frac{Y(s)}{R(s)} = \frac{\sum_{k=1}^n T_k \Delta_k}{\Delta}$$

Forward Paths:

$$\begin{aligned} T_1 &= G_1 G_2 G_3 G_4 G_5 G_6 \\ T_2 &= G_1 G_7 G_5 G_6 \end{aligned}$$

Feedback Loops:

$$\begin{aligned} L_1 &= H_1 \\ L_2 &= G_2 H_2 \\ L_3 &= G_3 H_3 \\ L_4 &= G_2 G_3 G_4 H_4 \\ L_5 &= G_6 H_5 \\ L_6 &= G_4 G_5 G_6 H_6 \end{aligned}$$

Non-touching Loop Pairs:

$$\begin{aligned} L_1 L_2 &= H_1 G_2 H_2 \\ L_1 L_3 &= H_1 G_3 H_3 \\ L_1 L_4 &= H_1 G_2 G_3 G_4 H_4 \\ L_1 L_5 &= H_1 G_6 H_5 \\ L_1 L_6 &= H_1 G_4 G_5 G_6 H_6 \\ L_2 L_5 &= G_2 H_2 G_6 H_5 \\ L_2 L_6 &= G_2 H_2 G_4 G_5 G_6 H_6 \\ L_3 L_5 &= G_3 H_3 G_6 H_5 \\ L_4 L_5 &= G_2 G_3 G_4 H_4 G_6 H_5 \end{aligned}$$

Three Non-touching Loops:

$$\begin{aligned}
L_1L_2L_5 &= H_1G_2H_2G_6H_5 \\
L_1L_2L_6 &= H_1G_2H_2G_4G_5G_6H_6 \\
L_1L_3L_5 &= H_1G_3H_3G_6H_5 \\
L_1L_4L_5 &= H_1G_2G_3G_4H_4G_6H_5
\end{aligned}$$

Delta:

$$\begin{aligned}
\Delta &= 1 - (L_1 + L_2 + L_3 + L_4 + L_5 + L_6) \\
&\quad + (L_1L_2 + L_1L_3 + L_1L_4 + L_1L_5 + L_1L_6 + L_2L_5 + L_2L_6 + L_3L_5 + L_4L_5) \\
&\quad - (L_1L_2L_5 + L_1L_2L_6 + L_1L_3L_5 + L_1L_4L_5)
\end{aligned}$$

Delta for Individual Forward Paths:

$$\begin{aligned}
\Delta_1 &= 1 \\
\Delta_2 &= 1
\end{aligned}$$

Transfer Function:

$$\frac{Y(s)}{R(s)} = \frac{T_1\Delta_1 + T_2\Delta_2}{\Delta}$$

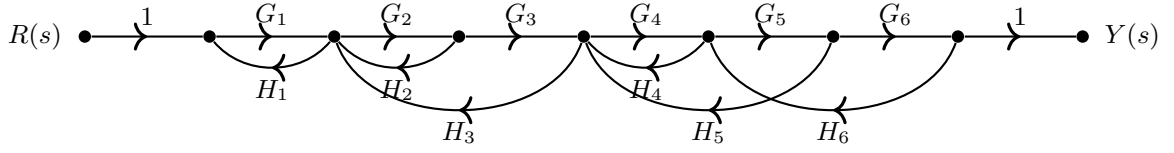
$$\begin{aligned}
\frac{Y(s)}{R(s)} &= \frac{G_1G_2G_3G_4G_5G_6 + G_1G_7G_5G_6}{1 - (H_1 + G_2H_2 + G_3H_3 + G_2G_3G_4H_4 + G_6H_5 + G_4G_5G_6H_6) \\
&\quad + (H_1G_2H_2 + H_1G_3H_3 + H_1G_2G_3G_4H_4 + H_1G_6H_5 + H_1G_4G_5G_6H_6 \\
&\quad + G_2H_2G_6H_5 + G_2H_2G_4G_5G_6H_6 + G_3H_3G_6H_5 + G_2G_3G_4H_4G_6H_5) \\
&\quad - (H_1G_2H_2G_6H_5 + H_1G_2H_2G_4G_5G_6H_6 + H_1G_3H_3G_6H_5 + H_1G_2G_3G_4H_4G_6H_5)}
\end{aligned}$$

Substituting the given values (using $H_1 = 2$ to avoid zero determinant):

$$\frac{Y(s)}{R(s)} = \frac{2(1) + 4(1)}{1 - 7 + 13 - 8} = \frac{6}{-1} = -6$$

Problem 49

Determine the transfer function $Y(s)/R(s)$ for the signal flow graph shown in the figure using Mason's Gain Formula.



Given:

$$\begin{aligned} G_1 &= 2 & G_5 &= 2 & H_3 &= 1 \\ G_2 &= 1 & G_6 &= 1 & H_4 &= 1 \\ G_3 &= 2 & H_1 &= 1 & H_5 &= 1 \\ G_4 &= 1 & H_2 &= 1 & H_6 &= 1 \end{aligned}$$

Required:

$$\text{Transfer function } \frac{Y(s)}{R(s)}.$$

Solution:

Using Mason's Rule:

$$\frac{Y(s)}{R(s)} = \frac{\sum_{k=1}^n T_k \Delta_k}{\Delta}$$

Forward Paths:

$$T_1 = G_1 G_2 G_3 G_4 G_5 G_6$$

Feedback Loops:

$$\begin{aligned} L_1 &= -G_1 H_1 \\ L_2 &= -G_2 H_2 \\ L_3 &= -G_2 G_3 H_3 \\ L_4 &= -G_4 H_4 \\ L_5 &= -G_4 G_5 H_5 \\ L_6 &= -G_5 G_6 H_6 \end{aligned}$$

Non-touching Loop Pairs:

$$\begin{aligned} L_1 L_4 &= G_1 G_4 H_1 H_4 \\ L_1 L_5 &= G_1 G_4 G_5 H_1 H_5 \\ L_1 L_6 &= G_1 G_5 G_6 H_1 H_6 \\ L_2 L_4 &= G_2 G_4 H_2 H_4 \\ L_2 L_5 &= G_2 G_4 G_5 H_2 H_5 \\ L_2 L_6 &= G_2 G_5 G_6 H_2 H_6 \\ L_3 L_6 &= G_2 G_3 G_5 G_6 H_3 H_6 \end{aligned}$$

Delta:

$$\begin{aligned} \Delta &= 1 - (L_1 + L_2 + L_3 + L_4 + L_5 + L_6) \\ &\quad + (L_1 L_4 + L_1 L_5 + L_1 L_6 + L_2 L_4 + L_2 L_5 + L_2 L_6 + L_3 L_6) \\ &= 1 - (-G_1 H_1 - G_2 H_2 - G_2 G_3 H_3 - G_4 H_4 - G_4 G_5 H_5 - G_5 G_6 H_6) \\ &\quad + (G_1 G_4 H_1 H_4 + G_1 G_4 G_5 H_1 H_5 + G_1 G_5 G_6 H_1 H_6 + G_2 G_4 H_2 H_4 \\ &\quad + G_2 G_4 G_5 H_2 H_5 + G_2 G_5 G_6 H_2 H_6 + G_2 G_3 G_5 G_6 H_3 H_6) \end{aligned}$$

Delta for Individual Forward Paths:

$$\Delta_1 = 1$$

Transfer Function:

$$\frac{Y(s)}{R(s)} = \frac{T_1 \Delta_1}{\Delta}$$

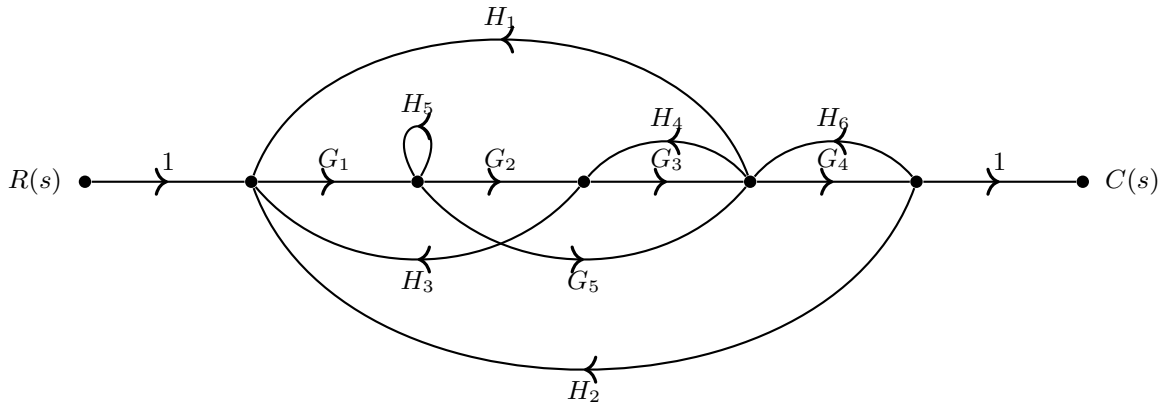
$$\frac{Y(s)}{R(s)} = \frac{G_1 G_2 G_3 G_4 G_5 G_6}{1 + G_1 H_1 + G_2 H_2 + G_2 G_3 H_3 + G_4 H_4 + G_4 G_5 H_5 + G_5 G_6 H_6 + G_1 G_4 H_1 H_4 + G_1 G_4 G_5 H_1 H_5 + G_1 G_5 G_6 H_1 H_6 + G_2 G_4 H_2 H_4 + G_2 G_4 G_5 H_2 H_5 + G_2 G_5 G_6 H_2 H_6 + G_2 G_3 G_5 G_6 H_3 H_6}$$

Substituting the given values:

$$\frac{Y(s)}{R(s)} = \frac{8}{1 - (-10) + 19} = \frac{8}{30} = \frac{4}{15}$$

Problem 50

Determine the transfer function $C(s)/R(s)$ for the signal flow graph shown in the figure using Mason's Gain Formula.



Given:

$$\begin{aligned} G_1 &= 2 & G_5 &= 2 & H_4 &= 1 \\ G_2 &= 1 & H_1 &= 1 & H_5 &= 1 \\ G_3 &= 2 & H_2 &= 1 & H_6 &= 1 \\ G_4 &= 1 & H_3 &= 1 \end{aligned}$$

Required:

$$\text{Transfer function } \frac{C(s)}{R(s)}.$$

Solution:

Using Mason's Rule:

$$\frac{C(s)}{R(s)} = \frac{\sum_{k=1}^n T_k \Delta_k}{\Delta}$$

Forward Paths:

$$\begin{aligned} T_1 &= G_1 G_2 G_3 G_4 \\ T_2 &= G_1 G_5 G_4 \end{aligned}$$

Feedback Loops:

$$\begin{aligned} L_1 &= G_1 G_2 G_3 H_1 \\ L_2 &= G_1 G_2 G_3 G_4 H_2 \\ L_3 &= G_1 G_5 G_4 H_2 \\ L_4 &= G_1 G_5 H_1 \\ L_5 &= G_1 G_2 H_3 \\ L_6 &= G_3 H_4 \\ L_7 &= H_5 \\ L_8 &= G_4 H_6 \end{aligned}$$

Non-touching Loop Pairs:

$$\begin{aligned} L_6 L_7 &= G_3 H_4 H_5 \\ L_7 L_8 &= H_5 G_4 H_6 \\ L_5 L_8 &= G_1 G_2 H_3 G_4 H_6 \end{aligned}$$

Delta:

$$\begin{aligned}\Delta &= 1 - (L_1 + L_2 + L_3 + L_4 + L_5 + L_6 + L_7 + L_8) + (L_6L_7 + L_7L_8 + L_5L_8) \\ &= 1 - (G_1G_2G_3H_1 + G_1G_2G_3G_4H_2 + G_1G_5G_4H_2 + G_1G_5H_1 + G_1G_2H_3 + G_3H_4 + H_5 + G_4H_6) \\ &\quad + (G_3H_4H_5 + H_5G_4H_6 + G_1G_2H_3G_4H_6)\end{aligned}$$

Delta for Individual Forward Paths:

$$\Delta_1 = 1$$

$$\Delta_2 = 1$$

Transfer Function:

$$\frac{C(s)}{R(s)} = \frac{T_1\Delta_1 + T_2\Delta_2}{\Delta}$$

$$\frac{C(s)}{R(s)} = \frac{G_1G_2G_3G_4 + G_1G_5G_4}{1 - (G_1G_2G_3H_1 + G_1G_2G_3G_4H_2 + G_1G_5G_4H_2 + G_1G_5H_1 + G_1G_2H_3 + G_3H_4 + H_5 + G_4H_6) + (G_3H_4H_5 + H_5G_4H_6 + G_1G_2H_3G_4H_6)}$$

Substituting the given values:

$$\frac{C(s)}{R(s)} = \frac{4 + 4}{1 - 22 + 5} = \frac{8}{-16} = -\frac{1}{2}$$